

A Heterogeneous Moran Process for the Analysis of Public Goods Games

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1 Introduction

2 Literature Review

Evolutionary game theory has long been at the forefront of the study of emergent cooperation, showing how a strategy that is seemingly unfavourable for the individual, but beneficial to the collective, can become dominant in systems of rational actors. This is often shown through the study of games such as the iterated prisoner’s dilemma[7, 1] and the snowdrift game[17]. These examples, however, only allow for pairwise interactions, which does not accurately represent many real-world scenarios. Therefore, we look at a common game used to model the sharing of resources and interactions between multiple people - the public goods game.

In order to further tailor the idea of a public goods game to real world scenarios, we introduce the idea of heterogeneity to the system. Often in the study of games we consider each individual to exist in the same conditions - having the same utility and probability of transitioning to a different action type. However, this does not accurately reflect many of the interactions that occur in the real world, where individuals may have different factors affecting which strategies they are able to take. For example, in a public goods game, one player may wish to contribute x units to the public good, however they do not have enough wealth available to spare such a contribution. Heterogeneity allows us to give varying attributes to the different actors in order to simulate such situations. These attributes can be encoded into the players themselves as separate to the game, for example the reputation in [10], they may exist within the strategies of the players [9, 5], or they may exist within the actual payoffs which the players receive.

In much of the literature, we see heterogeneous public goods game being played graphically. By this I mean that in much research, players are placed on graphs (often regular square lattices, as in [10, 5]) and participate in multiple public goods games at the same time, with their payoff being the combined payoff from all their games. This gives even more weight to the decision to cooperate or defect, and allows players to interact in scenarios with many different levels of these heterogeneous attributes. We also see many ideas for what these attributes could be based upon. [10] takes the idea of that players could build a reputation through their decisions across multiple generations, with players refusing to contribute higher amounts to groups made up of those who have a reputation for defection. While this is a dynamically updating heterogeneous attribute, some have proposed static attributes based on inherent properties of the individuals, such as in [9] where the players participate in different amounts of games, and contribute different amounts based on this factor.

In both of the above cases, it is shown that an increased scrutiny of heterogeneous attributes in a spatial public goods game is beneficial to the emergence of cooperation, though for different reasons. In [10], we see that increasing such scrutiny obviously harms defectors, as they receive reduced payoffs based on their low reputation due to being unable to coerce high contributions from other players. [9], however, finds that if we encourage players to cooperate with those in a similar number of games to them, here done by tuning the β parameter, the high-degree defector nodes are unable to gather a large payoff, and so will be unable to spread their defection to their neighbours. If you encourage players to interact with those with different degrees, however, then middle-degree defectors will make heavy losses and be unable to survive.

[5] shows how the benefits of this heterogeneity are not always as pronounced as it may seem. When the contribution is not calculated by some attribute, but rather is an attribute in and of itself (essentially becoming a new strategy) then even in systems where contribution dominates, we will see low-value contributions act as a sort of defection against the most globally beneficial strategies. Therefore, we can see that the emergence of cooperation is not the end of the story in some cases, and we must look at what sort of cooperation it is that we have fostered the emergence of.

One common theme in the spatial public goods game is the manner in which cooperation spreads. An initial “invasion” of defectors leads to cooperation mainly existing within small clusters. These clusters, however, are much more profitable than their defecting neighbours. Therefore, the cooperating clusters will influence the nearby defectors to join in, which

eventually spreads throughout the system. This relies on the ability of the clusters to resist the initial invasion of defectors - an occurrence which relies on a high enough r value (the positive multiplier of the contribution to the public good).

We also see that a potential climate club may soon be forming within the European Union. While it functions slightly differently to the classic example in [12], it follows the same principles of encouraging countries to join the group by punishing those outside of the club, in order to attempt to reduce overall carbon emissions. This is the “Carbon Border Adjustment Mechanism” [4]. Currently, all goods within the EU require taxation based on their carbon footprint, however imported goods from outside cannot be subject to such taxes. CBAM aims to prevent foreign goods from gaining benefits due to environmentally harmful practices by enforcing that EU companies report the carbon content of imported goods, and purchase certificates to cover the environmental cost of said goods. However, if a country already levies a tax on their own goods (similar to the EU’s carbon tax), then this cost will be taken into consideration, and fewer certificates must be purchased. This discourages the importation of goods from countries which do not levy a similar tax on their own items, forming a structure which behaves very similarly to the “climate clubs” in [12].

3 The Model

In this section, similarly to [2, 3] we examine the underlying framework modelling dynamic processes on a fully heterogeneous population. We begin by defining a population of individuals and the actions which they may take, and then we see different methods by which we can model the evolution of a population over time.

3.1 An Initial System

- N **ordered** individuals who play actions from action sets $A_i = (a_{i,1}, a_{i,2}, \dots, a_{i,k})$. In this paper, we shall assume that all action sets contain the same possible actions for each player.
- A state space S given by the set of ordered N -tuples with entries $\mathbf{a} = (a_1, a_2, a_3, \dots) \in A_1 \times A_2 \times A_3 \times \dots$
- A fitness function $S \rightarrow \mathbb{R}^N$. We denote by π the payoff function for a given game. We denote the i^{th} entry of $\pi(\mathbf{a})$ by $\pi_i(\mathbf{a})$.

Now let $h(\mathbf{a}, \mathbf{b})$ denote the Hamming distance between two states $\mathbf{a}, \mathbf{b} \in S$ [8]. The set of states $\mathbf{b}$ such that $h(\mathbf{a}, \mathbf{b}) = 1$ is defined as $\text{Neb}(\mathbf{a})$, the “Neighbourhood set of a state” [3]. Our system will be in the form of a Markov chain [15] which moves between states with a probability > 0 if and only if $h(\mathbf{a}, \mathbf{b}) \leq 1$. We call the index at which $\mathbf{a}$ and $\mathbf{b}$ differ the *index of difference*, denoted $I(\mathbf{ab})$ [3]. We shall look at different population dynamics by which we may model such a Markov chain.

3.2 Population Dynamics

In this section, we give an overview of various population dynamics. This includes the following well-known dynamics, adapted to a fully heterogeneous population:

- The Moran process, first described in [11]
- Fermi Imitation Dynamics, a common choice of dynamic often adapted to the study of cooperation on a graph [10, 9, 16]
- Aspiration dynamics, a model first defined in [?]
- Introspection Dynamics, a more modern dynamic defined in [2]

We also introduce a new dynamic called “Introspective Imitation Dynamics”. This is a novel dynamic particularly suited to the process of learning from others in a heterogeneous population.

3.2.1 Choice and Selection intensity

In Moran-process-type dynamics, which model biological evolution, we introduce a parameter ϵ referred to as *selection intensity* via the mapping

$$f_i(\mathbf{a}) = 1 + \epsilon \pi_i(\mathbf{a}).$$

Appropriate bounds on ϵ ensure that the fitness values f_i remain non-negative, which is necessary for the reproduction probabilities in the Moran process to be well-defined. Smaller values of ϵ correspond to weaker selection, placing less

emphasis on payoff differences in the evolutionary update. In the limiting case $\epsilon = 0$, the process reduces to neutral drift, in which reproduction is independent of payoff. In contrast, population dynamics based on the Fermi update rule [16] employ a parameter β , which we refer to as the *choice intensity*. This parameter determines how sensitively individuals respond to payoff differences when making decisions. Depending on the specific dynamic under consideration, these payoff differences may arise from comparing an individual's current payoff with that of another individual in the population, or from comparing their current payoff with the payoff they would obtain by switching strategy.

3.2.2 The Moran Process

The Moran process [13] is a dynamic corresponding to the following algorithm:

1. A player is chosen to be duplicated with a probability proportional to their fitness in the population
2. A player is chosen with probability $\frac{1}{N}$ to be removed from the population

Let (a_{-j}) denote the state from the perspective of a_j , so that $\mathbf{a} = (a_j, a_{-j})$. Then, the entry $T_{\mathbf{ab}}$ of a transition matrix T in which each row and column correspond to a state in S is defined as the probability of a transition $\mathbf{a} = (a_1, a_2, \dots, a_N) \rightarrow \mathbf{b} = (b_1, b_2, \dots, b_N)$. For the Moran process, this is as follows:

$$T_{\mathbf{ab}} = \begin{cases} \frac{\sum_{a_i=b_{I(\mathbf{ab})}} f(a_i)}{N \sum_{a_j} f(a_j)} & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}), \text{ differing at position } I(\mathbf{ab}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \quad (1)$$

An important case which proceeds from (1) is that of a transition $\mathbf{a} \rightarrow \mathbf{b}$ where $\mathbf{b}$ contains an individual of a *type* not found in $\mathbf{a}$. For example, the transition $(0, 1) \rightarrow (0, 2)$. This would be forbidden by the intuition of a standard Moran process, as the new individual is the duplication of another individual in $\mathbf{a}$. However, we can see that the standard formula yields $p_{v,u} = 0$ because there does not exist $a_i \in \mathbf{a} : a_i = b_i^*$.

We can now discuss one of the key ideas in the Moran process. As no new action types can be introduced to the population once they are fully removed, we have a set of absorbing states $\mathbf{a}$ such that $a_1 = a_2 = \dots$, for which we have $p_{\mathbf{a} \rightarrow \mathbf{a}} = 1$. We denote the set of absorbing states S^γ . We therefore have that the Moran process forms an absorbing Markov chain [15], and we can investigate the probability of the population being absorbed into each absorbing state.

Consider a game with two players, each playing one of two actions. We have a state space $\mathbf{a} = (0, 0), \mathbf{b} = (0, 1), \mathbf{c} = (1, 0), \mathbf{d} = (1, 1)$, and our transition matrix T would take the form

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{f_1(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} & 1 - \frac{f_1(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} - \frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} & 0 & \frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} \\ \frac{f_2(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} & 0 & 1 - \frac{f_1(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} - \frac{f_2(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} & \frac{f_1(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2)$$

Our transition matrix represents the probability of transitioning from the state represented by the row to the state represented by the column. Here our columns represent $\mathbf{a}, \mathbf{b}, \mathbf{c}$ and $\mathbf{d}$ from left to right, and the rows represent these states in the same order from top to bottom.

Now, by taking the fundamental matrix of T and performing a right multiplication by R , the matrix with entries which represent transitions from transitive states to absorbing states [15], we can obtain the absorption matrix for this system.

For the above game, we find the following equation:

$$\left(I - \begin{bmatrix} 1 - \frac{f_1(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} - \frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} & 0 \\ 0 & 1 - \frac{f_1(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} - \frac{f_2(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} \end{bmatrix} \right)^{-1} \begin{bmatrix} \frac{f_1(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} & \frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} \\ \frac{f_2(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} & \frac{f_1(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} \end{bmatrix} \quad (3)$$

And solve to find our absorption matrix:

$$\begin{bmatrix} \frac{\frac{f_1(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} + \frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})}}{(2f_1(\mathbf{b})+2f_2(\mathbf{b}))} & \frac{\frac{f_1(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})} + \frac{f_2(\mathbf{b})}{2f_1(\mathbf{b})+2f_2(\mathbf{b})}}{(2f_1(\mathbf{b})+2f_2(\mathbf{b}))} \\ \frac{\frac{f_1(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} + \frac{f_2(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})}}{(2f_1(\mathbf{c})+2f_2(\mathbf{c}))} & \frac{\frac{f_1(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})} + \frac{f_2(\mathbf{c})}{2f_1(\mathbf{c})+2f_2(\mathbf{c})}}{(2f_1(\mathbf{c})+2f_2(\mathbf{c}))} \end{bmatrix} \quad (4)$$

In this matrix, the rows represent our initial transitive state (in our system, row 1 represents $\mathbf{b}$ and row 2 represents $\mathbf{c}$), and our columns represent which state the system is absorbed into (with $\mathbf{a}$ on the left, and $\mathbf{d}$ on the right). It is immediately clear from this result that the probability of the system ending up in a particular absorbing state does not rely on the fitness of individuals in the absorbing state, but rather their fitness in the transitive states. Thus, players can remain in a state which could be bettered by a single change, however they will not make such a change if no individual of another type exists in the population.

3.2.3 Fermi Imitation Dynamics

The Fermi Imitation Dynamic is another method of modelling evolution in a population. It models the following process:

- Two players, i and j are selected at random. Unlike [16], the player j does not have to be the neighbour of player i .
- Player i copies the action type of the player j with a probability according to the Fermi function $\phi(\pi_i(\mathbf{a}) - \pi_j(\mathbf{a})) = \frac{1}{1+e^{(\beta(\pi_i(\mathbf{a}) - \pi_j(\mathbf{a})))}}$

Where β is the choice intensity of the system, representing how sensitive our players' choice is to the difference between their current payoff, and the payoff which they are considering. This method has been commonly used in studies of evolutionary games, for example [10, 9, 6, 16].

Unlike a Moran Process, we can see that the players are not selected with a probability proportional to their fitness. However, Fermi Imitation Dynamics compared the fitness after the selection of players. This means that in Fermi Imitation Dynamics, there is not necessarily a player changing action in each timestep, whereas in a Moran process, there is always a player changing action type (although it may be a change to the same action type that they are currently performing). We now define our transition matrix according to the Fermi function:

$$T_{\mathbf{ab}} = \begin{cases} \left(\frac{1}{N(N-1)} \sum_{a_j=b_I(\mathbf{ab})} \phi(\pi_{I(\mathbf{ab})}(a) - \pi_j(\mathbf{a})) \right) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \quad (5)$$

Here we see how the Fermi function can be used to define the transition probability between states in a Markov chain. Importantly, we see that the first case of equation 5 is always in the interval $(0, 1)$, as the Fermi function always takes a value on $(0, 1)$, and the summation term is at most $N - 1$ values of said function, and thus takes a value $< N - 1$.

Now, let us look once more at our state space $\mathbf{a} = (0, 0)$, $\mathbf{b} = (0, 1)$, $\mathbf{c} = (1, 0)$, $\mathbf{d} = (1, 1)$. We can define a transition matrix:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{2}\phi(\pi_2(\mathbf{b}) - \pi_1(\mathbf{b})) & \frac{1}{2} & 0 & \frac{1}{2}\phi(\pi_1(\mathbf{b}) - \pi_2(\mathbf{b})) \\ \frac{1}{2}\phi(\pi_1(\mathbf{c}) - \pi_2(\mathbf{c})) & 0 & \frac{1}{2} & \frac{1}{2}\phi(\pi_2(\mathbf{c}) - \pi_1(\mathbf{c})) \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (6)$$

and using our previous method, we can acquire our absorption matrix:

$$\begin{bmatrix} \phi(\pi_2(\mathbf{b}) - \pi_1(\mathbf{b})) & \phi(\pi_1(\mathbf{b}) - \pi_2(\mathbf{b})) \\ \phi(\pi_1(\mathbf{c}) - \pi_2(\mathbf{c})) & \phi(\pi_2(\mathbf{c}) - \pi_1(\mathbf{c})) \end{bmatrix} \quad (7)$$

This shows that, similarly to the Moran process, our absorption probabilities rely on the fitness of players in transitive states and not those of players in the absorbing states. Thus, this dynamic is still not fully suited for modelling a heterogeneous system as a player can commonly copy a high-fitness strategy without considering the impact on their own fitness, and in some cases may enter an absorbing state with a transition which worsens their position with a high probability.

3.2.4 Aspiration Dynamics

Instead of considering the difference between two players' fitness, aspiration dynamics models an update rule based on how far away a player's performance in a game is from the fitness which they desire. It follows the algorithm:

1. A player i is chosen with probability $\frac{1}{N}$ to reconsider their strategy $a_{i,k}$
2. This player changes strategy with a probability $\phi(\pi_i(\mathbf{a}) - A)$

Here, A is player i 's *aspiration*, meaning the payoff which they desire to make from a given game. A player will accept a new strategy with a higher than random probability if their current payoff is lower than their aspired payoff. This dynamic is almost always only defined for two actions - traditionally the actions to either cooperate or defect. The related transition matrix can be seen below:

$$T_{\mathbf{ab}} = \begin{cases} \frac{1}{N} \phi(\pi_{I(\mathbf{ab})}(\mathbf{a}) - A_{I(\mathbf{ab})}) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \text{ and } j = I(\mathbf{ab}), \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \quad (8)$$

Now, consider the transition matrix for our general four-state system:

$$\begin{bmatrix} 1 - \frac{1}{2(1+e^{-\beta(A_2 - f_2(a))})} - \frac{1}{2(1+e^{-\beta(A_1 - f_1(a))})} & \frac{e^{A_2\beta}}{2(e^{A_2\beta} + e^{\beta f_2(a)})} & \frac{e^{A_1\beta}}{2(e^{A_1\beta} + e^{\beta f_1(a)})} & 0 \\ \frac{e^{A_2\beta}}{2(e^{A_2\beta} + e^{\beta f_2(b)})} & 1 - \frac{1}{2(1+e^{-\beta(A_2 - f_2(b))})} - \frac{1}{2(1+e^{-\beta(A_1 - f_1(b))})} & 0 & \frac{e^{A_1\beta}}{2(e^{A_1\beta} + e^{\beta f_1(b)})} \\ \frac{e^{A_1\beta}}{2(e^{A_1\beta} + e^{\beta f_1(c)})} & 0 & 1 - \frac{1}{2(1+e^{-\beta(A_2 - f_2(c))})} - \frac{1}{2(1+e^{-\beta(A_1 - f_1(c))})} & \frac{e^{A_2\beta}}{2(e^{A_2\beta} + e^{\beta f_2(c)})} \\ 0 & \frac{e^{A_1\beta}}{2(e^{A_1\beta} + e^{\beta f_1(d)})} & \frac{e^{A_2\beta}}{2(e^{A_2\beta} + e^{\beta f_2(d)})} & 1 - \frac{1}{2(1+e^{-\beta(A_2 - f_2(d))})} - \frac{1}{2(1+e^{-\beta(A_1 - f_1(d))})} \end{bmatrix} \quad (9)$$

We can see that as our players no longer consider the actions of the rest of the population when considering their decision, we no longer have an absorbing Markov chain. Thus, we no longer consider an absorption matrix, but instead we consider the steady state of our Markov chain [15] This is the proportion of time spent in each state when the Markov chain reaches a stable distribution across the states. It is well proven that any irreducible, finite Markov chain eventually reaches its unique stationary distribution. This is also independent of the starting state of the distribution. [?]. For our four state system, we acquire the following result for a stationary distribution:

3.2.5 Introspection Dynamics

In introspection dynamics, the following process take place [2]

1. A player i is chosen with probability $\frac{1}{N}$ to reconsider their strategy $a_{i,k}$
2. Player i chooses another possible strategy, $a_{i,l}$, from the set of all possible strategies.
3. The chosen player replaces their strategy with probability $p_{a_{I(\mathbf{ab})} \rightarrow b_{I(\mathbf{ab})}}(a_{-I(\mathbf{ab})}) = \phi(\Delta\pi_{I(\mathbf{ab})})$, where $\Delta\pi_{I(\mathbf{ab})} = \pi_{I(a,b)}(\mathbf{a}) - \pi_{I(a,b)}(\mathbf{b})$ is the difference between the possible fitness of the new strategy and the fitness of the player's current strategy.

[2] considers this process for the two player case, and it is expanded to multiple players in [3].

This process is well suited to a heterogeneous population. This is because in some cases using imitation dynamics, a player may copy a strategy due to its fitness when played by another individual in the population, despite the fact that such a strategy actually has a lower payoff for the focal player than their current strategy. In introspection dynamics, players do not consider the fitness of other individuals, and so cannot be fooled by high fitness strategies with attributes different to that of our focal player.

The transition matrix for the introspection dynamics process [3] is given by:

$$T_{\mathbf{ab}} = \begin{cases} \frac{1}{N(m_j - 1)} \phi(\Delta\pi_{I(\mathbf{ab})}) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \text{ and } j = I(\mathbf{ab}), \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \quad (10)$$

where m_j is the number of actions available to individual j . Since we assume that all individuals have the same action set with size k , we can replace $m_j - 1$ with $k - 1$.

Now, Couto also calculated in [2] the explicit strategy distribution for an asymmetric interaction between two players. We can adapt these results to define our transition matrix for our example state space **a,b,c,d**:

$$\begin{bmatrix} 1 - \frac{1}{2(e^{\beta(\pi_2(a)-\pi_2(b))+1})} - \frac{1}{2(e^{\beta(\pi_1(a)-\pi_1(c))+1})} & \frac{1}{2(e^{\beta(\pi_2(a)-\pi_2(b))+1})} & \frac{1}{2(e^{\beta(\pi_1(a)-\pi_1(c))+1})} & 0 \\ \frac{1}{2(e^{\beta(-\pi_2(a)+\pi_2(b))+1})} & 1 - \frac{1}{2(e^{\beta(\pi_1(b)-\pi_1(d))+1})} - \frac{1}{2(1+e^{-\beta(\pi_2(a)-\pi_2(b))})} & 0 & \frac{1}{2(e^{\beta(\pi_1(b)-\pi_1(d))+1})} \\ \frac{1}{2(e^{\beta(-\pi_1(a)+\pi_1(c))+1})} & 0 & 1 - \frac{1}{2(e^{\beta(\pi_2(c)-\pi_2(d))+1})} - \frac{1}{2(1+e^{-\beta(\pi_1(a)-\pi_1(c))})} & \frac{1}{2(e^{\beta(\pi_2(c)-\pi_2(d))+1})} \\ 0 & \frac{1}{2(e^{\beta(-\pi_1(b)+\pi_1(d))+1})} & \frac{1}{2(e^{\beta(-\pi_2(c)+\pi_2(d))+1})} & 1 - \frac{1}{2(1+e^{-\beta(\pi_2(c)-\pi_2(d))})} - \frac{1}{2(1+e^{-\beta(\pi_1(b)-\pi_1(d))})} \end{bmatrix} \quad (11)$$

We then consider the steady state vectors for this system. For this, we take the left eigenvector of the transition matrix, which is given by:

Unlike in the case of imitation dynamics, there are no absorbing states for a Markov chain transitioning according to imitation dynamics. Therefore, we define a steady state vector **u** according to the method Couto uses in [2].

3.2.6 Introspective Imitation Dynamics

Introspection dynamics allows players to learn based on their own fitness, resulting in a reduced probability of state changes that worsen the focal individual's fitness in a heterogeneous population. Now we consider a population dynamic that utilises this property of introspection dynamics but within the context of an imitative process. Consider a population of N players with the same action set. We define the introspective imitation process as follows:

1. A focal player is chosen at random to reconsider their strategy
2. Another player a_j in the population is chosen with a probability proportional to their fitness within the population to have their strategy considered
3. The focal player changes their strategy with probability $\phi(\Delta(f_{I(\mathbf{ab})}))$

We can define a transition matrix T of an introspective imitation process as follows:

$$T_{\mathbf{ab}} = \begin{cases} \frac{1}{N} \frac{\sum_{a_j=b_{I(\mathbf{a},\mathbf{b})}} f_j(\mathbf{a})}{\sum_k f_k(\mathbf{a})} \phi(\Delta(f_{I(\mathbf{ab})})) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \quad (12)$$

We see that players can only copy the actions present in the state, however they determine whether or not to copy a player based on their own fitness rather than by considering the fitness of another player. This models a behaviour where individuals learn from each other, but do not blindly trust that the strategy will work for themselves just because it worked for another player. Instead, players look to others to see what action they *could* play, but judge based on their own fitness to decide what action they *should* play.

Note that Introspective Imitation Dynamics takes into account both selection intensity and choice intensity. This means that both the level to which a player's fitness is considered when their strategy is being chosen for consideration, and also how much it is considered when choosing whether to accept the new strategy, is present in the dynamic.

Let us now see this model applied to our ongoing example in the state space $S = \{\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}\}$. Let $\eta_{i,j}(\mathbf{x}, \mathbf{y}) = 1 + e^{\beta((f_i(\mathbf{x}) - f_j(\mathbf{y})))}$ be the denominator of the Fermi function. We first see our transition matrix:

$$\begin{bmatrix} \frac{f_1(b)}{2\eta_{2,2}(\mathbf{b},\mathbf{a})(f_1(b)+f_2(b))} & 1 - \frac{f_2(b)}{\eta_{1,1}(\mathbf{b},\mathbf{d})(2f_1(b)+2f_2(b))} - \frac{f_1(b)}{\eta_{2,2}(\mathbf{b},\mathbf{a})(2f_1(b)+2f_2(b))} & 0 & 0 \\ \frac{f_2(c)}{2\eta_{1,1}(\mathbf{c},\mathbf{a})(f_1(c)+f_2(c))} & 0 & 1 - \frac{f_1(c)}{\eta_{2,2}(\mathbf{c},\mathbf{d})(2f_1(c)+2f_2(c))} - \frac{f_2(c)}{\eta_{1,1}(\mathbf{c},\mathbf{a})(2f_1(c)+2f_2(c))} & 0 \\ 0 & 0 & 0 & \frac{f_2(b)}{2\eta_{1,1}(\mathbf{b},\mathbf{d})(f_1(b)+f_2(b))} \\ 0 & 0 & \frac{f_1(c)}{2\eta_{2,2}(\mathbf{c},\mathbf{d})(f_1(c)+f_2(c))} & \frac{f_1(b)}{2\eta_{2,2}(\mathbf{b},\mathbf{a})(f_1(b)+f_2(b))} \end{bmatrix} \quad (13)$$

And, using the same method as in the other methods of imitation dynamics, we obtain the absorption matrix:

$$\begin{bmatrix} \frac{\eta_{1,1}(\mathbf{b},\mathbf{d})f_1(b)}{\eta_{1,1}(\mathbf{b},\mathbf{d})f_1(b)+\eta_{2,2}(\mathbf{b},\mathbf{a})f_2(b)} & \frac{\eta_{2,2}(\mathbf{b},\mathbf{a})f_2(b)}{\eta_{1,1}(\mathbf{b},\mathbf{d})f_1(b)+\eta_{2,2}(\mathbf{b},\mathbf{a})f_2(b)} \\ \frac{\eta_{2,2}(\mathbf{c},\mathbf{d})f_2(c)}{\eta_{1,1}(\mathbf{c},\mathbf{a})f_1(c)+\eta_{2,2}(\mathbf{c},\mathbf{d})f_2(c)} & \frac{\eta_{1,1}(\mathbf{c},\mathbf{a})f_1(c)}{\eta_{1,1}(\mathbf{c},\mathbf{a})f_1(c)+\eta_{2,2}(\mathbf{c},\mathbf{d})f_2(c)} \end{bmatrix} \quad (14)$$

Unlike other imitation dynamics, we can see that the absorption probability in an introspective imitation process relies on the fitness that at least one individual will possess in the absorbing state. This means that players will enter absorbing states with a higher probability if it improves their own fitness, despite any heterogeneity between payoffs for the same action played by different players. This solves the problem of players making worsening moves based on the fitness of another individual in the population, while still retaining the property of imitation dynamics that new strategies cannot be introduced to the system.

3.3 Mutation

We can also model how individuals may mutate within the system. This is the process by which after a state transition, the chosen player will instead mutate to a different action type with some small probability. We then have two possible feasible transitions:

- If $h(\mathbf{ab}) = 0$, then the only way in which the transition can be achieved is through the transition of a player $i \rightarrow i$ and then no mutation occurring.
- If $h(\mathbf{ab}) = 1$, then there are two ways in which the transition can occur. Either, the player i which is different between $\mathbf{a}$ and $\mathbf{b}$ transitions by the standard process, with no mutation occurring, or no transition occurs, but player i mutates into the type which they possess in $\mathbf{b}$

We let $\mu_{i,j}$ be the probability that individual i mutates to type j . Thus, given a transition matrix T , we can produce the related transition matrix with mutation as follows:

$$T_{\mathbf{ab}}^{(m)} = \begin{cases} T_{\mathbf{ab}}(1 - \sum_k \mu_{I(\mathbf{ab}),k}) + (\frac{\mu_{i,\mathbf{b}_I(\mathbf{ab})}}{N}) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \quad (15)$$

In all cases, this forms an ergodic Markov chain rather than an absorbing chain. Thus, we do not obtain absorption matrices for models which use mutation, but instead we obtain a steady state, which no longer depends on our starting state.

We will now see each of our processes for the case where mutation is considered.

3.3.1 The Moran Process

In the case of the Moran process, our transition matrix becomes:

$$T_{\mathbf{ab}}^{(m)} = \begin{cases} (\frac{\sum_{a_i=b_{I(\mathbf{ab})}} f(a_i)}{N \sum_{a_j} f(a_j)})(1 - \sum_k \mu_{I(\mathbf{ab}),k}) + (\frac{\mu_{i,\mathbf{b}_I(\mathbf{ab})}}{N}) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}), \text{ differing at position } I(\mathbf{ab}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \quad (16)$$

For the case of the two player game, we now have the transition matrix:

$$\begin{bmatrix} -\frac{\mu_{12}}{2} - \frac{\mu_{22}}{2} + 1 & \frac{\mu_{22}}{2} & 0 \\ \frac{\mu_{21}}{2} + \frac{(\epsilon f_1(b)+1)(-\mu_{21}-\mu_{22}+1)}{2(\epsilon f_1(b)+\epsilon f_2(b)+2)} & -\frac{\mu_{12}}{2} - \frac{\mu_{21}}{2} - \frac{(\epsilon f_1(b)+1)(-\mu_{21}-\mu_{22}+1)}{2(\epsilon f_1(b)+\epsilon f_2(b)+2)} - \frac{(\epsilon f_2(b)+1)(-\mu_{11}-\mu_{12}+1)}{2(\epsilon f_1(b)+\epsilon f_2(b)+2)} + 1 & \frac{\mu_{12}}{2} + \frac{(\epsilon f_2(b)+1)(-\mu_{11}-\mu_{12}+1)}{2(\epsilon f_1(b)+\epsilon f_2(b)+2)} \\ \frac{\mu_{11}}{2} + \frac{(\epsilon f_2(c)+1)(-\mu_{11}-\mu_{12}+1)}{2(\epsilon f_1(c)+\epsilon f_2(c)+2)} & 0 & -\frac{\mu_{11}}{2} - \frac{\mu_{22}}{2} - \frac{(\epsilon f_1(c)+1)(-\mu_{21}-\mu_{22}+1)}{2(\epsilon f_1(c)+\epsilon f_2(c)+2)} - \frac{(\epsilon f_2(c)+1)(-\mu_{11}-\mu_{12}+1)}{2(\epsilon f_1(c)+\epsilon f_2(c)+2)} + 1 \\ 0 & \frac{\mu_{11}}{2} & -\frac{\mu_{11}}{2} - \frac{\mu_{21}}{2} + 1 \end{bmatrix}$$

and the steady state:

3.3.2 Fermi Imitation Dynamics

When we introduce mutation to Fermi Imitation Dynamics, we obtain the following transition matrix:

$$T_{\mathbf{ab}}^{(m)} = \begin{cases} \left(\frac{1}{N(N-1)} \sum_{a_j=b_{I(\mathbf{ab})}} \phi(\pi_{I(\mathbf{ab})}(a) - \pi_j(\mathbf{a})) \right) (1 - \sum_k \mu_{I(\mathbf{ab}),k}) + \left(\frac{\mu_{i,\mathbf{b}_I(\mathbf{ab})}}{N} \right) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \quad (17)$$

In the case of the two player game, we obtain the transition matrix:

$$\begin{bmatrix} -\frac{\mu_{12}}{2} - \frac{\mu_{22}}{2} + 1 & \frac{\mu_{22}}{2} & 0 \\ \frac{\mu_{21}}{2} + \frac{-\mu_{21}-\mu_{22}+1}{2(e^{\beta(-f_1(b)+f_2(b))+1})} & -\frac{\mu_{12}}{2} - \frac{\mu_{21}}{2} + 1 - \frac{-\mu_{11}-\mu_{12}+1}{2(e^{\beta(f_1(b)-f_2(b))+1})} - \frac{-\mu_{21}-\mu_{22}+1}{2(e^{\beta(-f_1(b)+f_2(b))+1})} & \frac{\mu_{12}}{2} + \frac{-\mu_{11}-\mu_{12}+1}{2(e^{\beta(f_1(b)-f_2(b))+1})} \\ \frac{\mu_{11}}{2} + \frac{-\mu_{11}-\mu_{12}+1}{2(e^{\beta(f_1(c)-f_2(c))+1})} & 0 & -\frac{\mu_{11}}{2} - \frac{\mu_{22}}{2} + 1 - \frac{-\mu_{11}-\mu_{12}+1}{2(e^{\beta(f_1(c)-f_2(c))+1})} - \frac{-\mu_{21}-\mu_{22}+1}{2(e^{\beta(-f_1(c)+f_2(c))+1})} \\ 0 & \frac{\mu_{11}}{2} & -\frac{\mu_{11}}{2} - \frac{\mu_{21}}{2} + 1 \end{bmatrix}$$

And the steady state:

3.3.3 Aspiration Dynamics

As aspiration and introspection dynamics do not form absorbing Markov chains, we do not need mutation to be applied in order to form an ergodic chain. However, we will consider these cases for completeness, and thus we obtain the following transition matrix:

$$T_{\mathbf{ab}}^{(m)} = \begin{cases} \left(\frac{1}{N} \phi(\pi_{I(\mathbf{ab})} - A_i) \right) (1 - \sum_k \mu_{I(\mathbf{ab}),k}) + \left(\frac{\mu_{i,\mathbf{b}_I(\mathbf{ab})}}{N} \right) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \text{ and } j = I(\mathbf{ab}), \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases}$$

In the case of the general 2-player game, we get the following:

$$\begin{bmatrix} 1 - \frac{0.5}{1+e^{-\beta(A_2-f_2(a))}} - \frac{0.5}{1+e^{-\beta(A_1-f_1(a))}} & \frac{0.5e^{A_2\beta}}{e^{A_2\beta}+e^{\beta f_2(a)}} & 0 \\ \frac{0.5e^{A_2\beta}}{e^{A_2\beta}+e^{\beta f_2(b)}} & 1 - \frac{0.5}{1+e^{-\beta(A_2-f_2(b))}} - \frac{0.5}{1+e^{-\beta(A_1-f_1(b))}} & \frac{0.5e^{A_1\beta}}{e^{A_1\beta}+e^{\beta f_1(a)}} \\ \frac{0.5e^{A_1\beta}}{e^{A_1\beta}+e^{\beta f_1(c)}} & 0 & 0 \\ 0 & \frac{0.5e^{A_1\beta}}{e^{A_1\beta}+e^{\beta f_1(d)}} & 1 - \frac{0.5}{1+e^{-\beta(A_2-f_2(c))}} - \frac{0.5}{1+e^{-\beta(A_1-f_1(c))}} \\ & & \frac{0.5e^{A_2\beta}}{e^{A_2\beta}+e^{\beta f_2(d)}} \\ & & 1 - \frac{0.5}{1+e^{-\beta(A_2-f_2(d))}} - \frac{0.5}{1+e^{-\beta(A_1-f_1(d))}} \end{bmatrix}$$

3.3.4 Introspection Dynamics

As introspection dynamics does not form an absorbing Markov chain, we do not need mutation to form an ergodic chain. However, we will consider the following transition matrix for the case with mutation:

$$T_{\mathbf{ab}}^{(m)} = \begin{cases} \left(\frac{1}{N(m_j-1)} \phi(\Delta\pi_{I(\mathbf{ab})}) \right) (1 - \sum_k \mu_{I(\mathbf{ab}),k}) + \left(\frac{\mu_{i,\mathbf{b}_I(\mathbf{ab})}}{N} \right) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \text{ and } j = I(\mathbf{ab}), \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b}, \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases}$$

which, in the case of the two player game, gives us the following:

$$\begin{bmatrix} -\frac{\mu_{12}}{2} - \frac{\mu_{22}}{2} + 1 - \frac{-\mu_{21}-\mu_{22}+1}{2(e^{\beta(f_2(a)-f_2(b))+1})} - \frac{-\mu_{11}-\mu_{12}+1}{2(e^{\beta(f_1(a)-f_1(c))+1})} & \frac{\mu_{22}}{2} + \frac{-\mu_{21}-\mu_{22}+1}{2(e^{\beta(f_2(a)-f_2(b))+1})} & \frac{\mu_{12}}{2} + \frac{-\mu_{11}-\mu_{12}+1}{2(e^{\beta(f_1(a)-f_1(c))+1})} & 0 \\ \frac{\mu_{21}}{2} + \frac{-\mu_{21}-\mu_{22}+1}{2(e^{\beta(f_2(a)-f_2(b))+1})} & -\frac{\mu_{12}}{2} - \frac{\mu_{21}}{2} + 1 - \frac{-\mu_{21}-\mu_{22}+1}{2(e^{\beta(f_2(a)-f_2(b))+1})} - \frac{-\mu_{11}-\mu_{12}+1}{2(e^{\beta(f_1(b)-f_1(d))+1})} & 0 & \frac{\mu_{12}}{2} + \frac{-\mu_{11}-\mu_{12}+1}{2(e^{\beta(f_1(b)-f_1(d))+1})} \\ \frac{\mu_{11}}{2} + \frac{-\mu_{11}-\mu_{12}+1}{2(e^{\beta(f_1(c)-f_1(d))+1})} & 0 & -\frac{\mu_{11}}{2} - \frac{\mu_{22}}{2} + 1 - \frac{-\mu_{21}-\mu_{22}+1}{2(e^{\beta(f_2(c)-f_2(d))+1})} - \frac{-\mu_{11}-\mu_{12}+1}{2(e^{\beta(f_1(c)-f_1(c))+1})} & \frac{\mu_{22}}{2} + \frac{-\mu_{21}-\mu_{22}+1}{2(e^{\beta(f_2(c)-f_2(d))+1})} \\ 0 & \frac{\mu_{11}}{2} + \frac{-\mu_{11}-\mu_{12}+1}{2(e^{\beta(f_1(b)-f_1(d))+1})} & \frac{\mu_{21}}{2} + \frac{-\mu_{21}-\mu_{22}+1}{2(e^{\beta(f_2(c)-f_2(d))+1})} & -\frac{\mu_{11}}{2} - \frac{\mu_{21}}{2} + 1 - \frac{-\mu_{21}-\mu_{22}+1}{2(e^{\beta(f_2(c)-f_2(d))+1})} - \frac{-\mu_{11}-\mu_{12}+1}{2(e^{\beta(f_1(b)-f_1(d))+1})} \end{bmatrix}$$

3.3.5 Introspective Imitation Dynamics

Finally, for the case of introspective imitation dynamics, we get the following transition matrix with mutation:

$$T_{\mathbf{ab}}^{(m)} = \begin{cases} \left(\frac{1}{N} \frac{\sum_{a_j=b_{I(\mathbf{a},\mathbf{b})}} f_j(\mathbf{a})}{\sum_k f_k(\mathbf{a})} \phi(\Delta(f_{I(\mathbf{ab})})) (1 - \sum_k \mu_{I(\mathbf{ab}),k}) + \left(\frac{\mu_{i,\mathbf{b}_{I(\mathbf{ab})}}}{N} \right) \right) & \text{if } \mathbf{b} \in \text{Neb}(\mathbf{a}) \\ 0 & \text{if } \mathbf{b} \notin \text{Neb}(\mathbf{a}) \text{ and } \mathbf{a} \neq \mathbf{b} \\ 1 - \sum_{\mathbf{b} \in S \setminus \{\mathbf{a}\}} T_{\mathbf{ab}} & \text{if } \mathbf{a} = \mathbf{b} \end{cases} \quad (18)$$

For the two player game, we obtain the transition matrix:

$$\begin{bmatrix} -\frac{\mu_{11}^a - \mu_{21}^a}{2} + 1 & \frac{\mu_{21}^a}{2} & 0 \\ \frac{\mu_{21}^a}{2} + \frac{(c f_1(b)+1)(-p_{21}-p_{22}+1)}{2(e^{p_1(-f_2(c)+f_2(b))+1})f_1(b)+f_2(b)+2} & -\frac{\mu_{11}^a - \mu_{21}^a}{2} - \frac{(c f_1(b)+1)(-p_{21}-p_{22}+1)}{2(e^{p_1(-f_2(c)+f_2(b))+1})f_1(b)+f_2(b)+2} - \frac{(c f_2(b)+1)(-p_{11}-p_{12}+1)}{2(e^{p_1(f_1(b)-f_1(c))+1})f_1(b)+f_2(b)+2} + 1 & 0 \\ \frac{\mu_{11}^a}{2} + \frac{(c f_1(c)+1)(-p_{11}-p_{12}+1)}{2(e^{p_1(-f_1(c)+f_1(c))+1})f_1(c)+f_2(c)+2} & 0 & -\frac{\mu_{11}^a - \mu_{21}^a}{2} - \frac{(c f_1(c)+1)(-p_{21}-p_{22}+1)}{2(e^{p_1(f_2(c)-f_2(c))+1})f_1(c)+f_2(c)+2} - \frac{(c f_2(c)+1)(-p_{11}-p_{12}+1)}{2(e^{p_1(-f_1(c)+f_1(c))+1})f_1(c)+f_2(c)+2} + 1 \\ 0 & \frac{\mu_{11}^a}{2} & -\frac{\mu_{11}^a}{2} - \frac{\mu_{21}^a}{2} + 1 \end{bmatrix}$$

which gives us a steady state:

3.4 Classification of Population Dynamics

We will now classify our population dynamics into one of two categories. The first are population dynamics which accept strategies from the population based on the fitness of said strategies when played by other individuals. These dynamics are poorly suited to a heterogeneous population, as individuals do not consider how the change will affect them. However, these do offer a good model for biological processes such as the evolution of traits in nature, as in these cases the fitness of an individual causes them to pass on their traits. The second are population dynamics which consider their own change in payoff in the case that they accept a new strategy. These dynamics are more well suited to a heterogeneous population, modelling state changes where individuals make an active decision to change state in order to improve their own fitness.

Below, we see a diagram showing the classification of the population dynamics shown in this section:

3.5 The Public Goods Game

In this section, we shall see how different population dynamics can be applied to the public goods game. [14, 6] In this game, individuals choose whether or not to contribute to a public resource pool. In the end, the total resource is multiplied by some factor $r > 1$ and distributed equally between each player. This model often encourages a behaviour known as “free-riding”, where players refuse to contribute to the pool and simply take the benefit provided by other individuals’ contribution. The classic problem of a public goods game is to provide an environment where such free-riding is less profitable than contributing to the public good.

We can model a public goods game by providing the following payoff function:

$$\pi_i(\mathbf{a}) = \frac{r \sum_{j=1}^N \alpha_j}{N} - \alpha_i \quad (19)$$

(where α_j is the contribution by individual j.) In the homogeneous public goods game, we have that α_j can only take one of two values: some constant α if individual j cooperates, and 0 if not. However, the heterogeneous game will require a more tailored α_j . Thus, we consider two cases of contribution distributions. Let M be the total possible contribution when all players cooperate. The first case, we consider a linear wealth distribution, under which a player i contributes $\frac{2Mi}{N(N+1)}$. In this population, each player contributes more than the last. The second population we call the binomial population. In this population, the first $n < N$ players contribute some low amount α_L , and the final $N - n$ players contribute some greater amount α_H .

4 Results

First, let us see the overall distribution of p_C across all models.

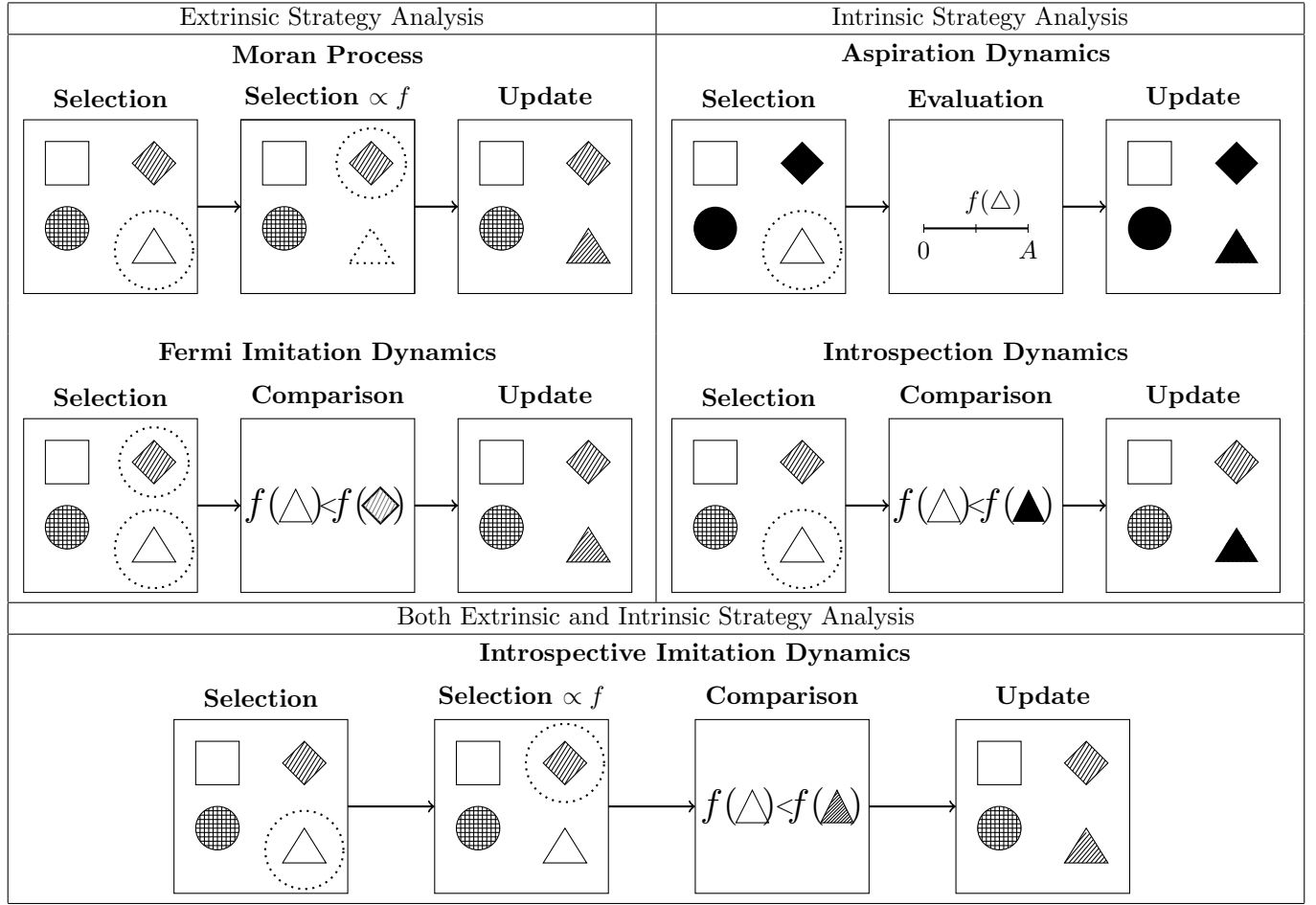
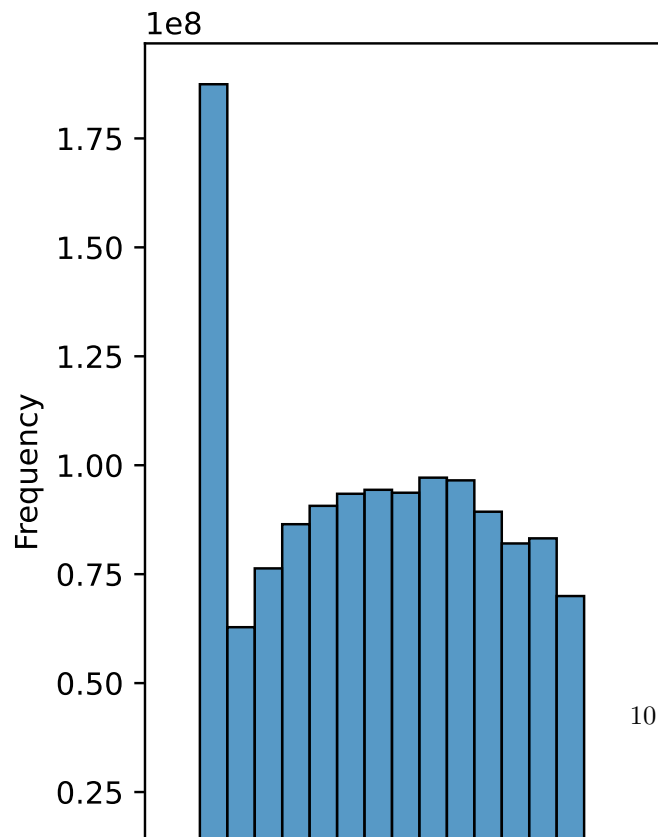
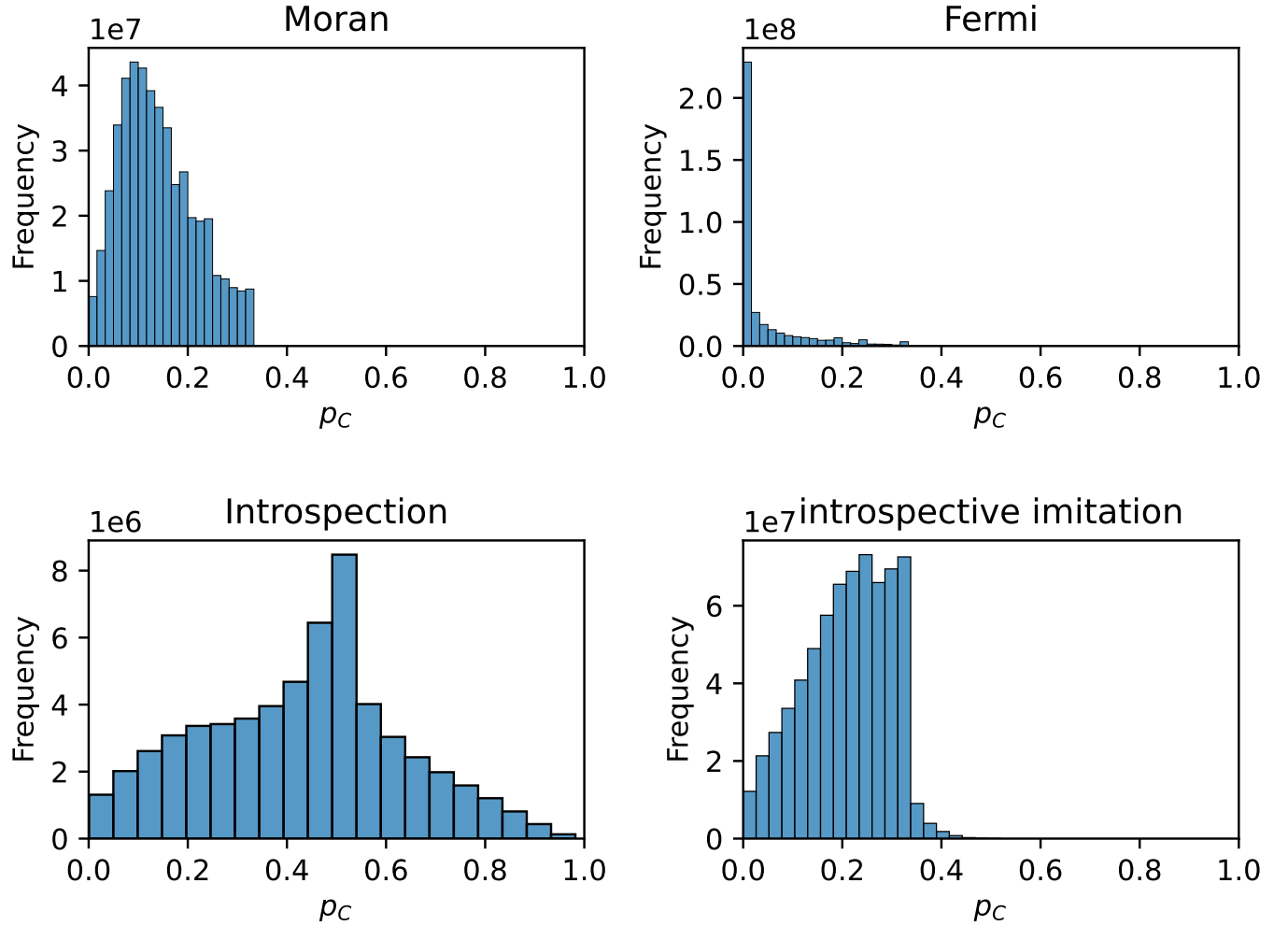


Table 1: A comparison of different population dynamics. We assume in all cases that the triangle is chosen for replacement with a probability $\frac{1}{N}$ and the diamond is chosen with a varying probability depending on the process. In the case of Introspection dynamics, we assume that the strategy represented by the shape being filled in is chosen with probability $\frac{1}{k-1}$, and for aspiration dynamics there is only one other strategy to accept.

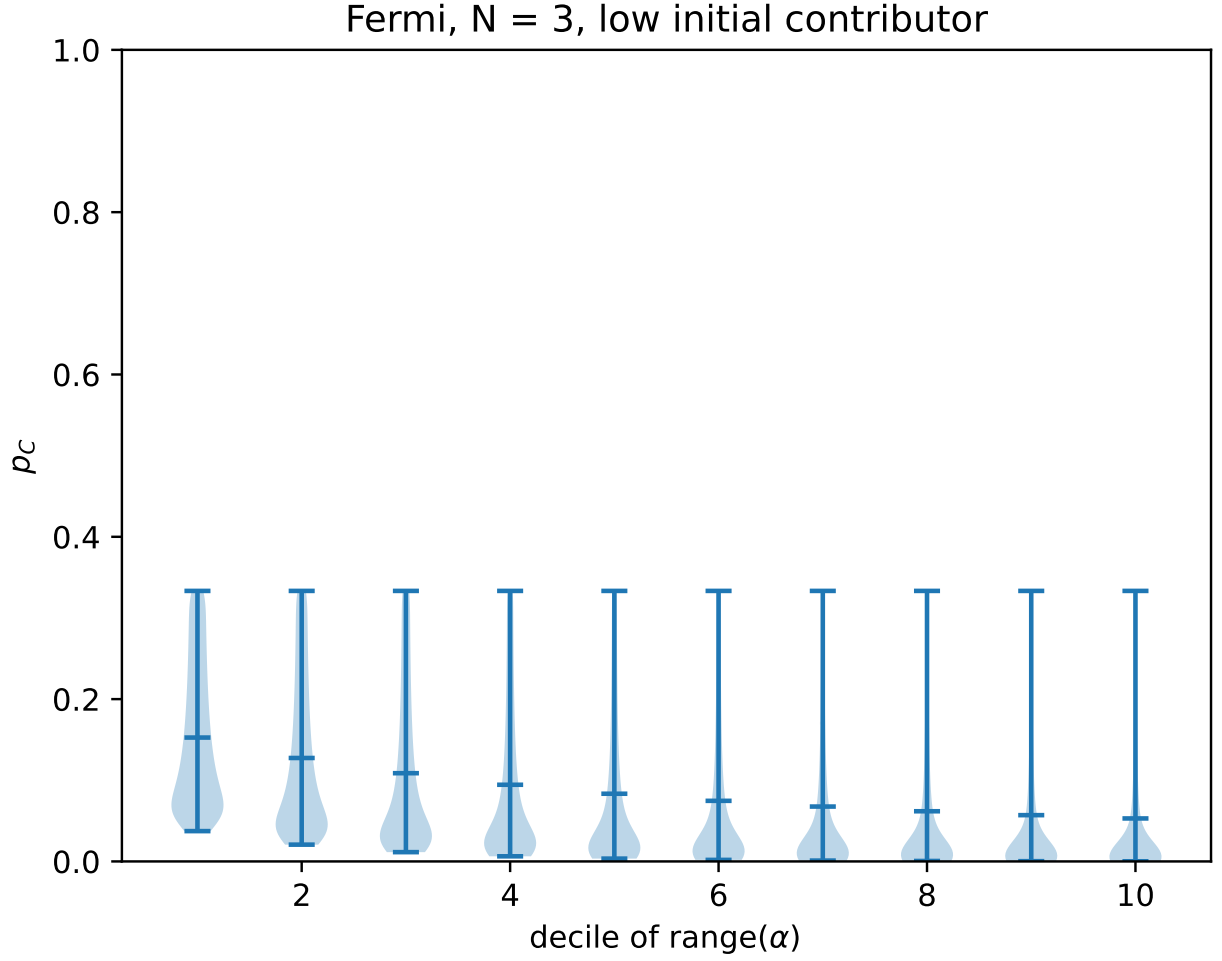


We see that there most systems do not produce a high value of p_C , however we can encourage cooperation under some conditions. Let us begin to investigate these conditions by seeing the distribution of p_C across our population dynamics.

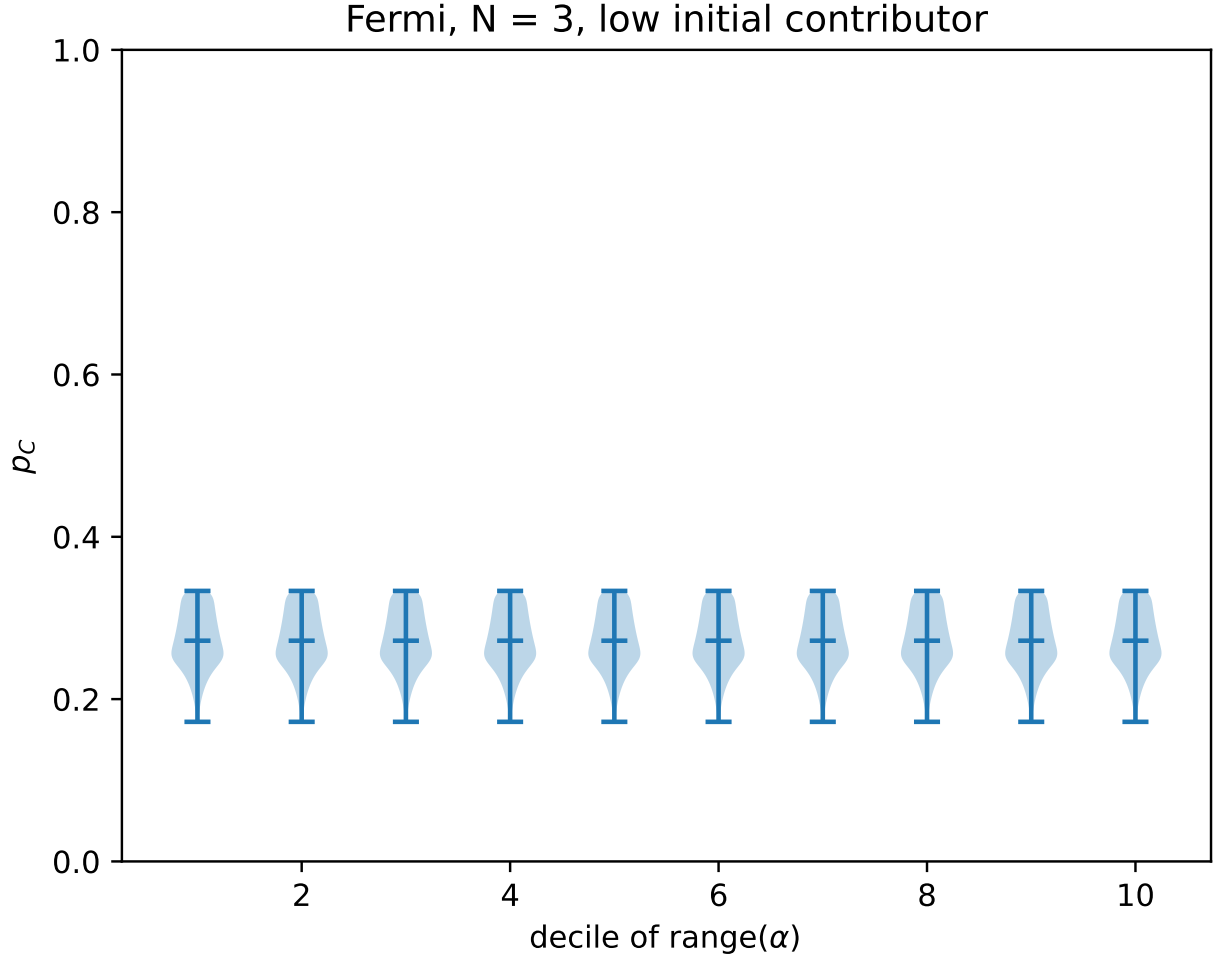


There is clearly a large difference in p_C values in different population dynamics, with Fermi imitation dynamics producing a very low value of p_C , and introspective imitation dynamics giving the highest values of any absorbing Markov chain. We now see how different parameters affect p_C .

First, we shall see how the heterogeneity of α effects the probability of cooperation with a low-contributing mutant in a linear population, and we shall ask the question “Is there a process which maximises p_C ?”

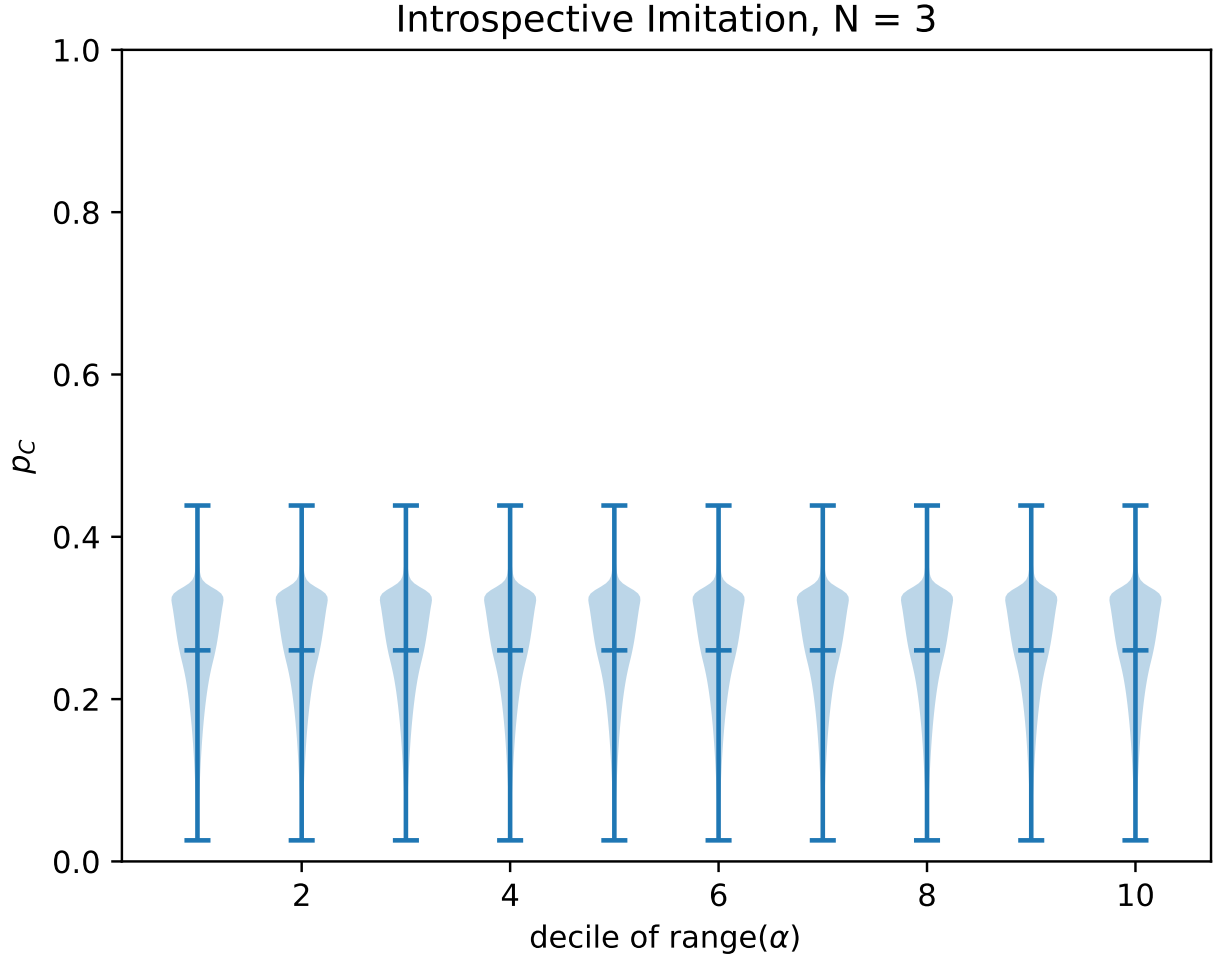


In Fermi imitation dynamics, we see that an increase in heterogeneity actually decreases the probability of cooperation. This is because with a higher heterogeneity, high contributing individuals are much harder to maintain, since the difference between their payoff and the payoff for a defector in the same state will be greater. This is because the high values of α_i are greater, and thus the probability of a defector copying the high contributors is lower, and the probability of a high contributor copying a defector is greater.

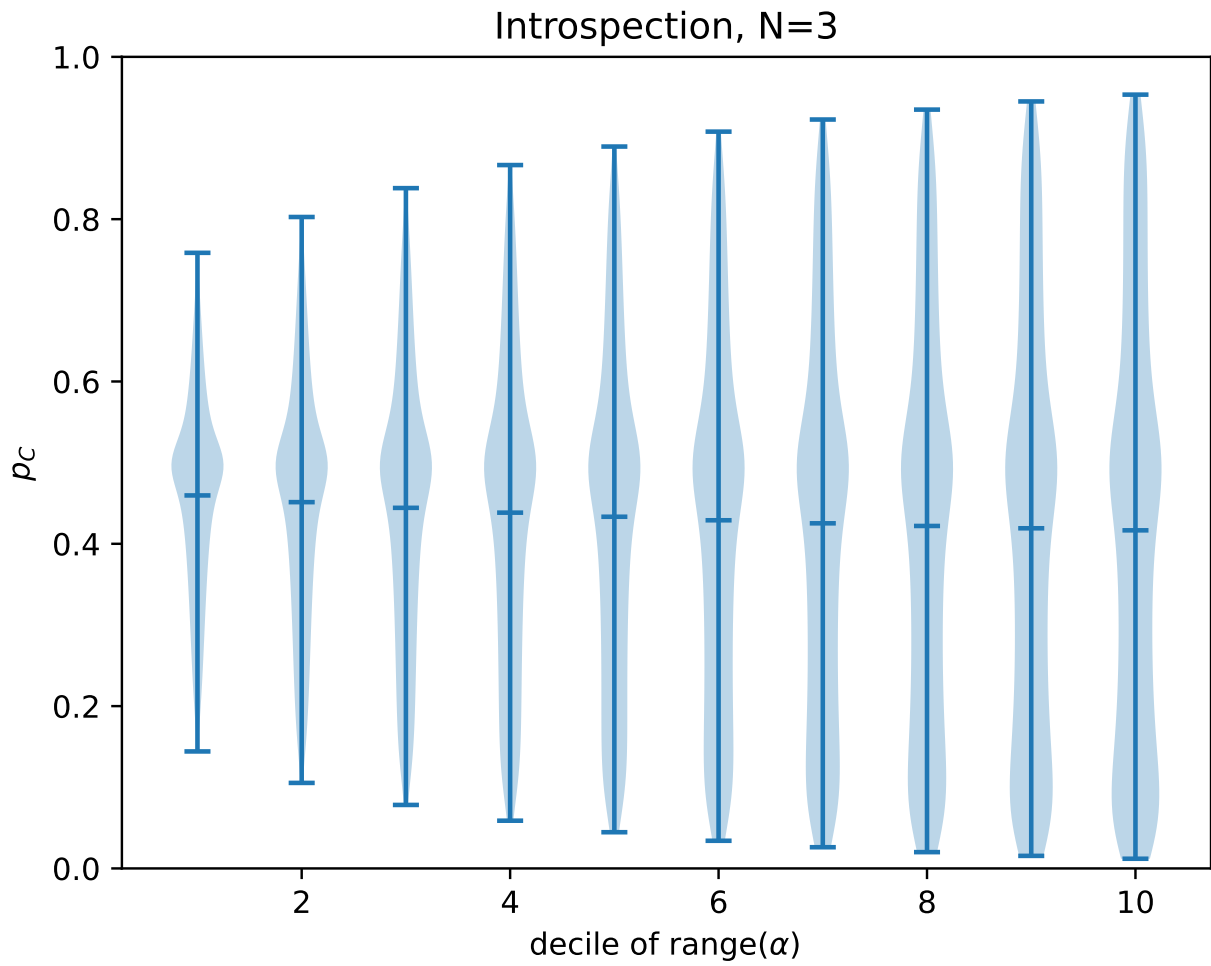


In the Moran process, we see no effect from the heterogeneity of α_i .

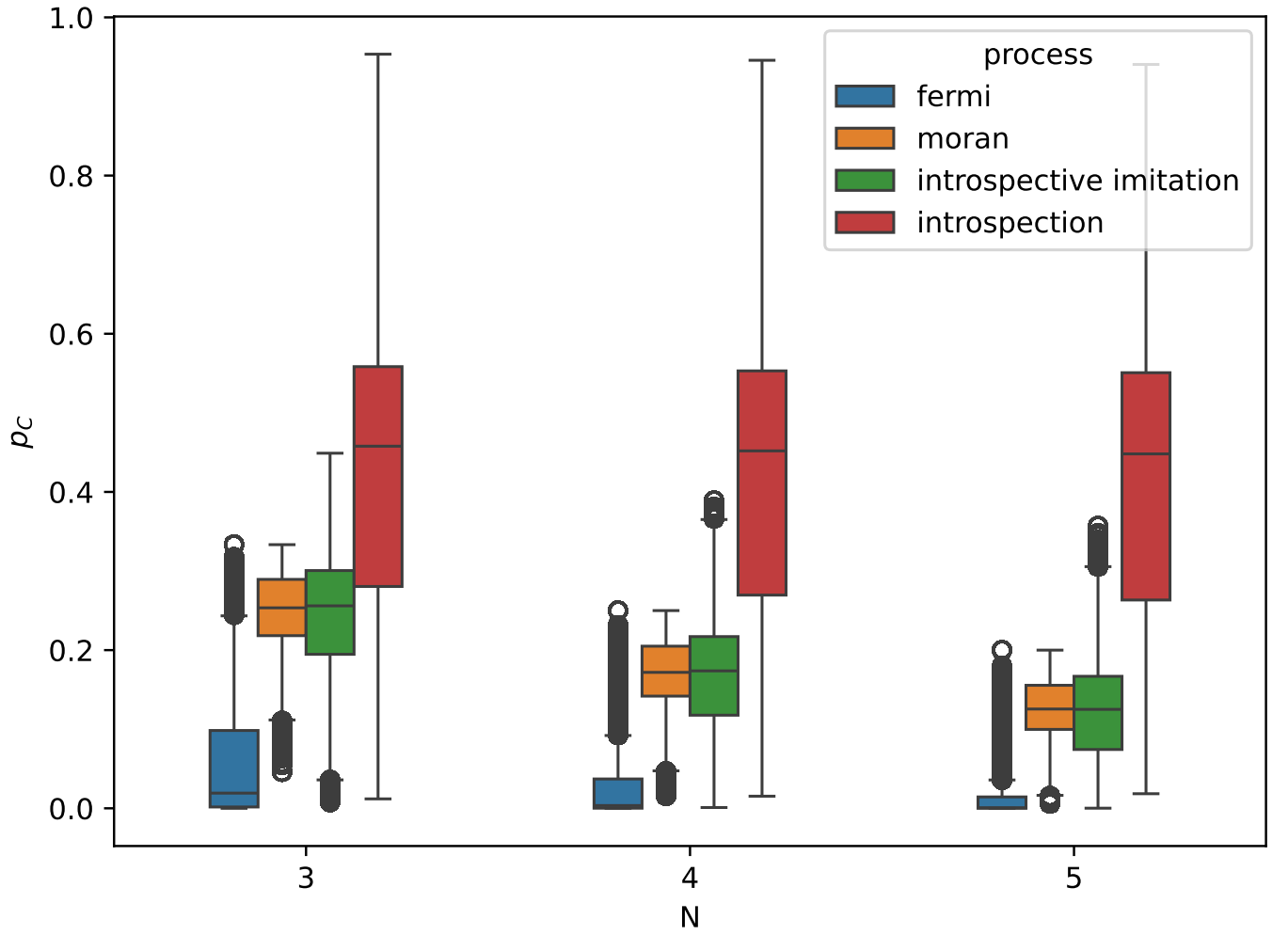
However, we do notice that in both of the above dynamics, there is a limit of $p_C = \frac{1}{3}$. This is because it is impossible for a state to have a contributor with a higher payoff than a defector if $\alpha_i > 0$, and thus in a purely extrinsic dynamic, the probability of a player accepting a defecting strategy will always be greater than the probability of a player accepting a cooperative strategy. Thus, we cannot achieve a value of p_C greater than that where players decide on strategies randomly, which would give us $p_C \leq \frac{1}{N}$



Here we see that introspective imitation is able to exceed the limit imposed on purely extrinsic processes. The introspective step is able to have a higher probability of accepting cooperation than defection if a player will have a higher payoff by cooperating than by defecting, even if in a given state the defectors will always have a higher payoff. Thus, it is possible to obtain higher values of p_C in introspective imitation dynamics than in purely extrinsic dynamics. It is also important to note that there is no result to show that the maximum value of p_C could not be improved from those shown in this data.

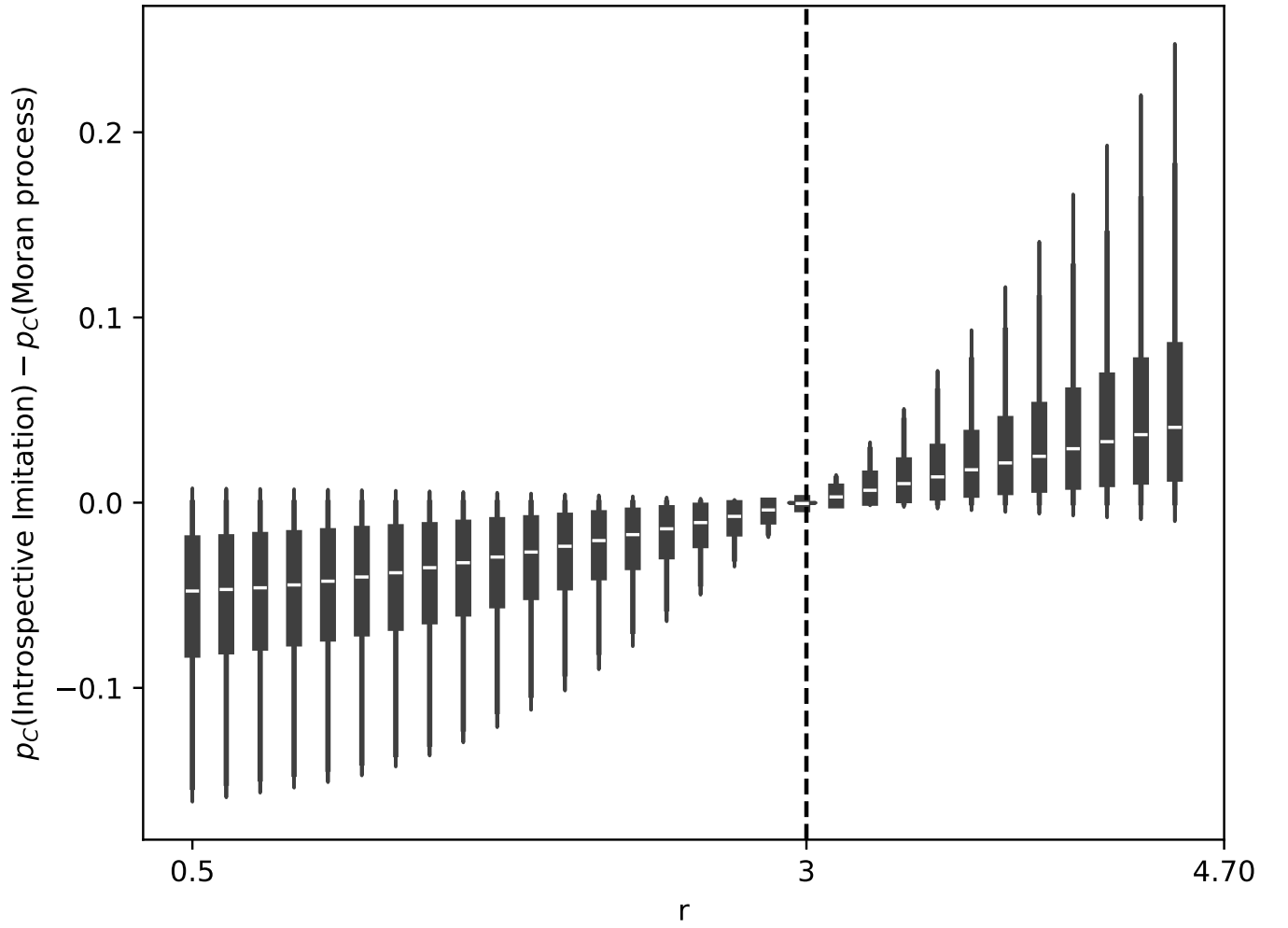


We see that this result also exists for introspection dynamics, though due to the different interpretation of p_C , this must be taken lightly.

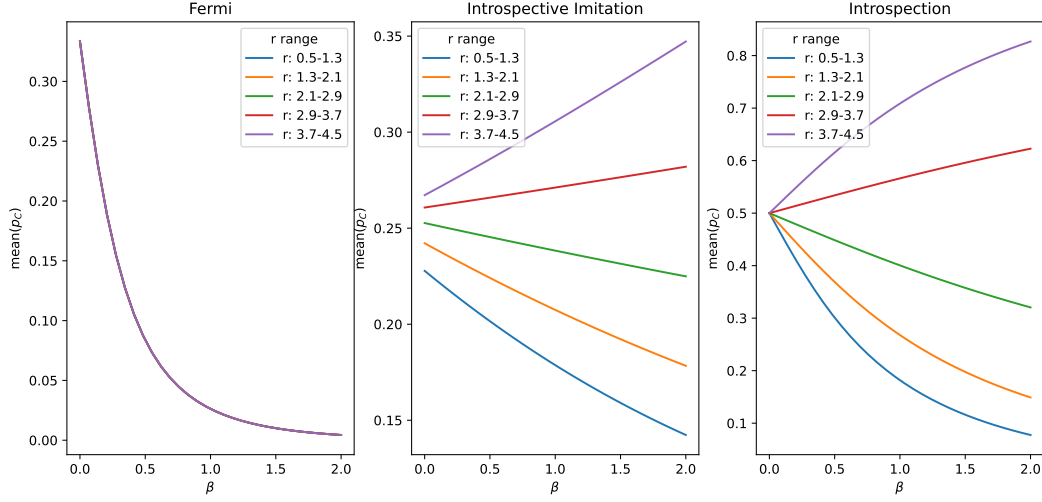


This result is not just true for the $N = 3$ case, but is also clearly true accross multiple values of N . This result is also true across different populations and even higher values of N , as can be seen in the supplementary information.

So to answer our previous question, intrinsic processes can in fact provide us with a greater probability of cooperation than purely extrinsic processes. In the next section, we will see under which conditions an intrinsic process begins to produce higher values of p_C than a purely extrinsic process. To do this, we compare the values of p_C in the Moran process and introspective imitation dynamics. This is because introspective imitation dynamics takes the form of the Moran process with an additional introspective step.



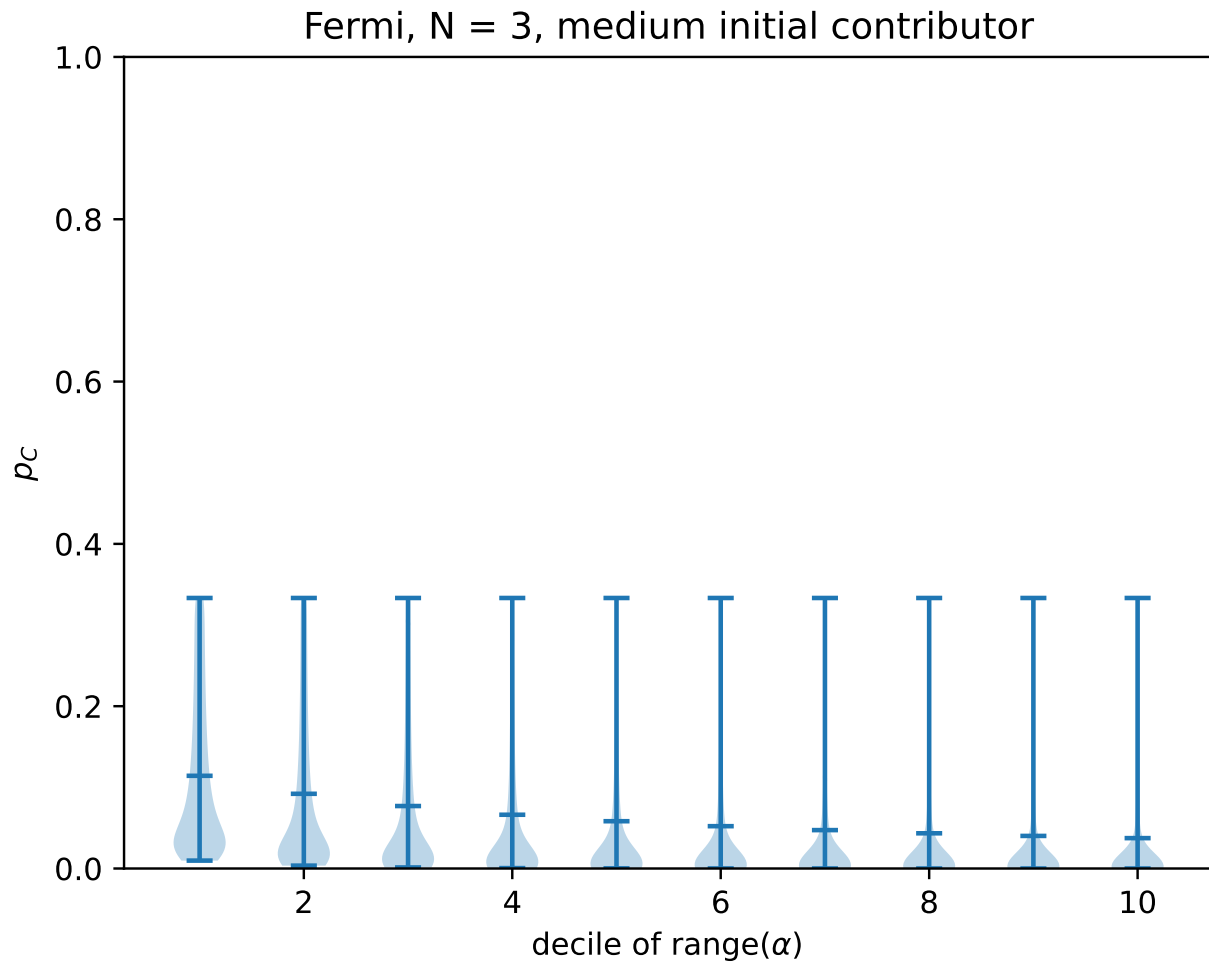
We can see that the parameter $r = N$ provides an exact point where the intrinsic processes begin to provide high p_C values. This is because when $r > N$, all players will gain a higher payoff by contributing than by defecting, regardless of the actions of the rest of the state, as we have $\frac{r\alpha_i}{N} - \alpha_i > 0$ for any value of α_i . Thus, this is the point at which the intrinsic step of introspective imitation dynamics begins to favour cooperation. For $r < N$, the intrinsic step favours defection, and thus the intrinsic step actually reduces p_C . We see in the supplementary information that this is indeed the point at which intrinsic processes begin to dominate.

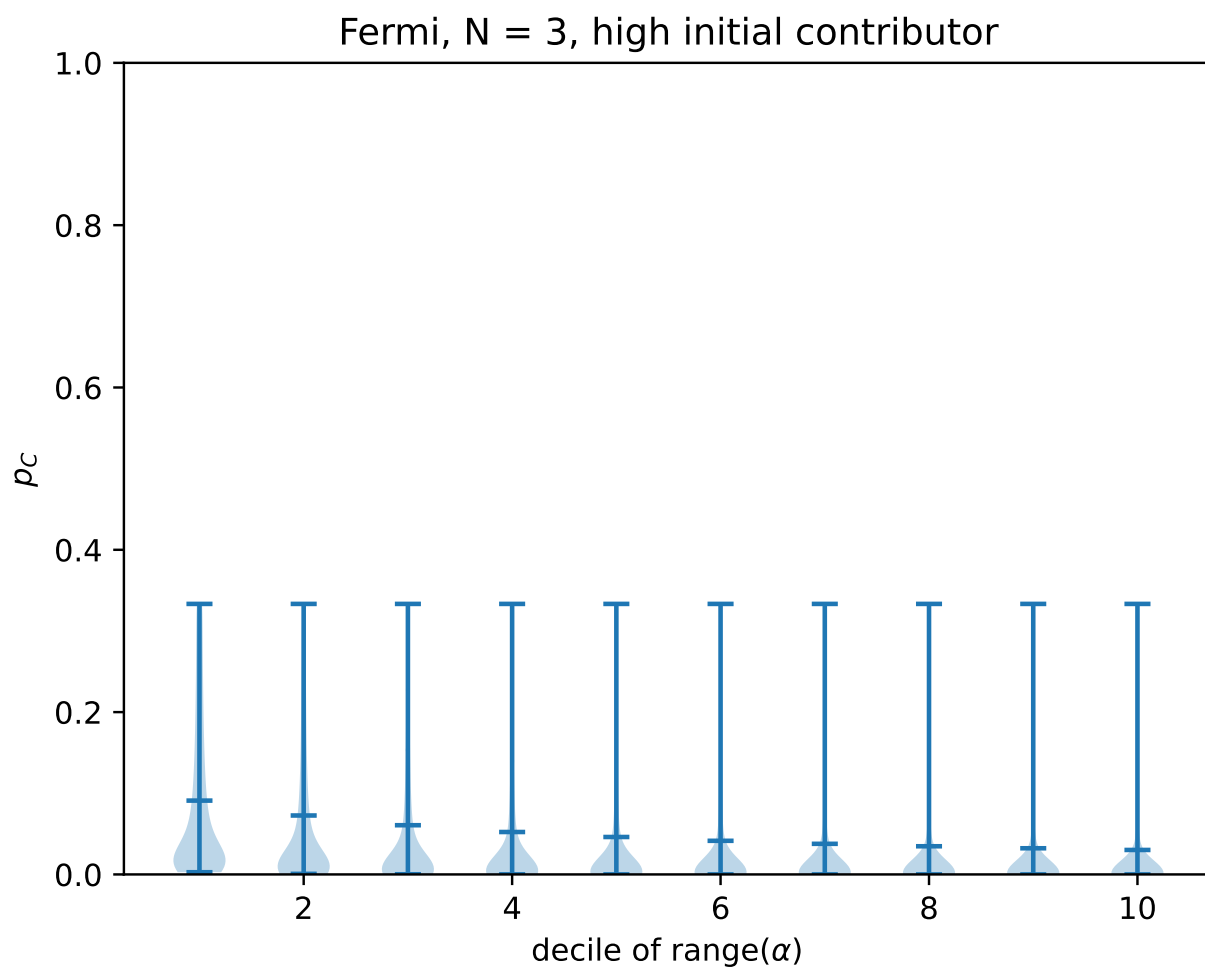


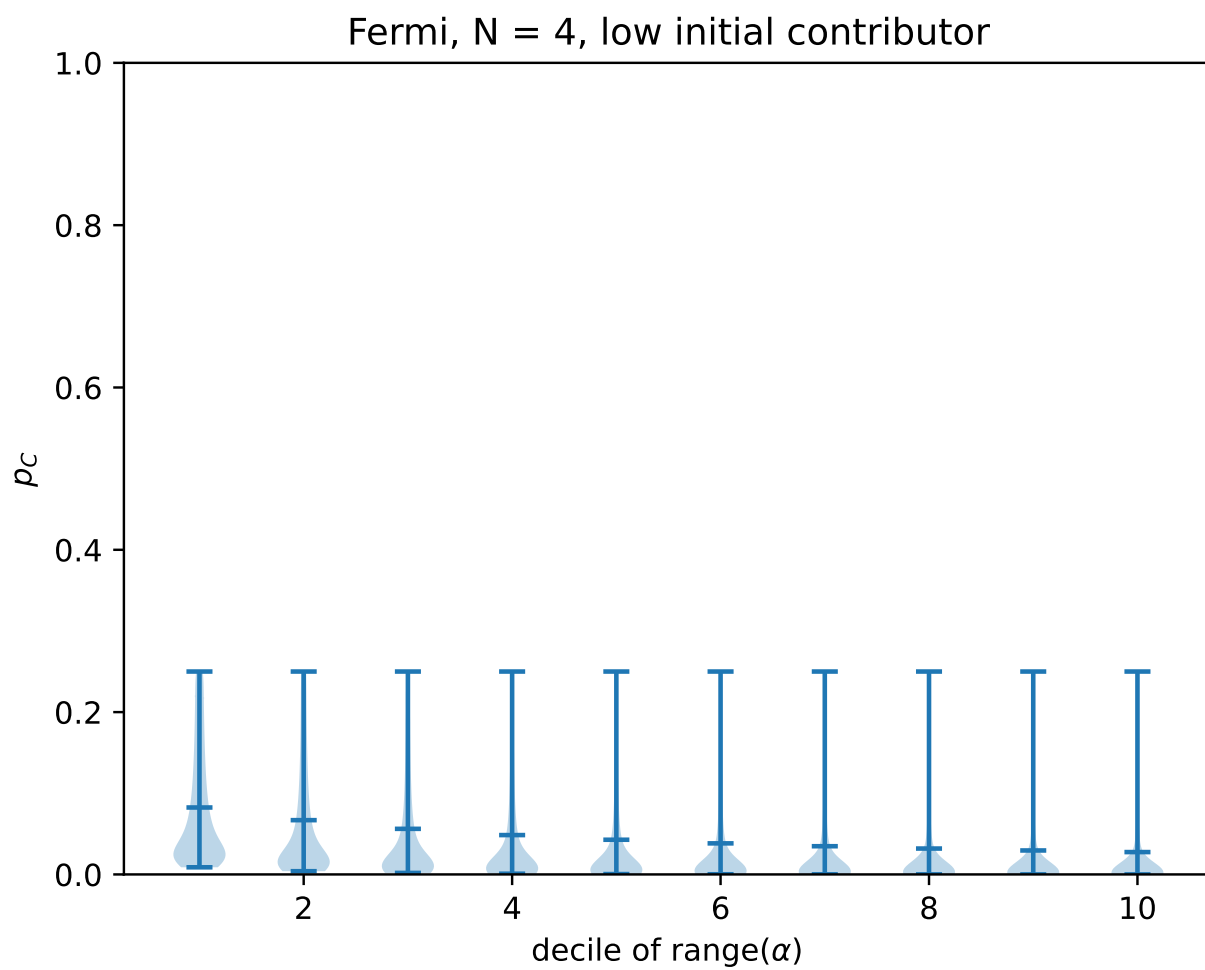
We also see that as the choice intensity β of a population dynamic increases, an intrinsic dynamic is able to facilitate an increase in p_C , whereas even for high r values, purely extrinsic processes will always have a negative correlation between β and p_C . This is because a purely extrinsic process will always see a defector as performing better than a cooperator, so a high choice intensity will always favour defection. However, in an intrinsic process, we can see that under certain conditions (specifically, the condition $r > N$ as before) the comparison between a cooperator and a defector may favour cooperation by comparing a defector to their own possible payoff by cooperating, and thus a higher p_C may be obtained by a higher value of β in an intrinsic process.

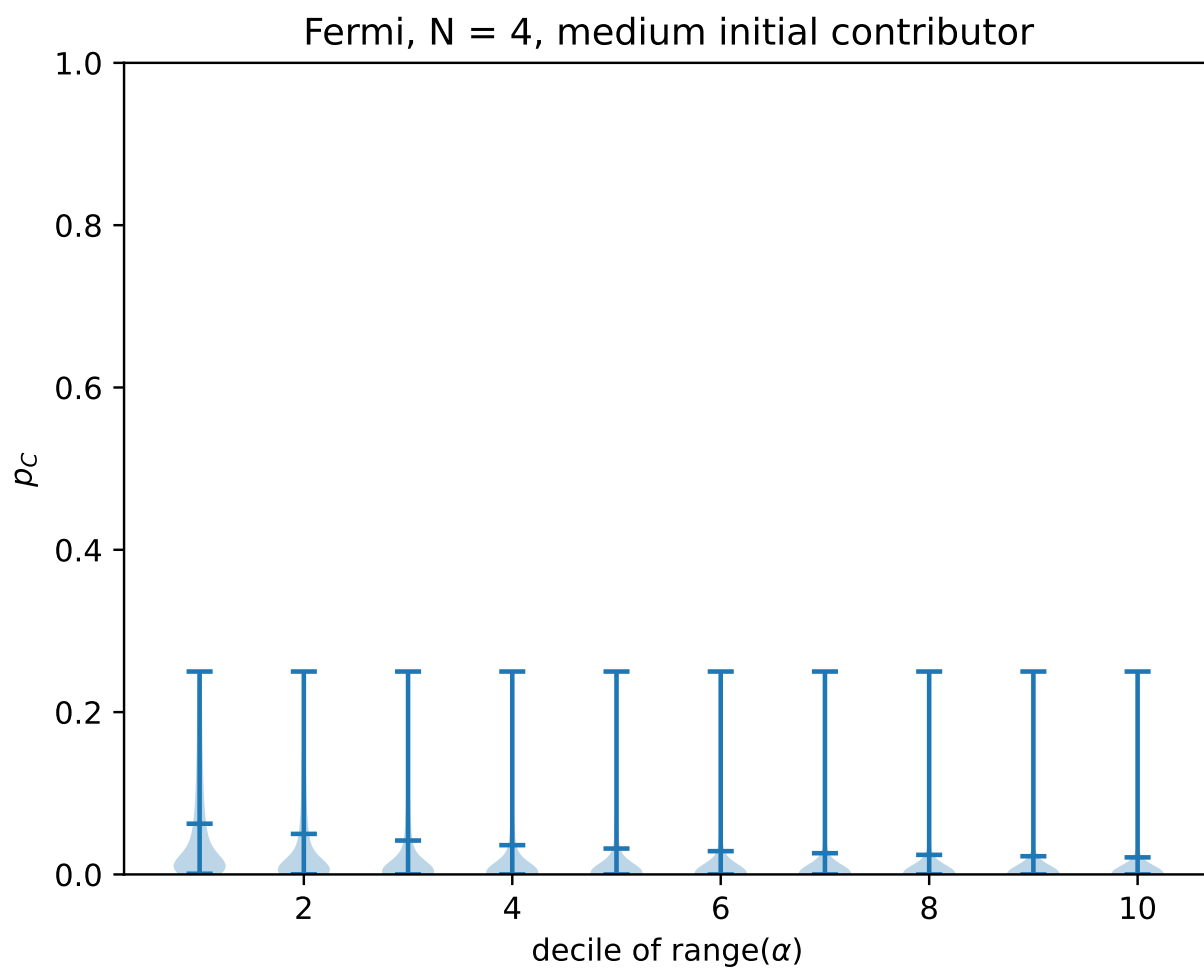
5 supplementary information

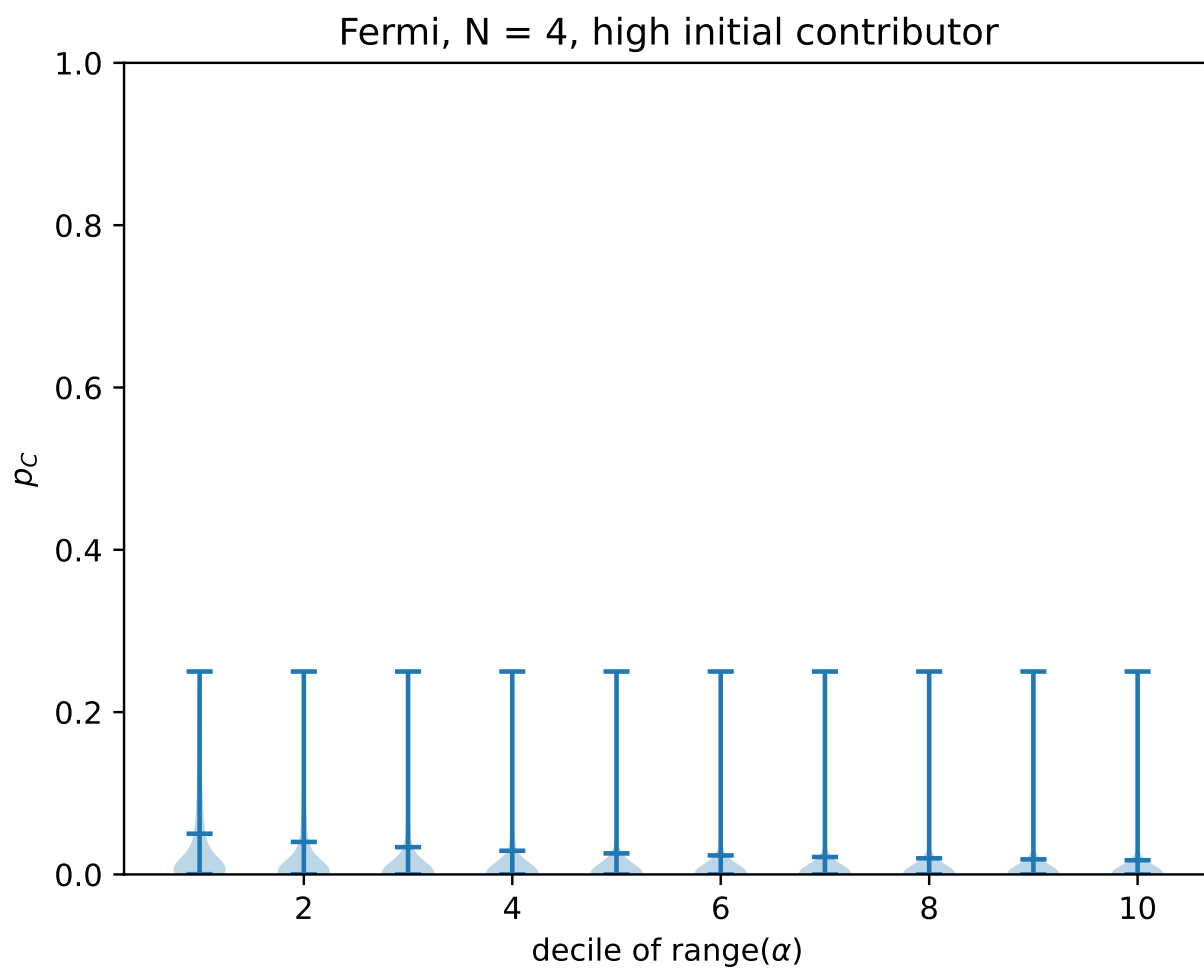
5.1 Higher N and other populations

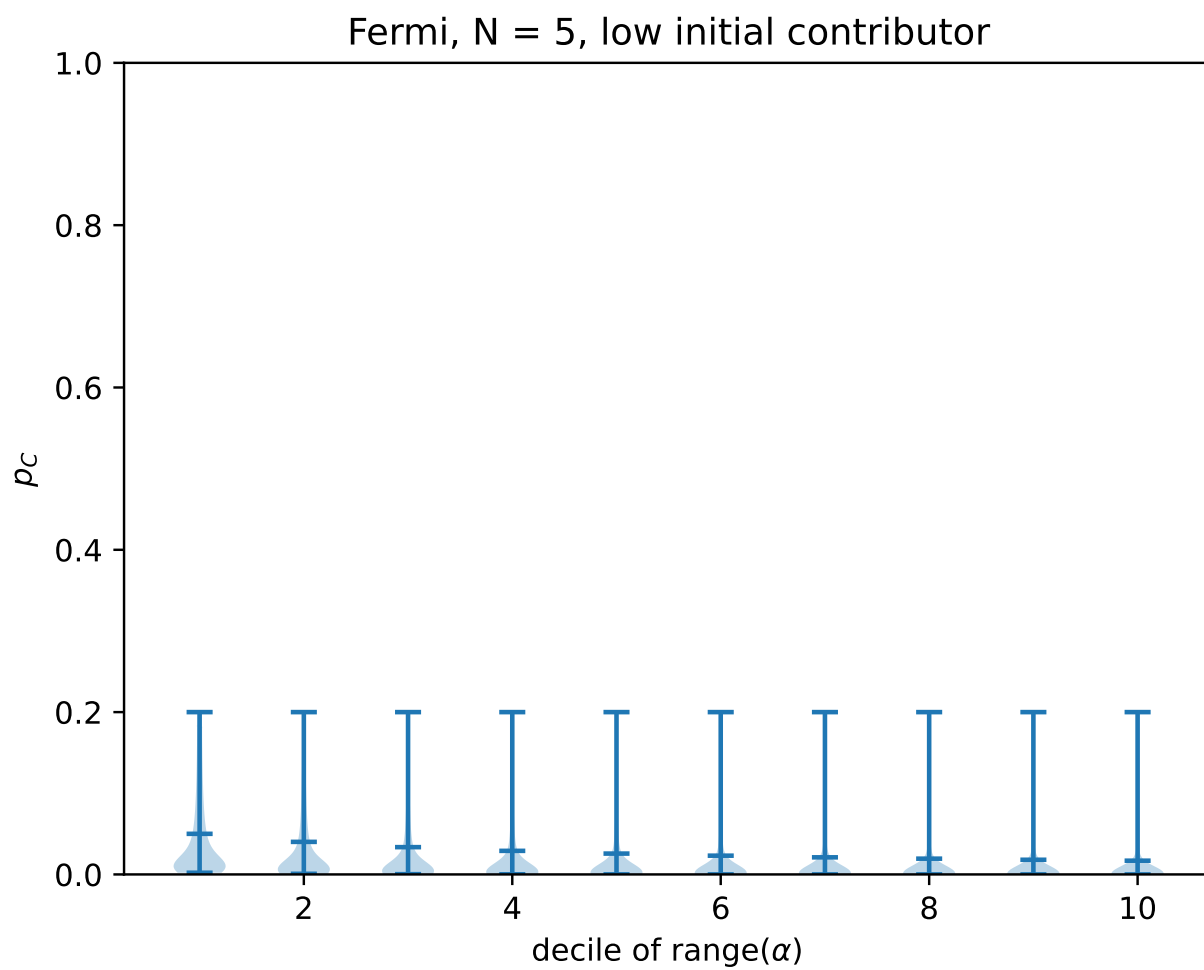


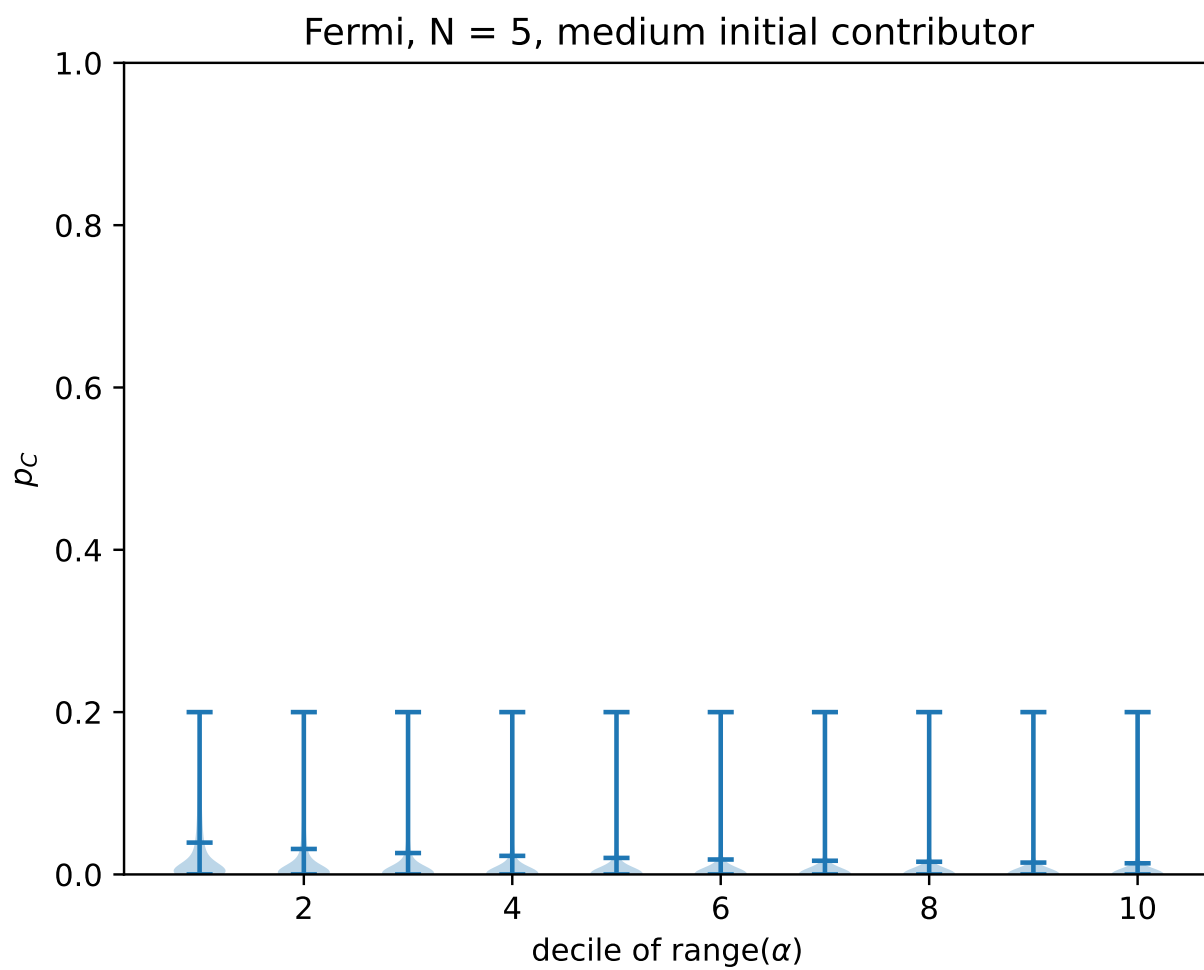


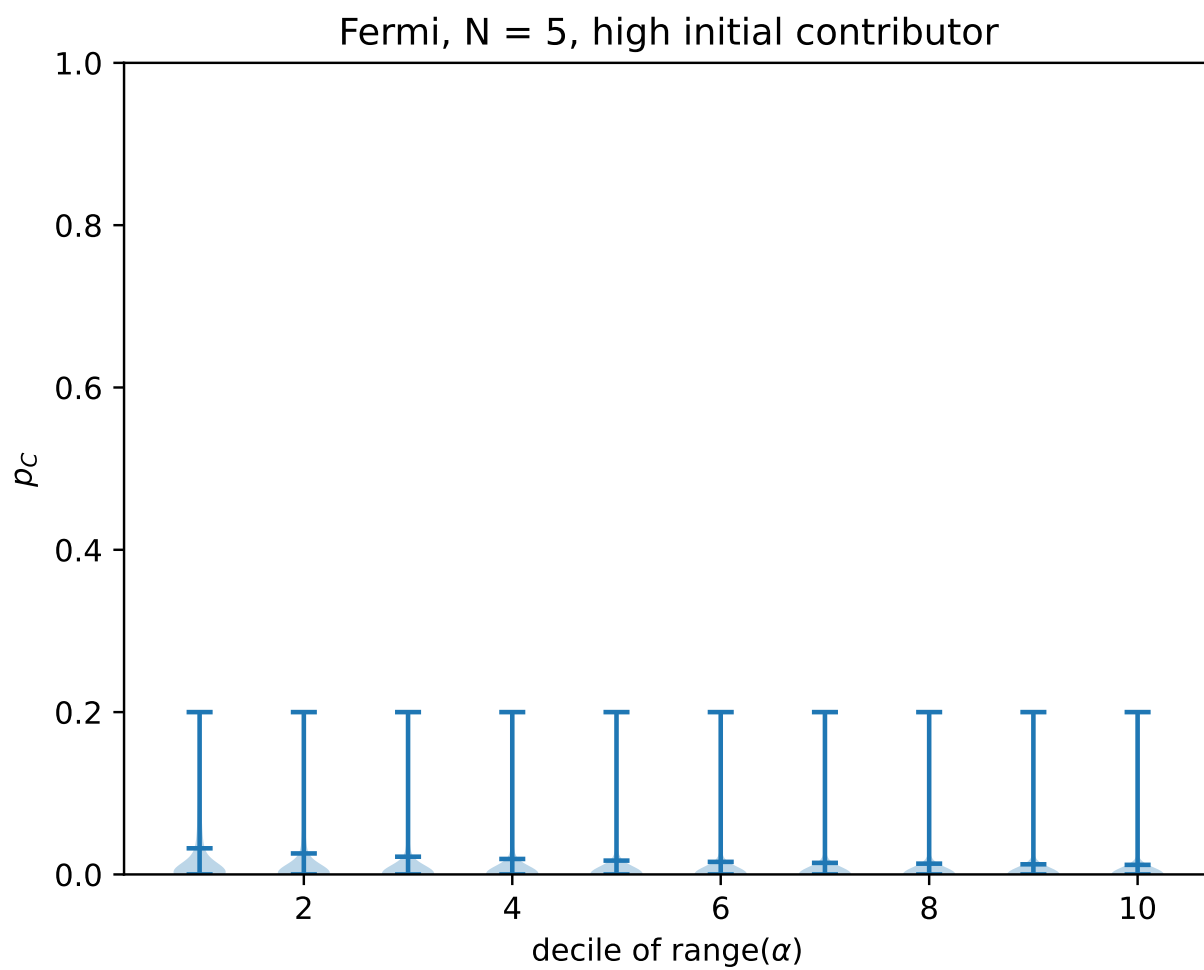


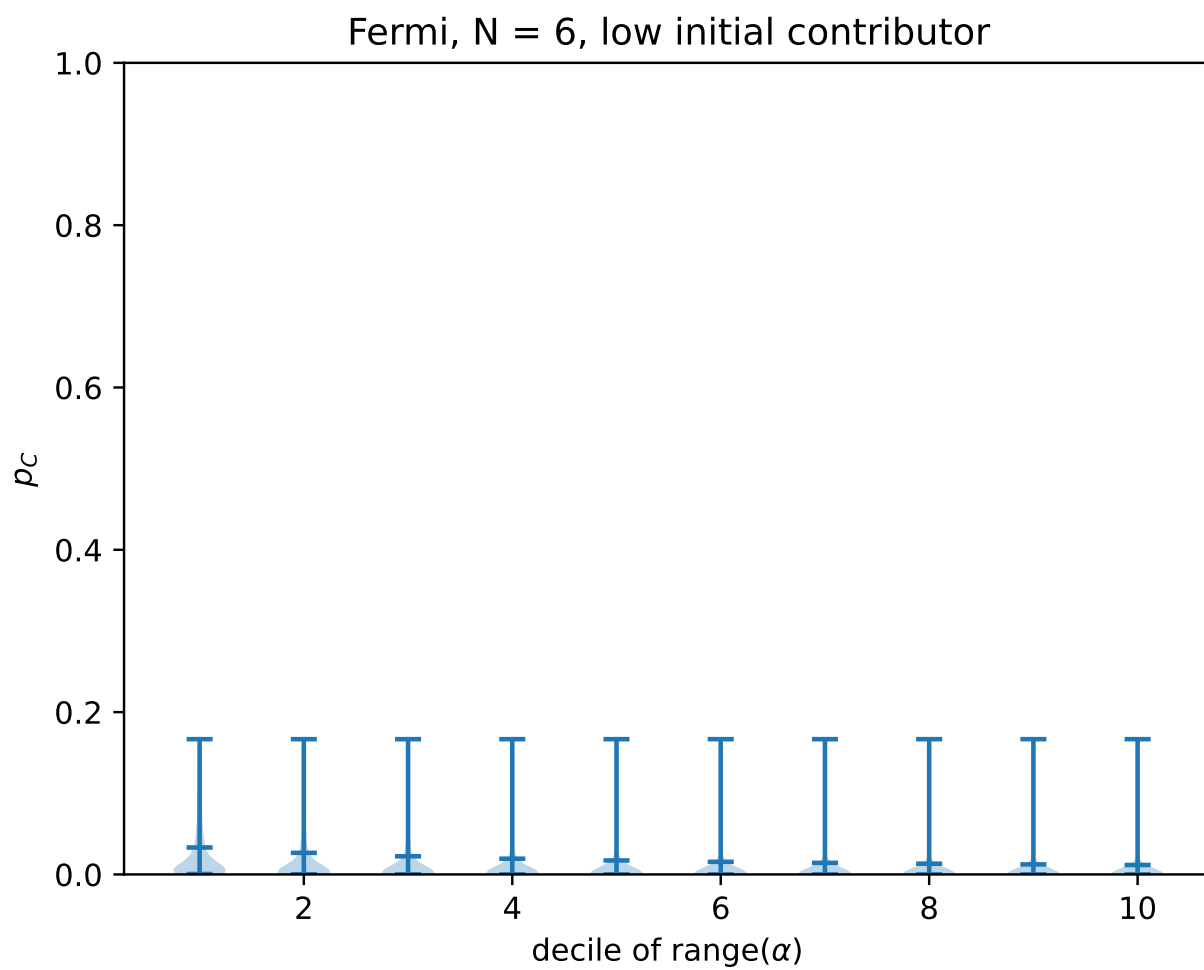


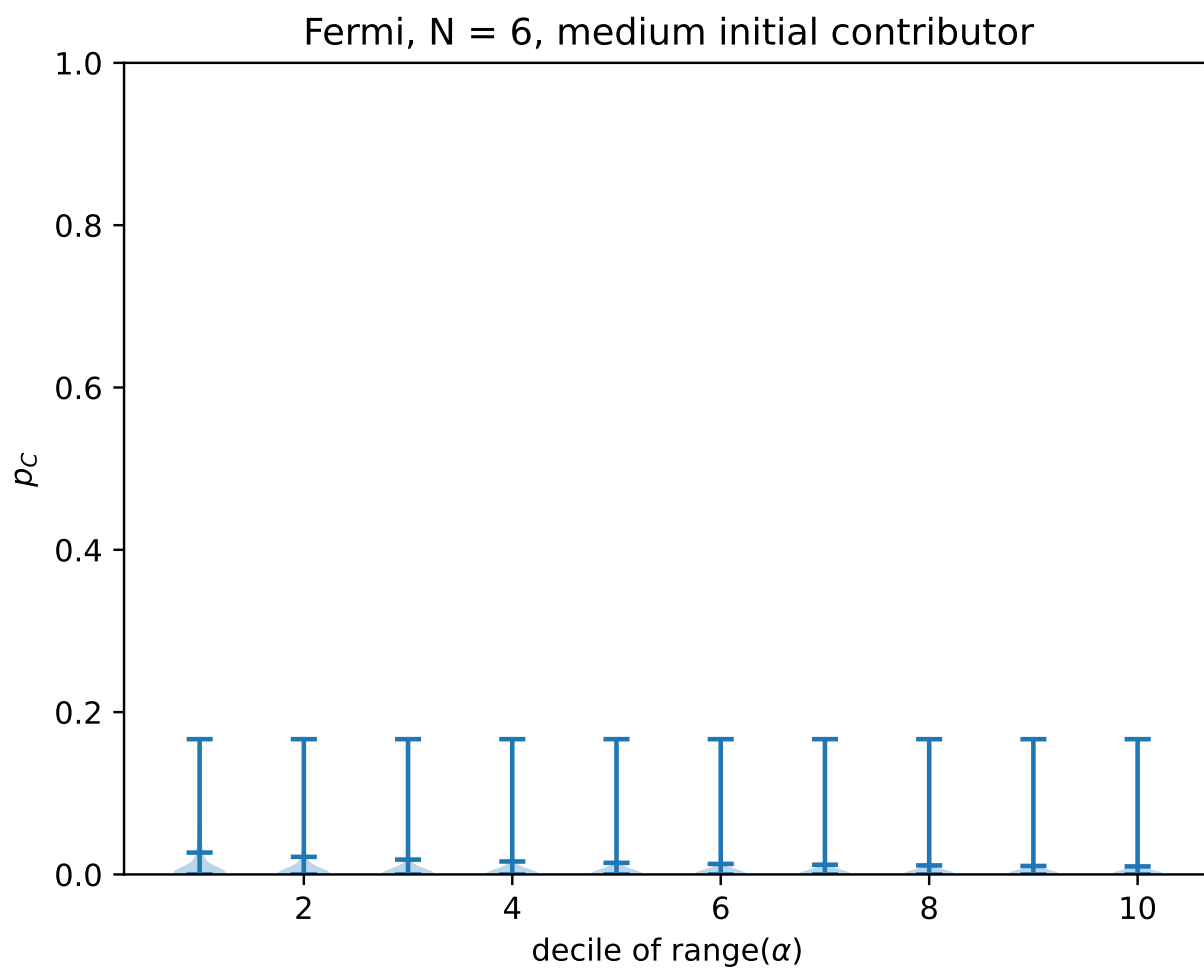


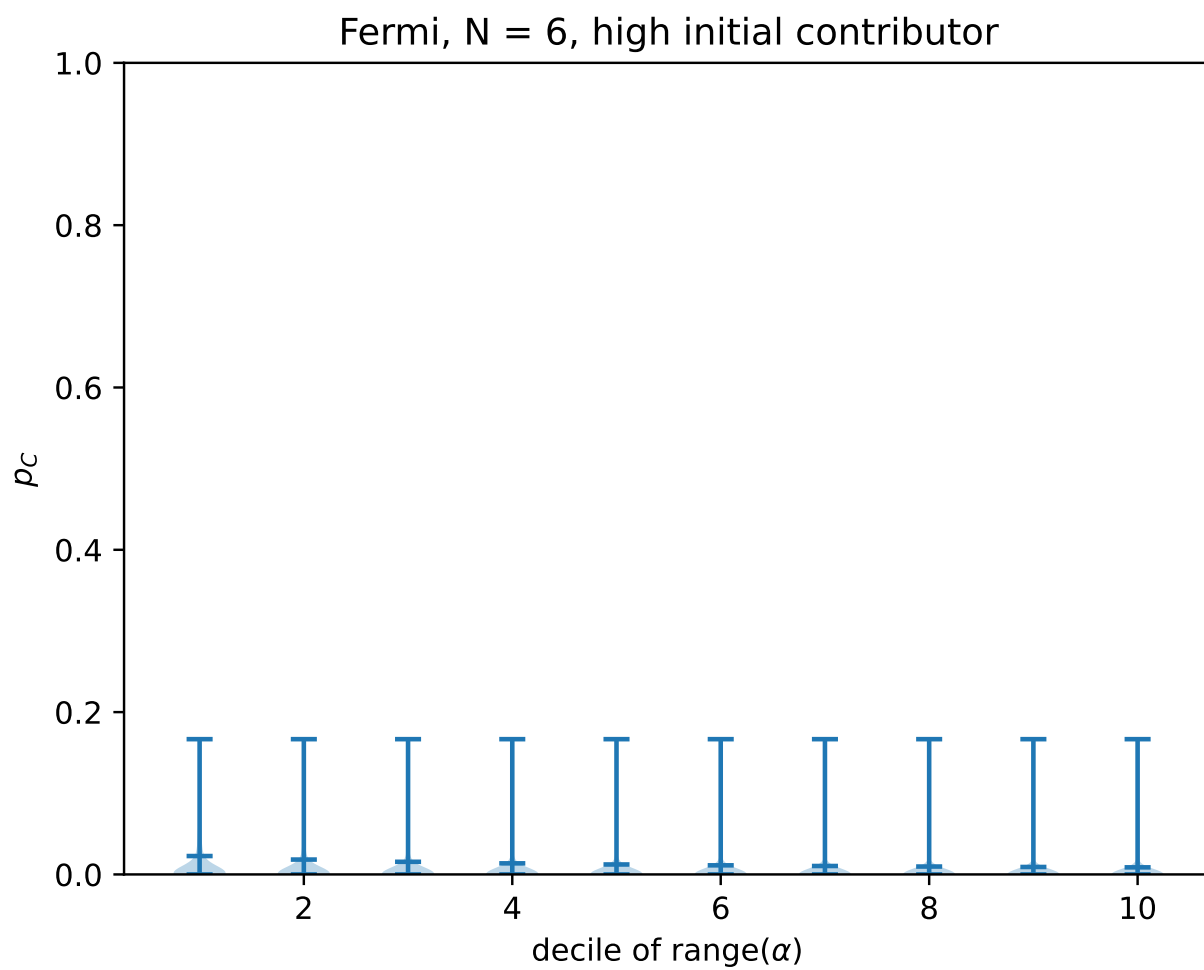


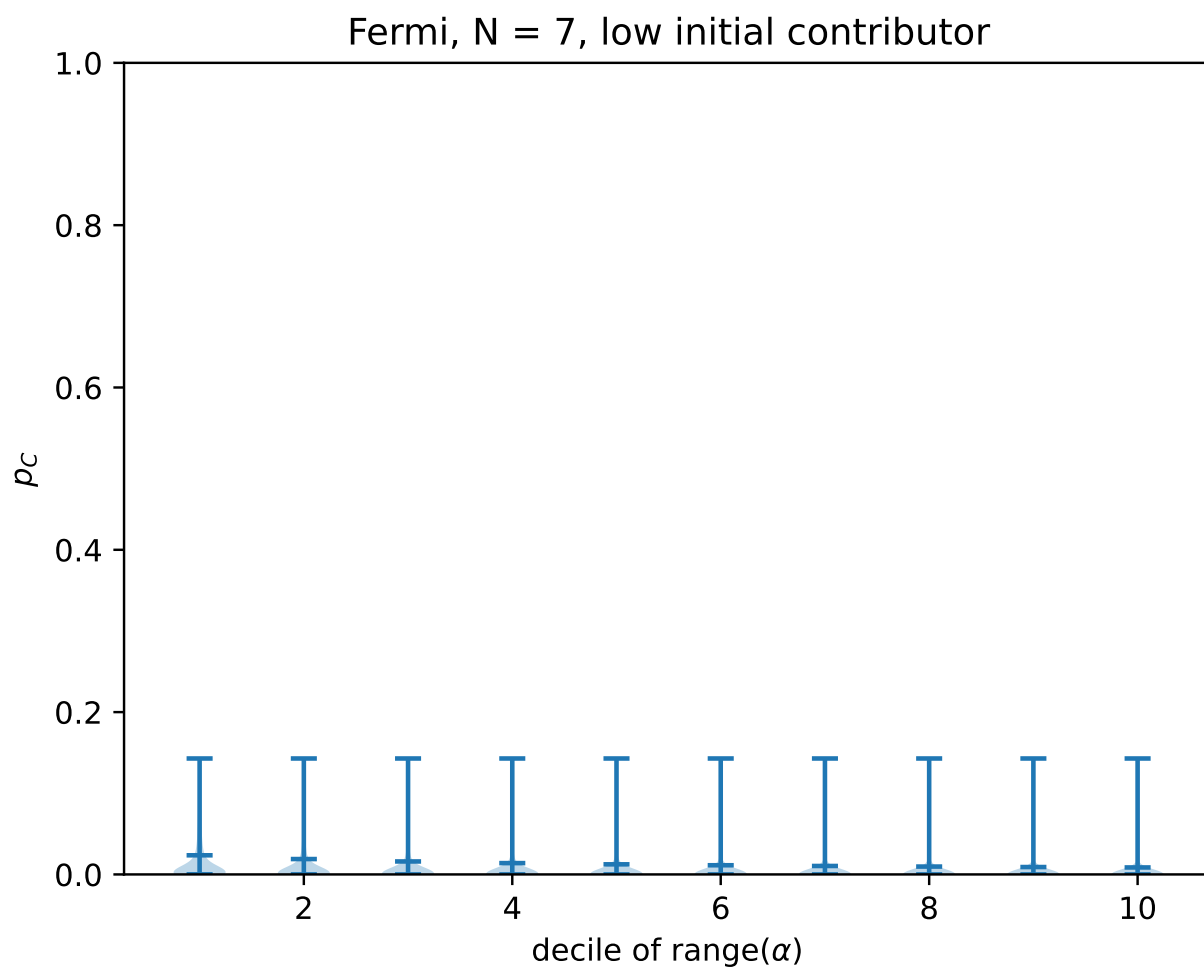


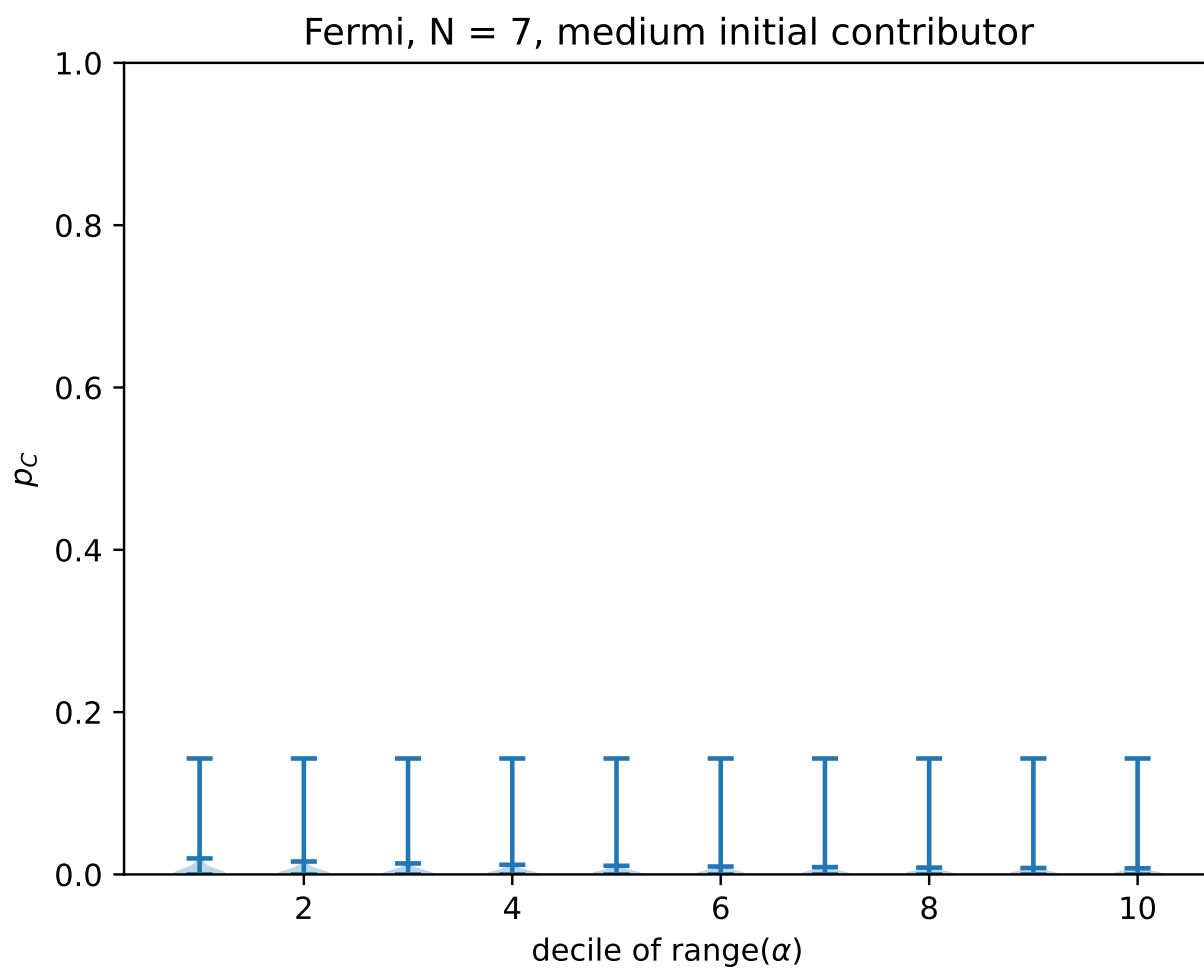


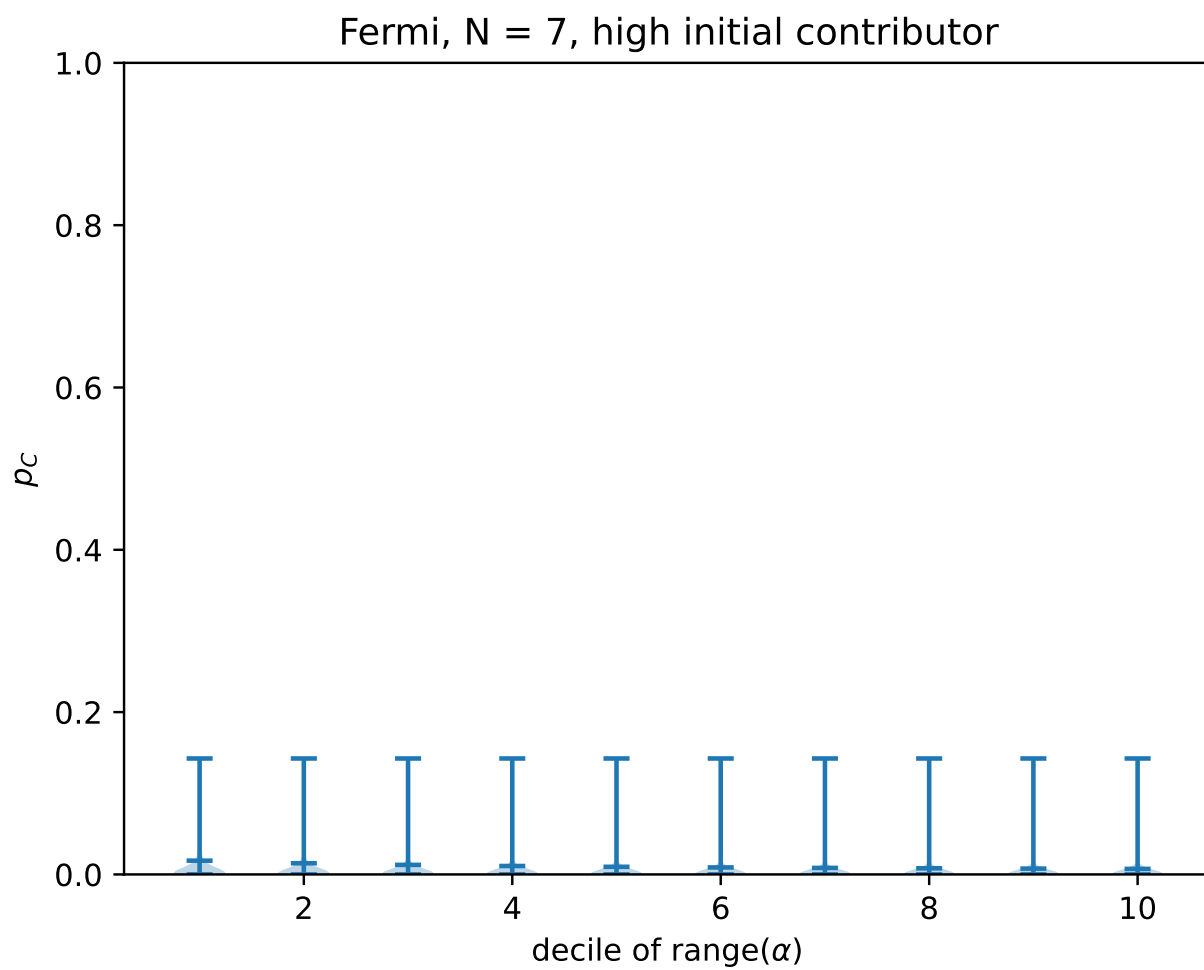


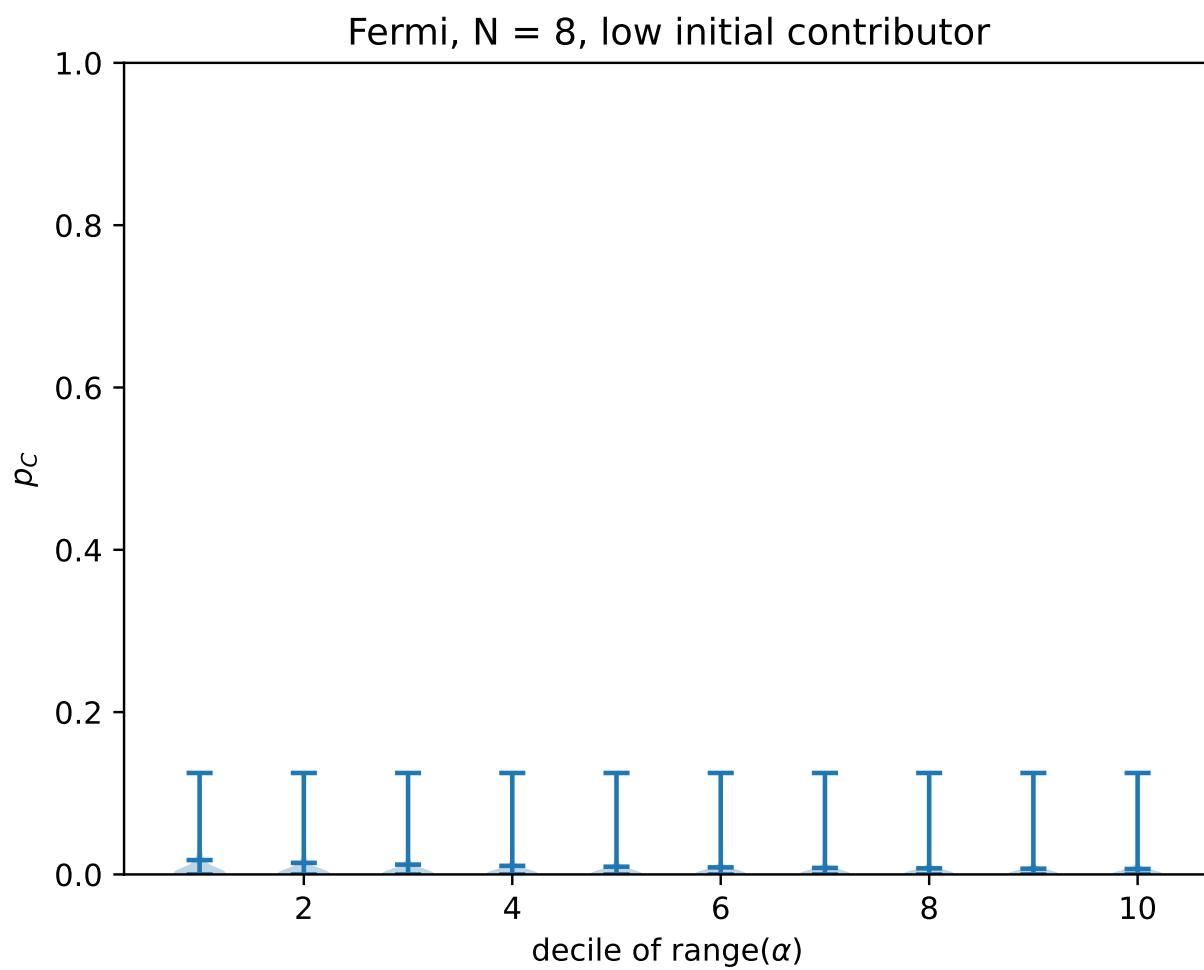


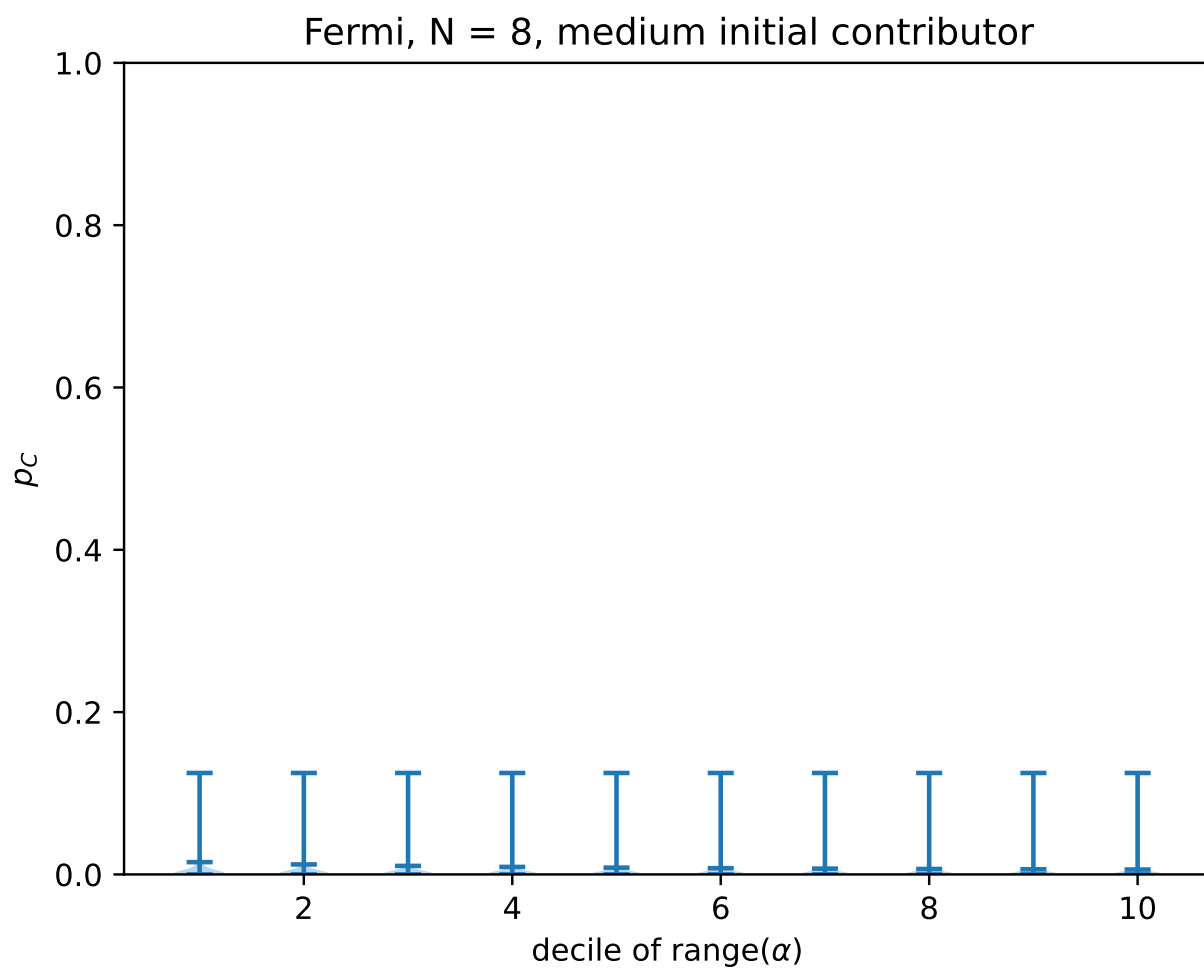


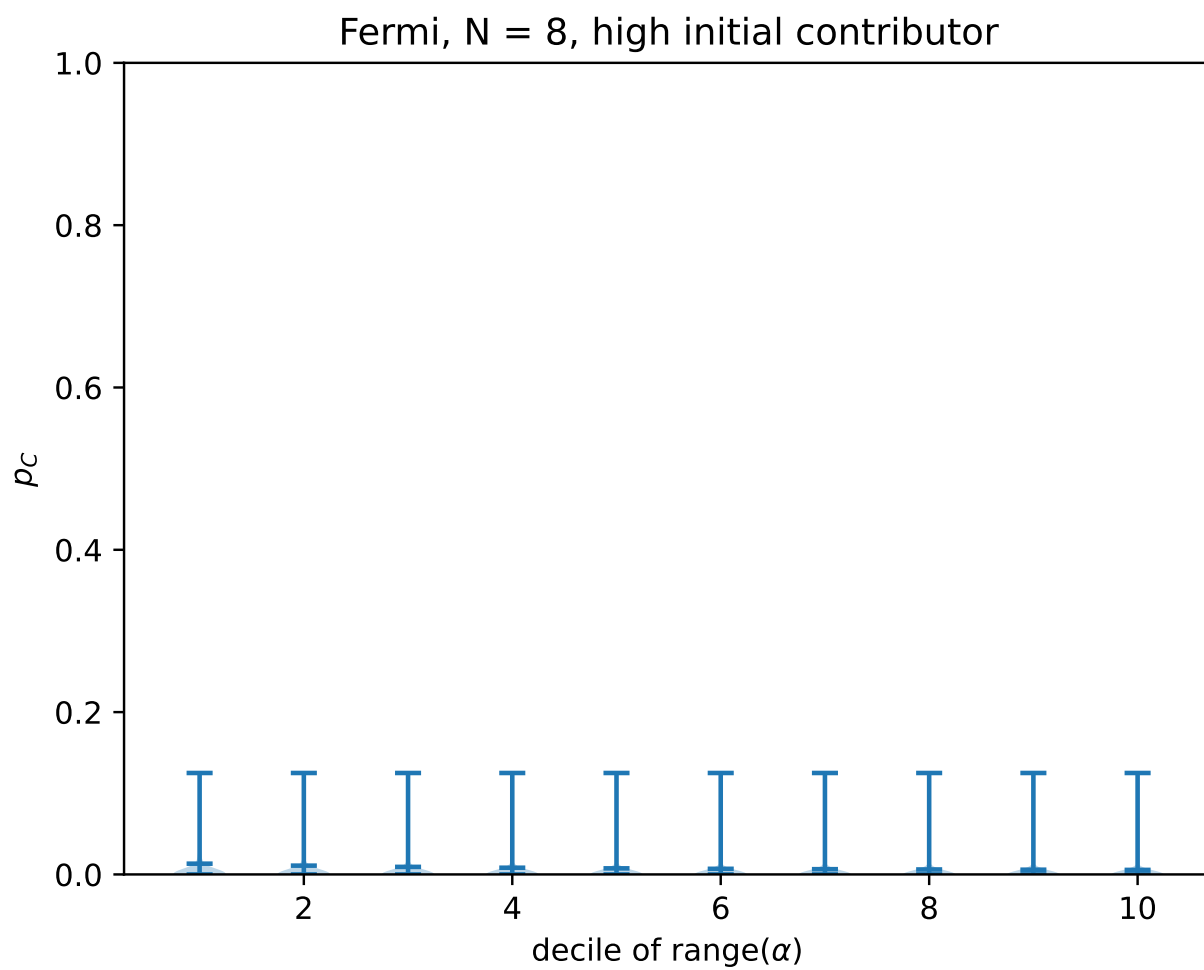


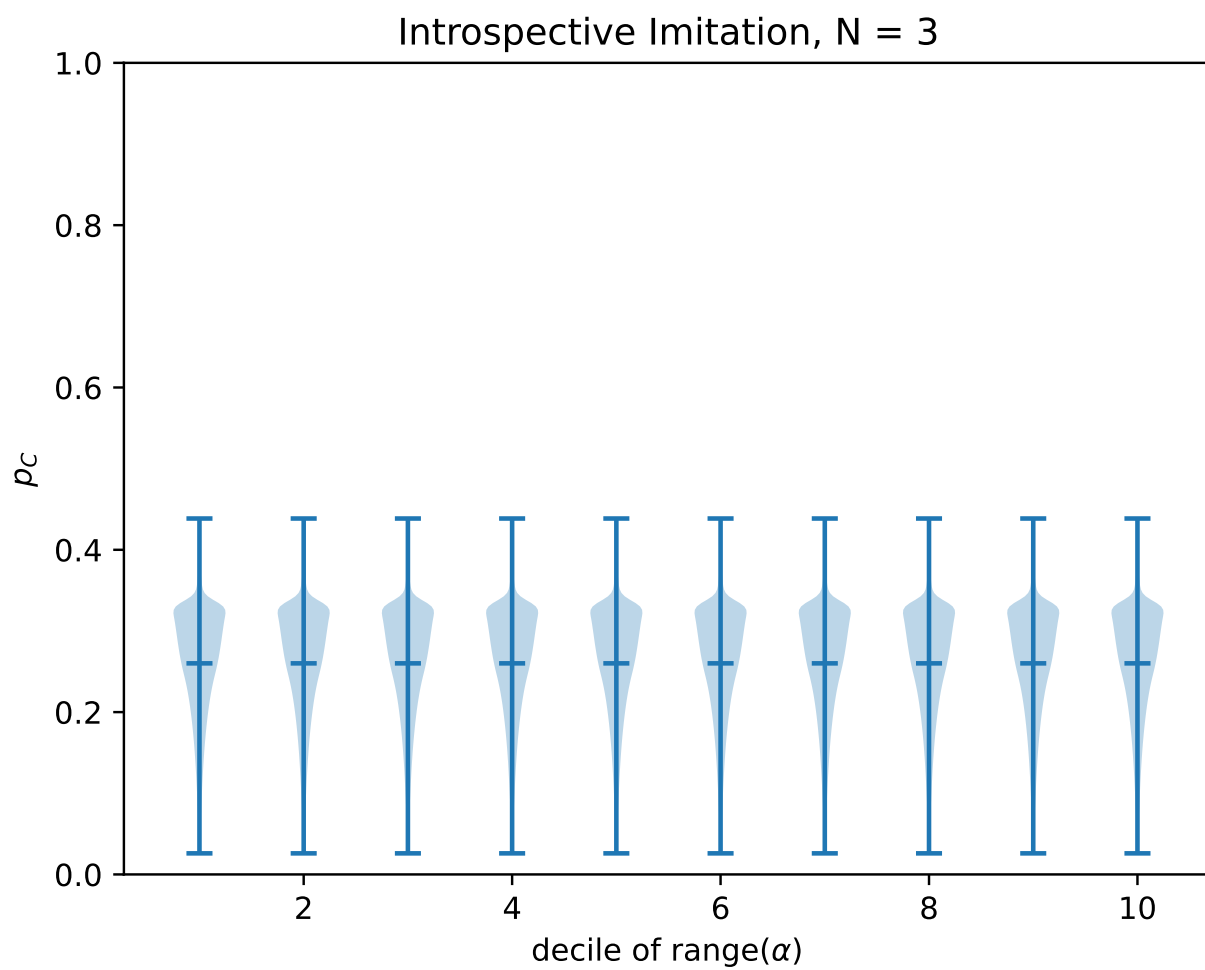


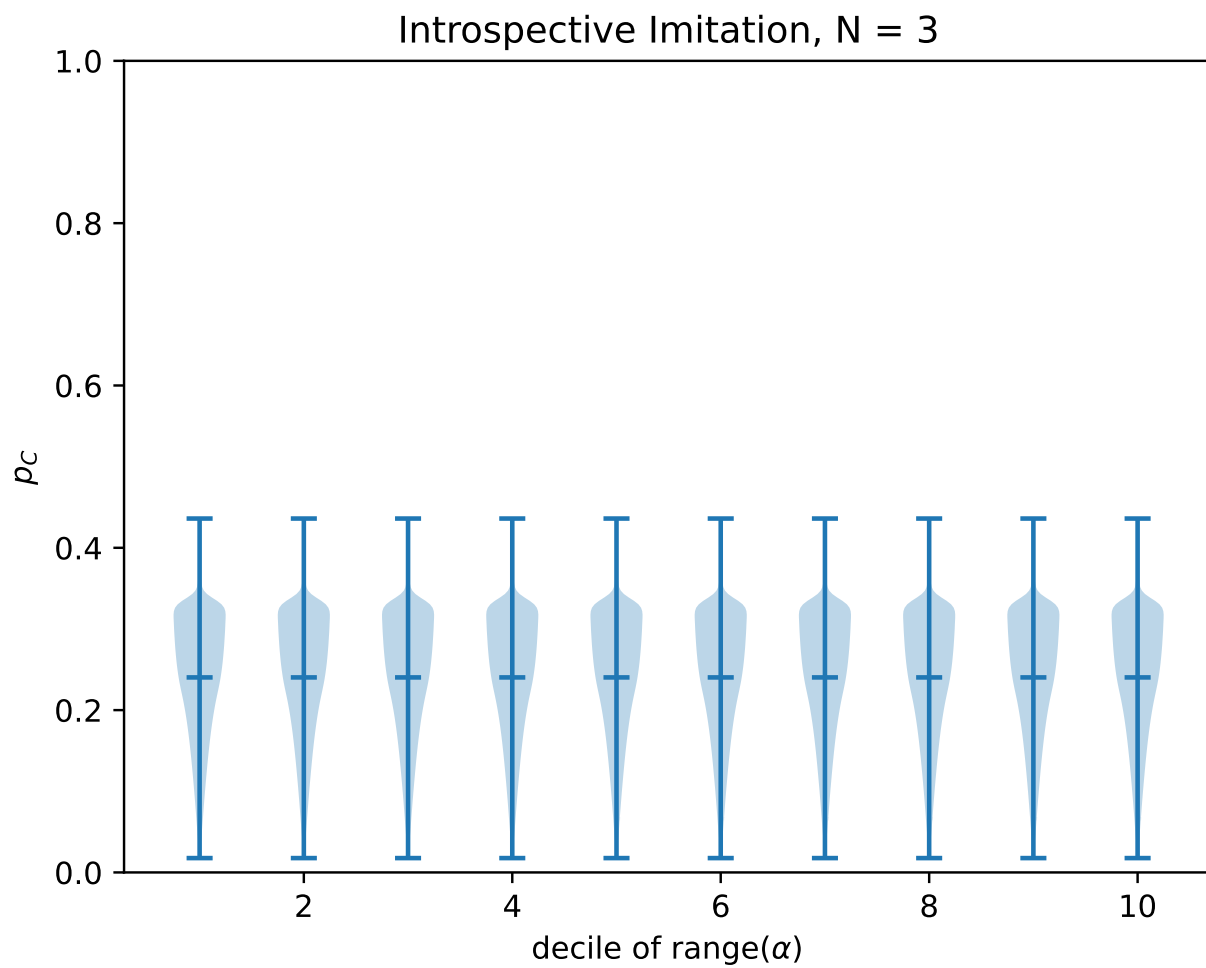


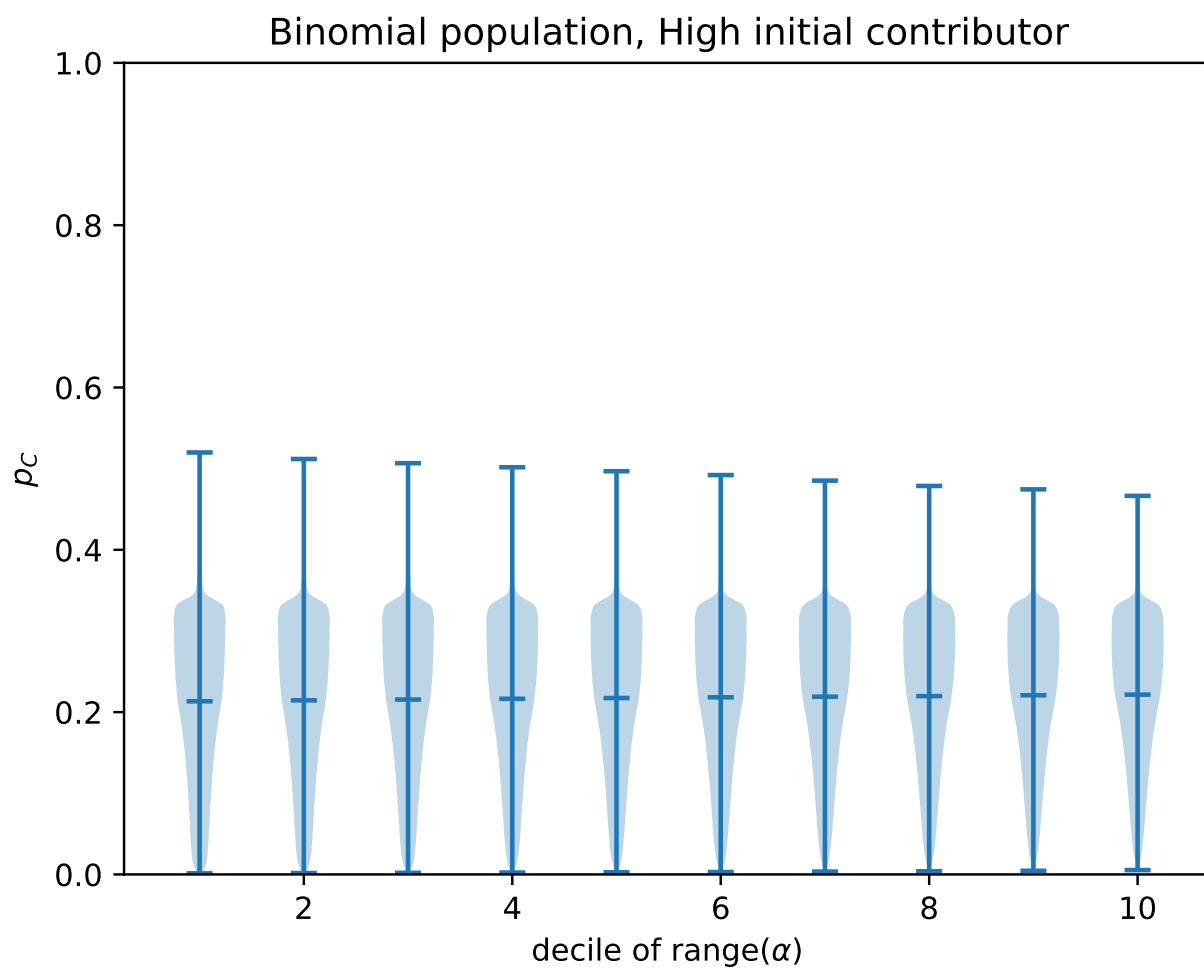


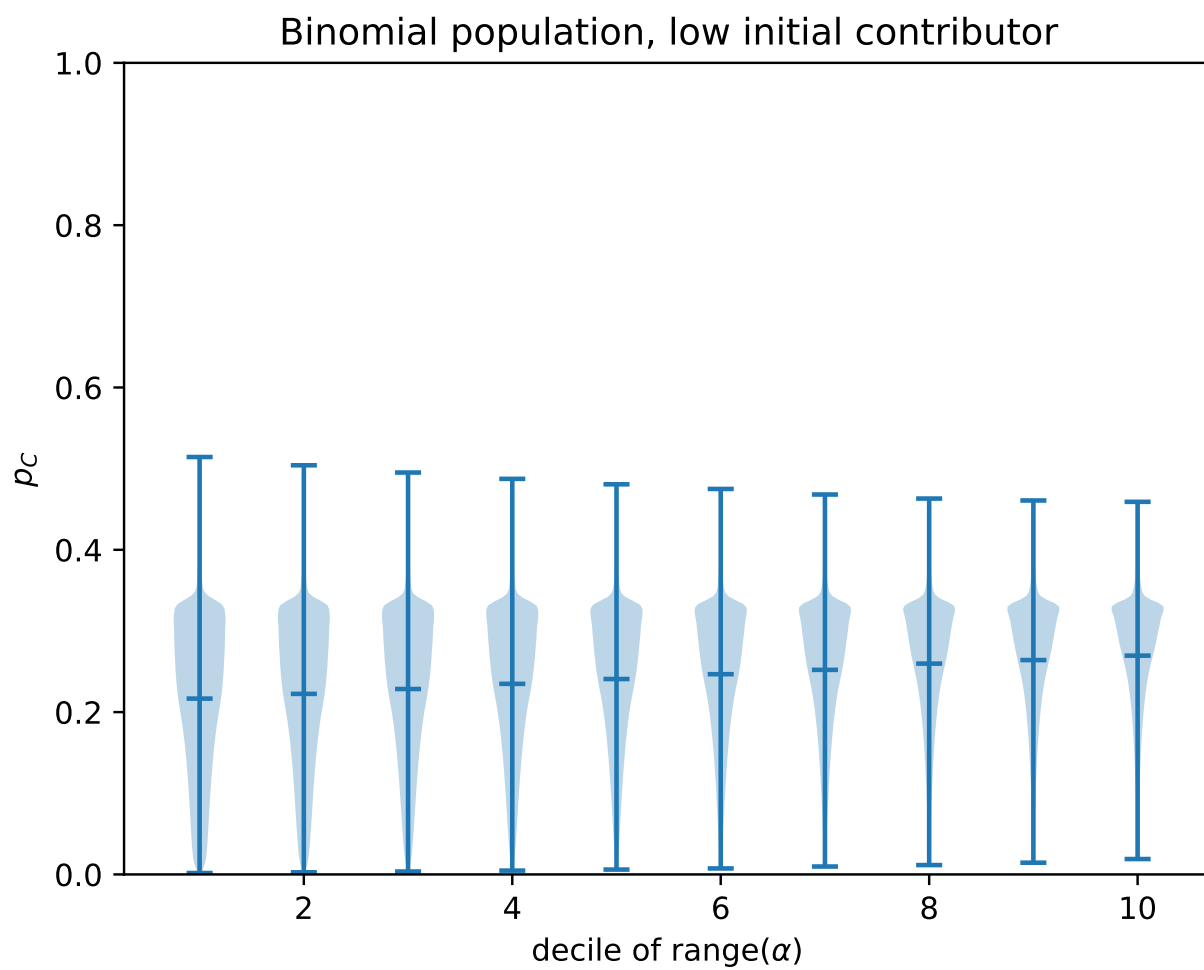


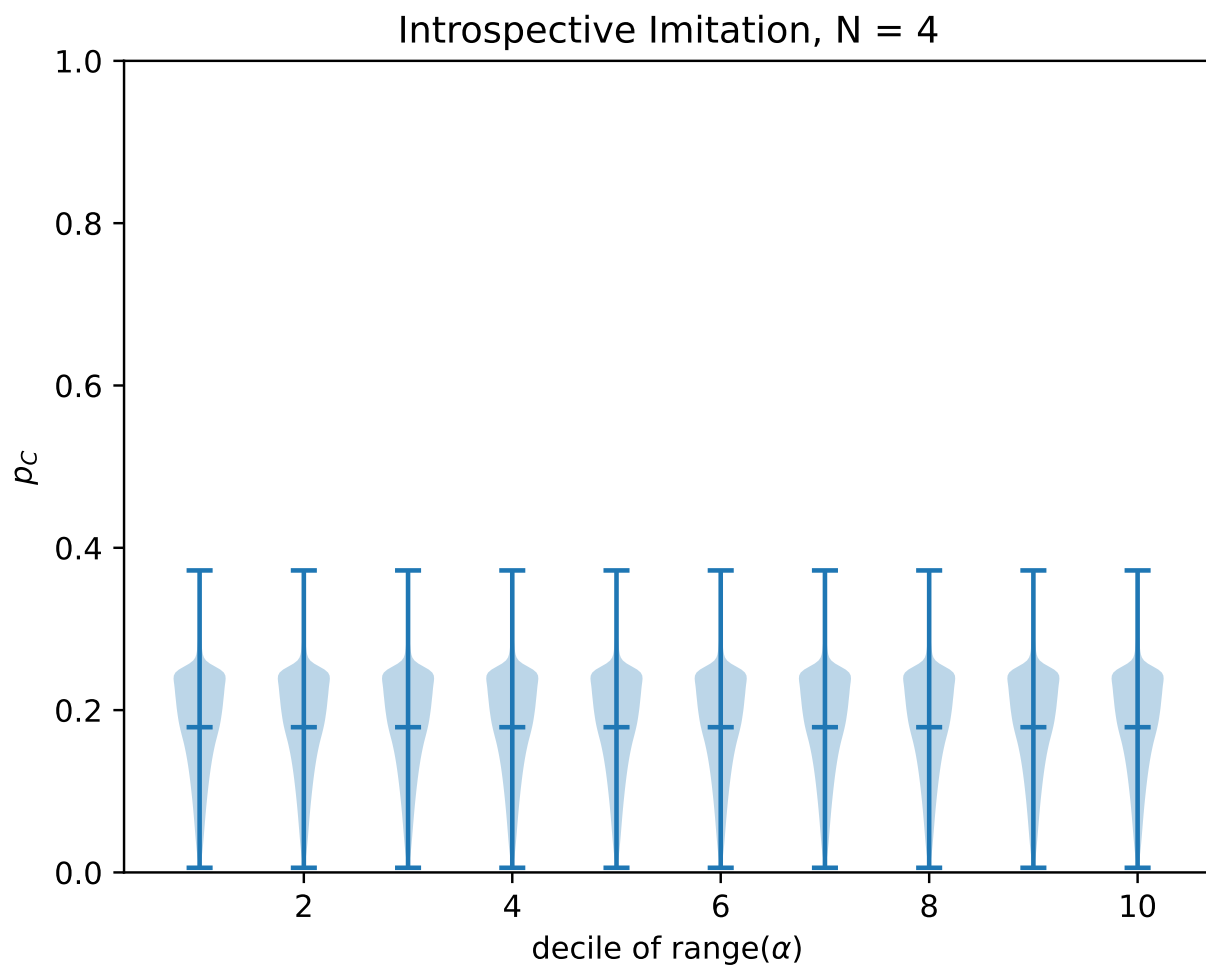


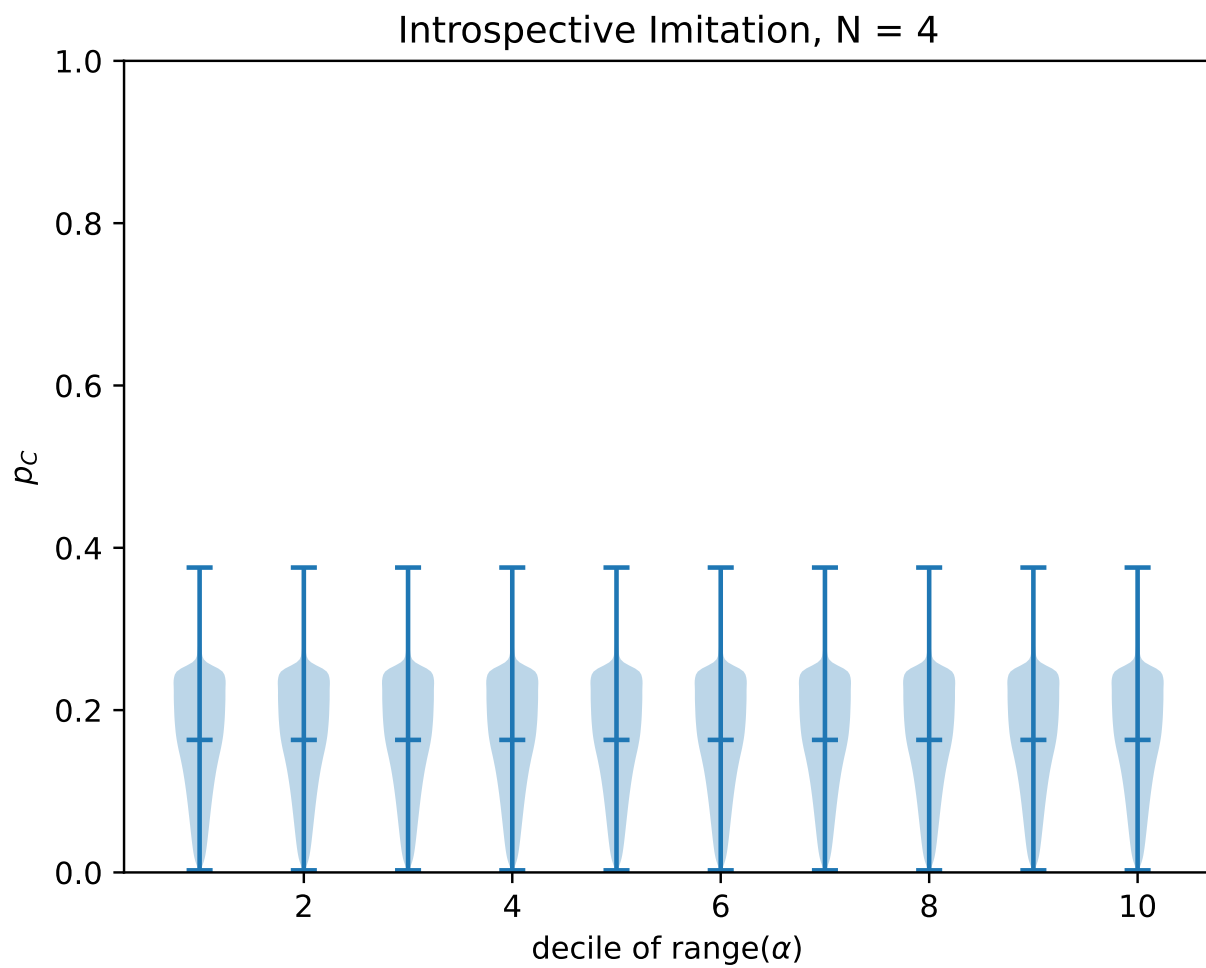


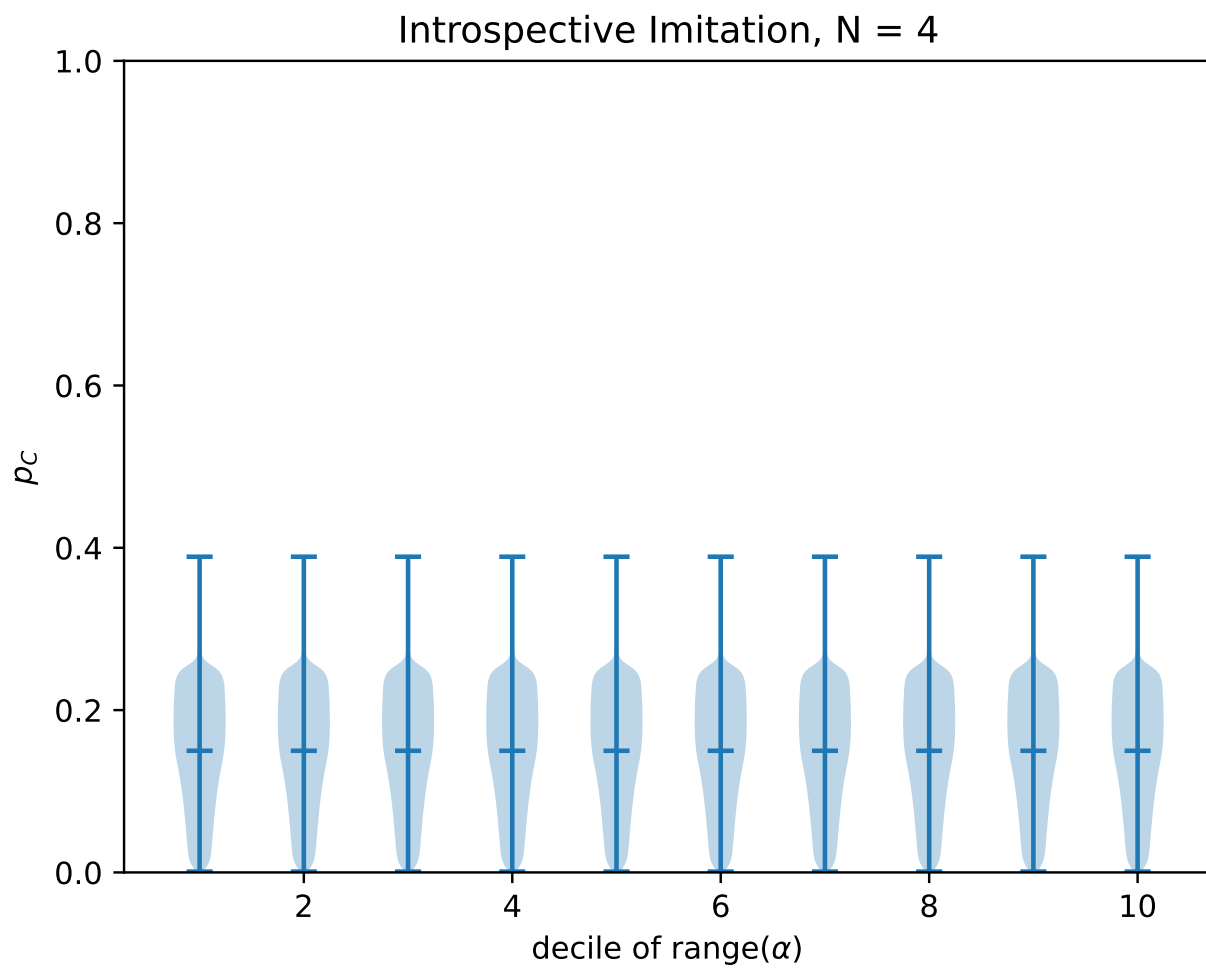


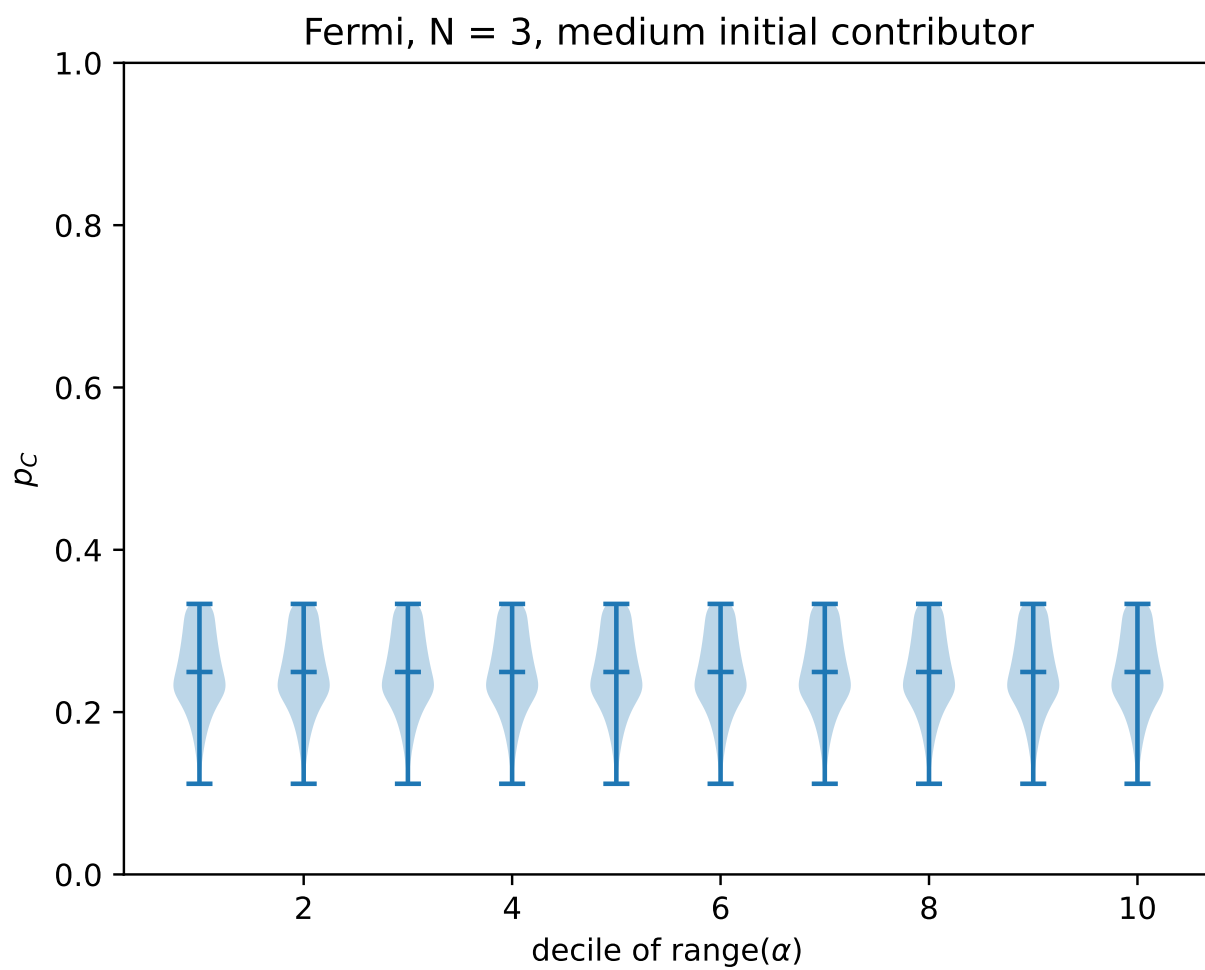


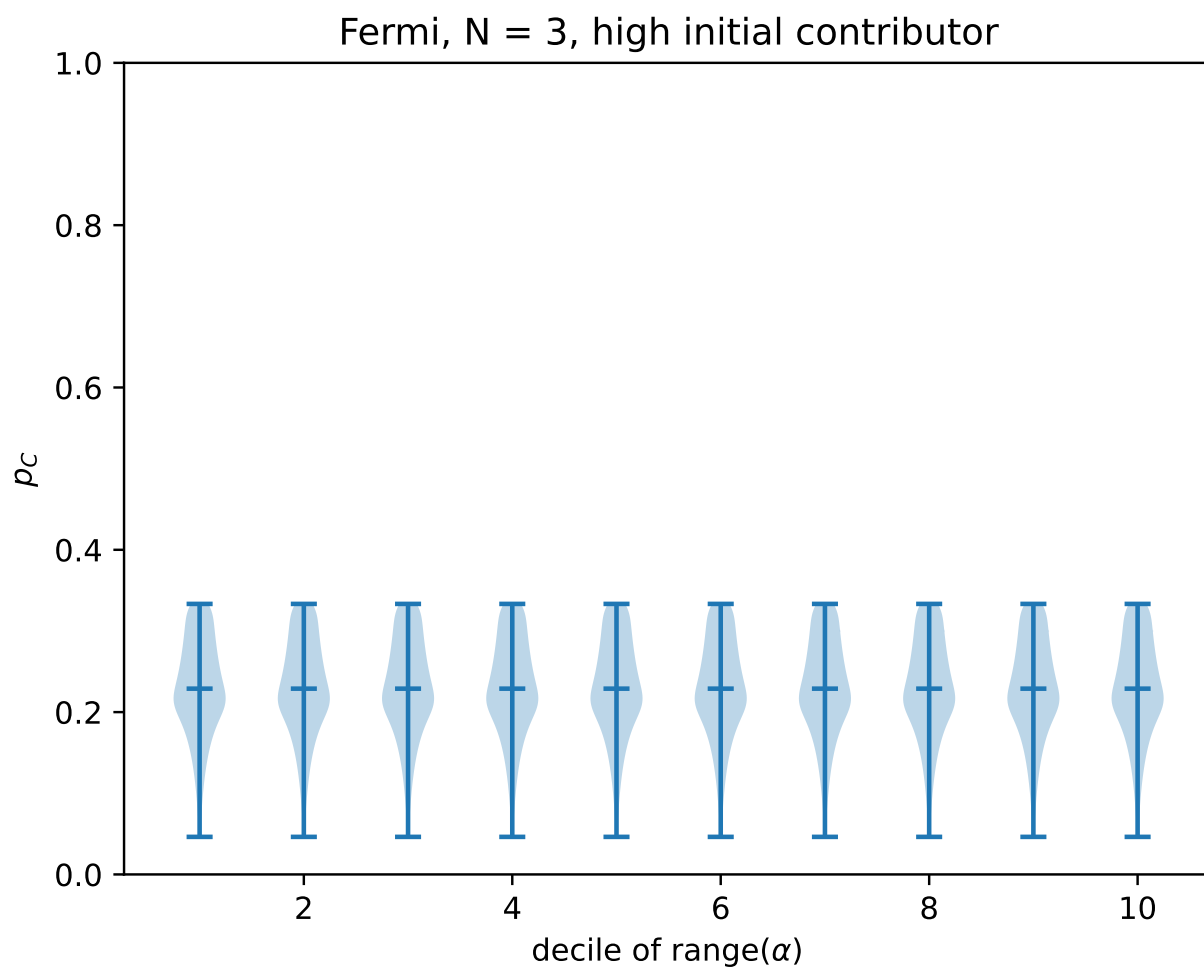


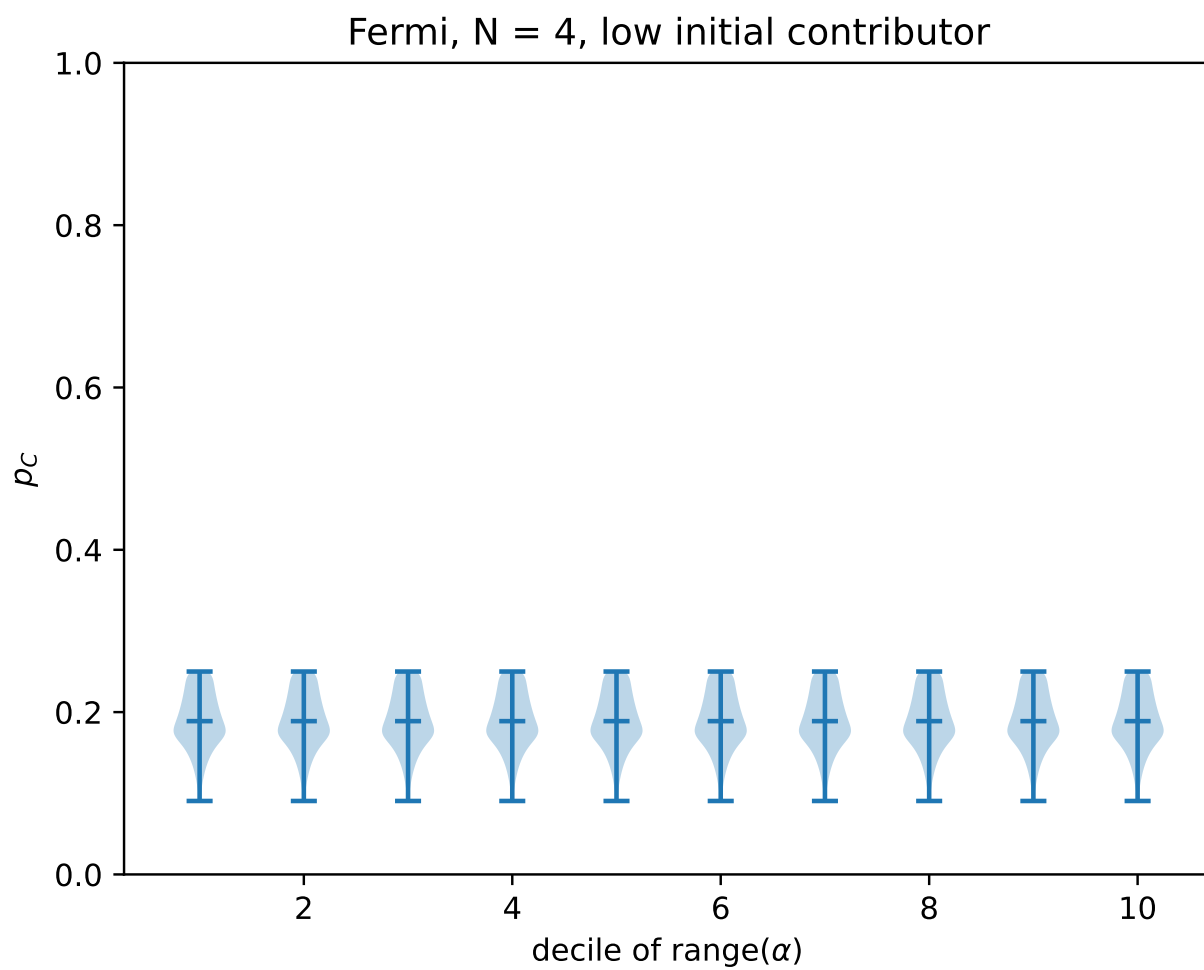


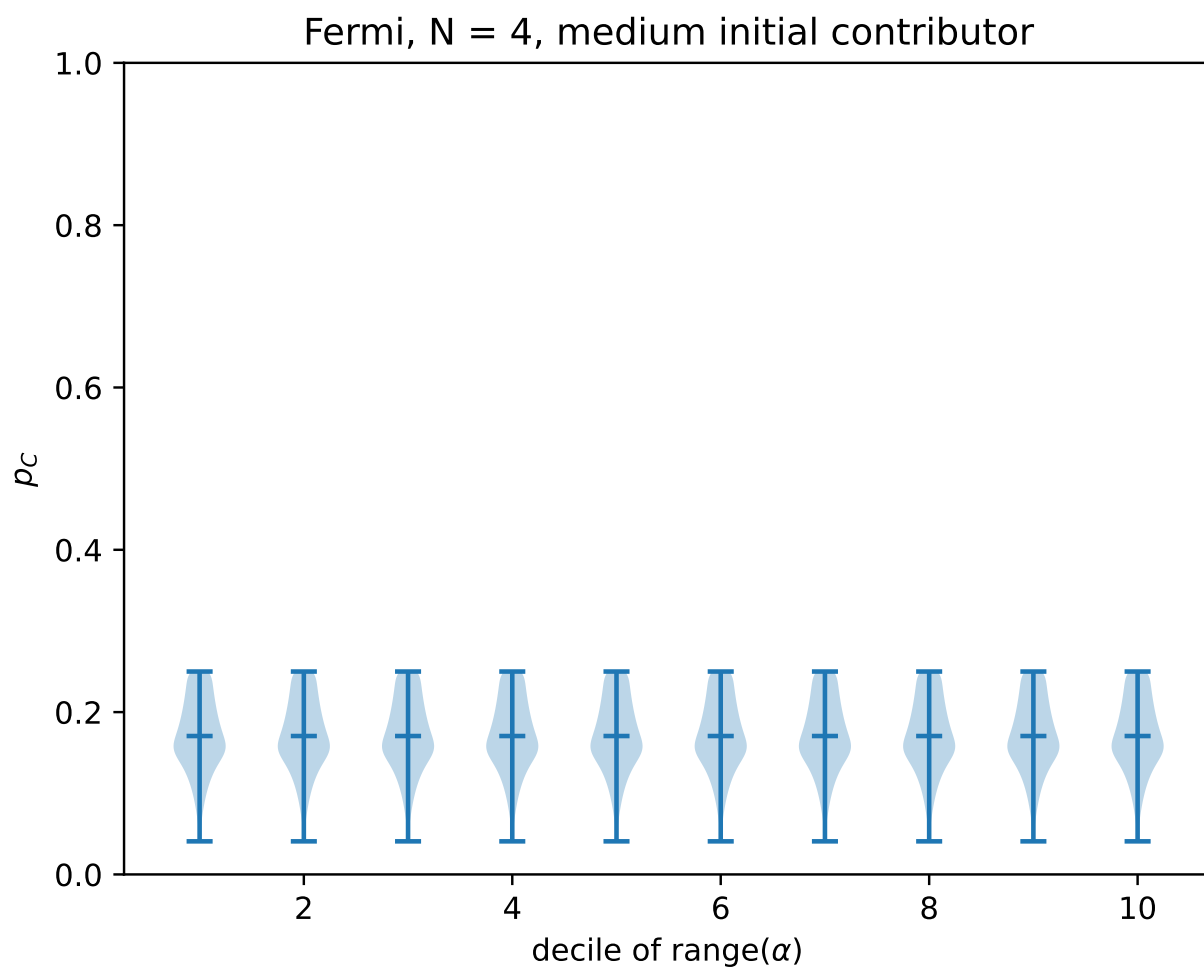


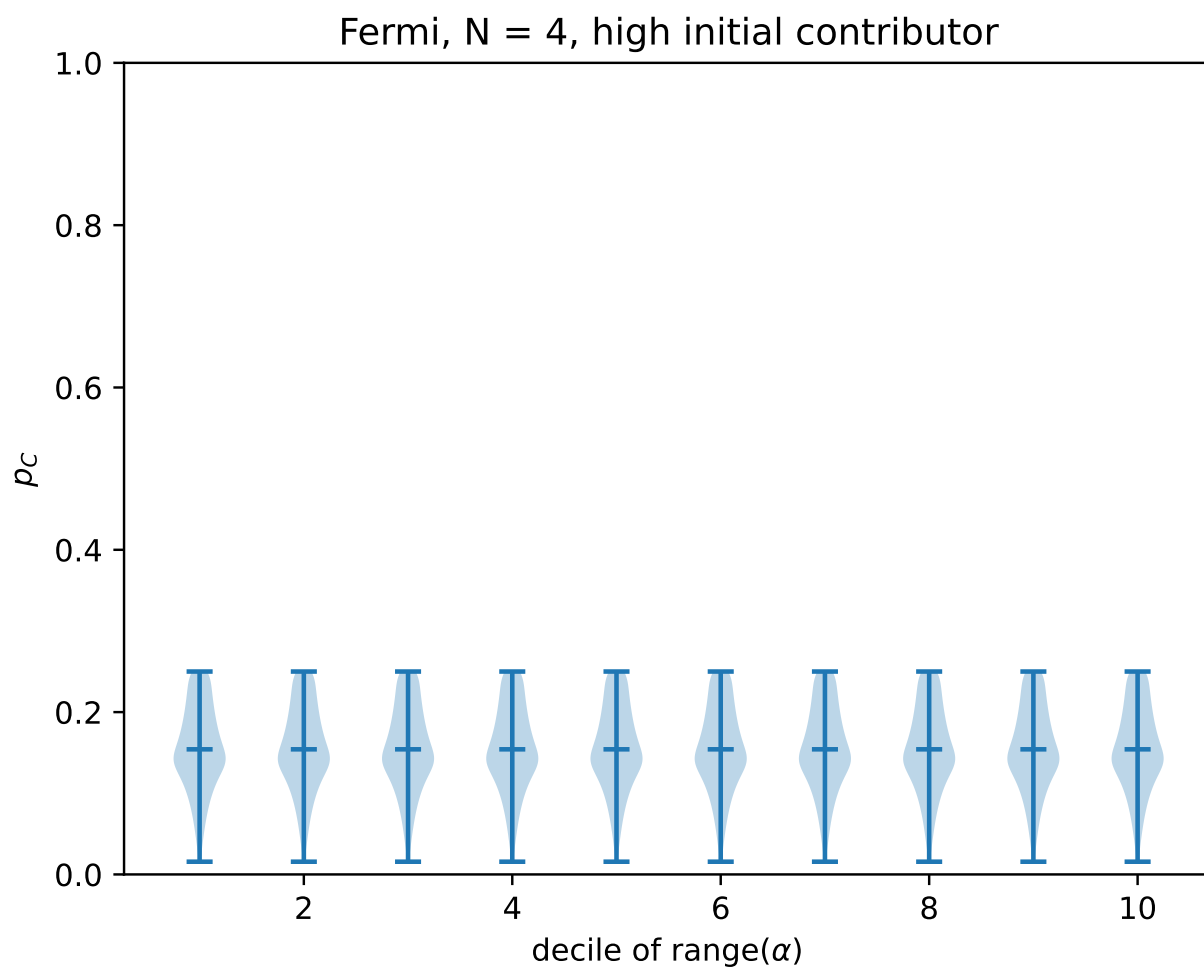


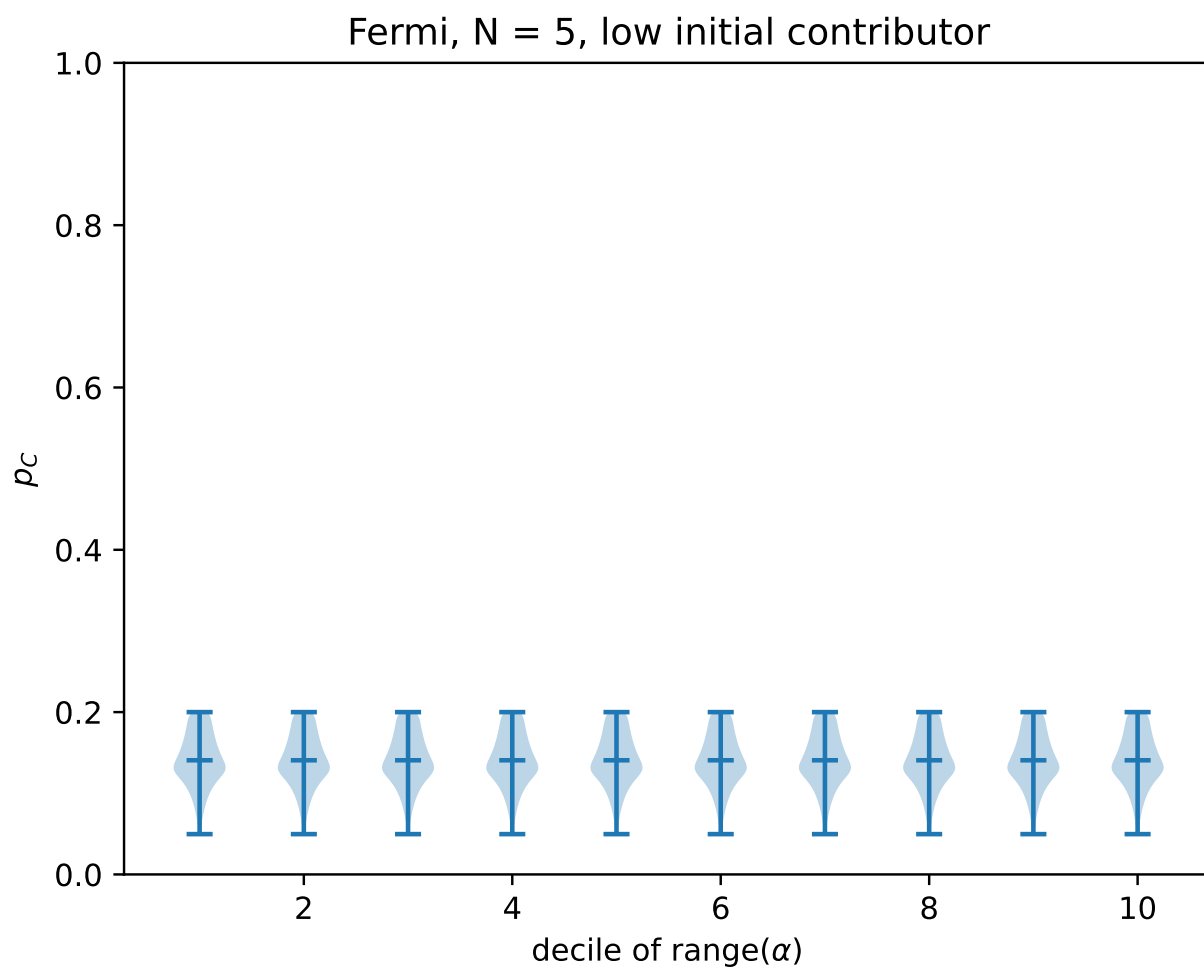


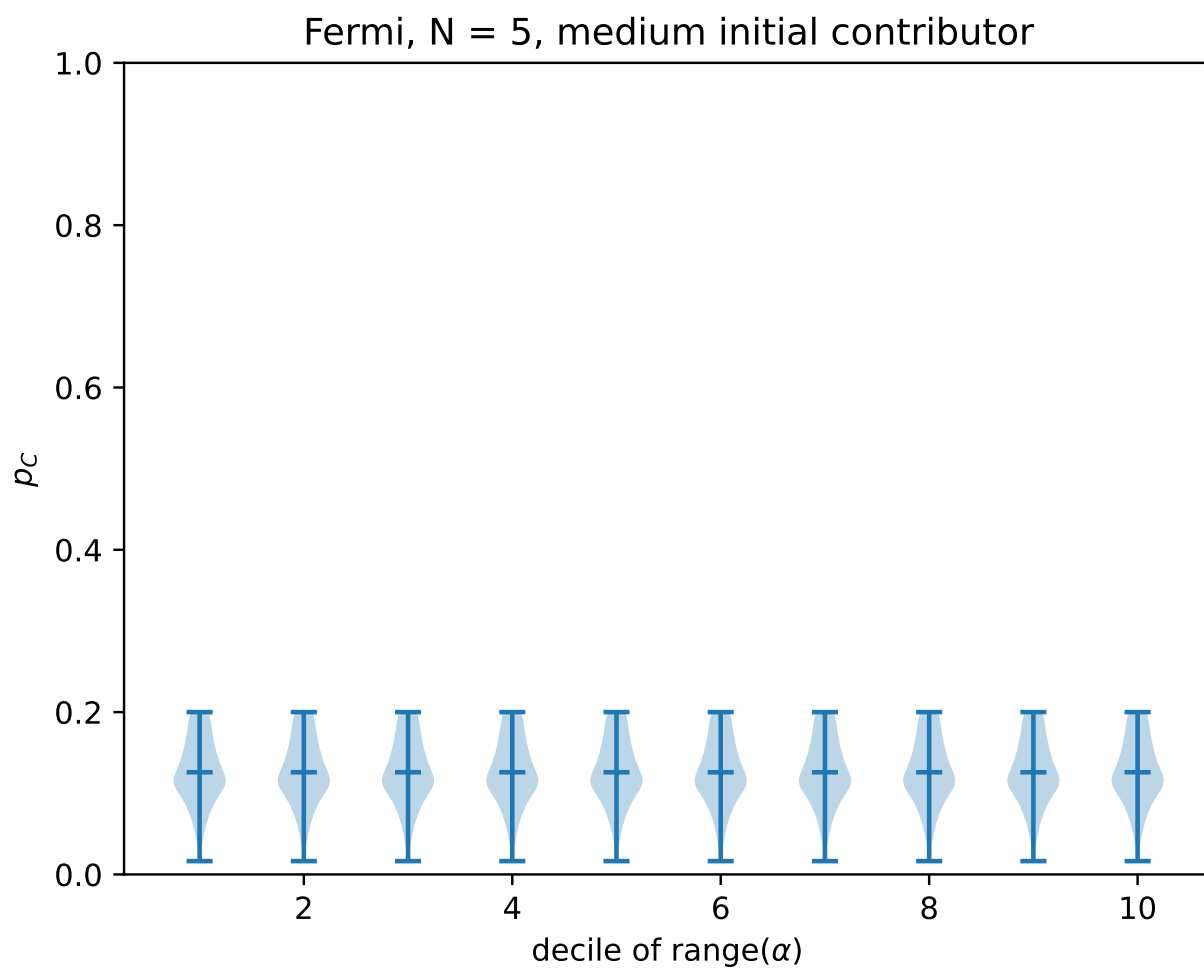


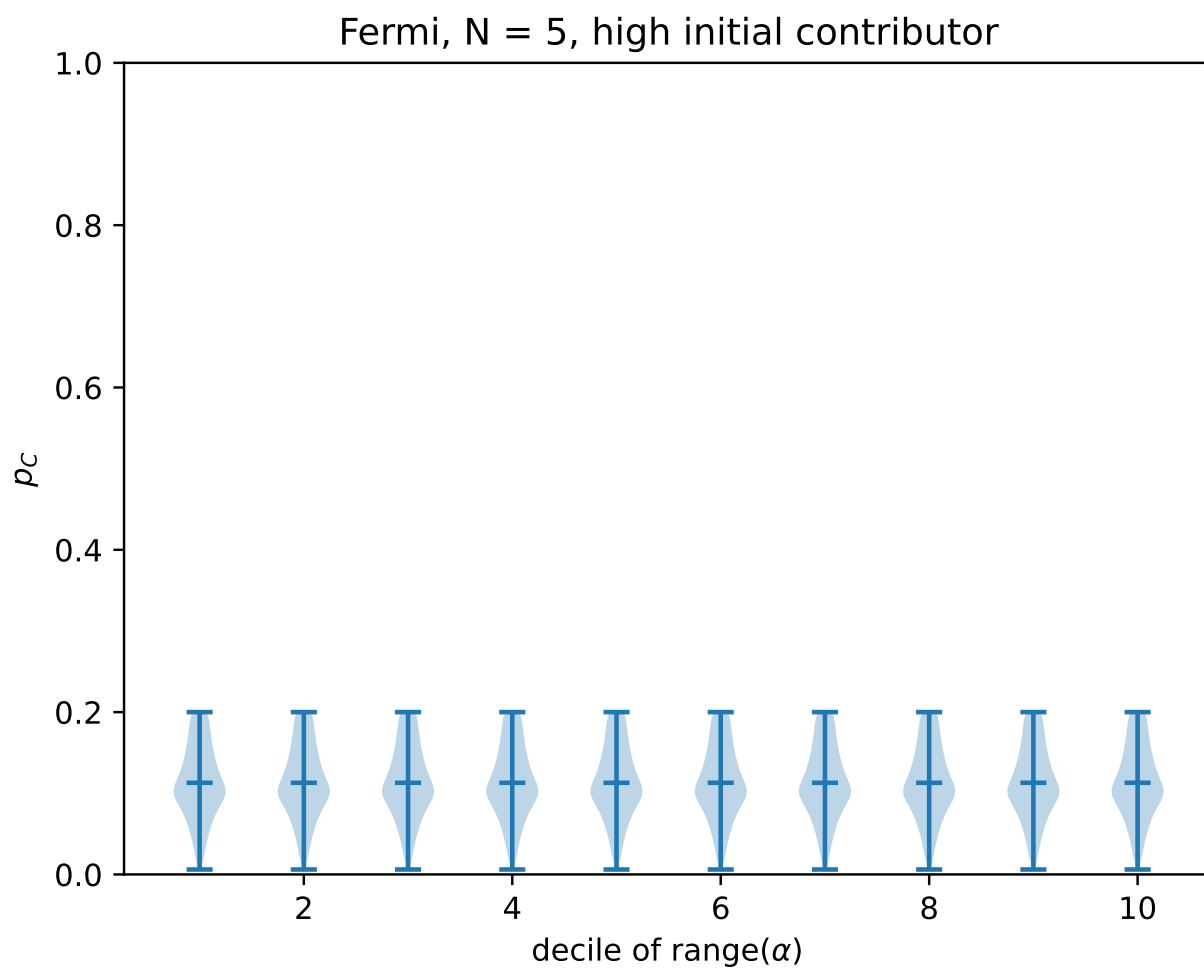


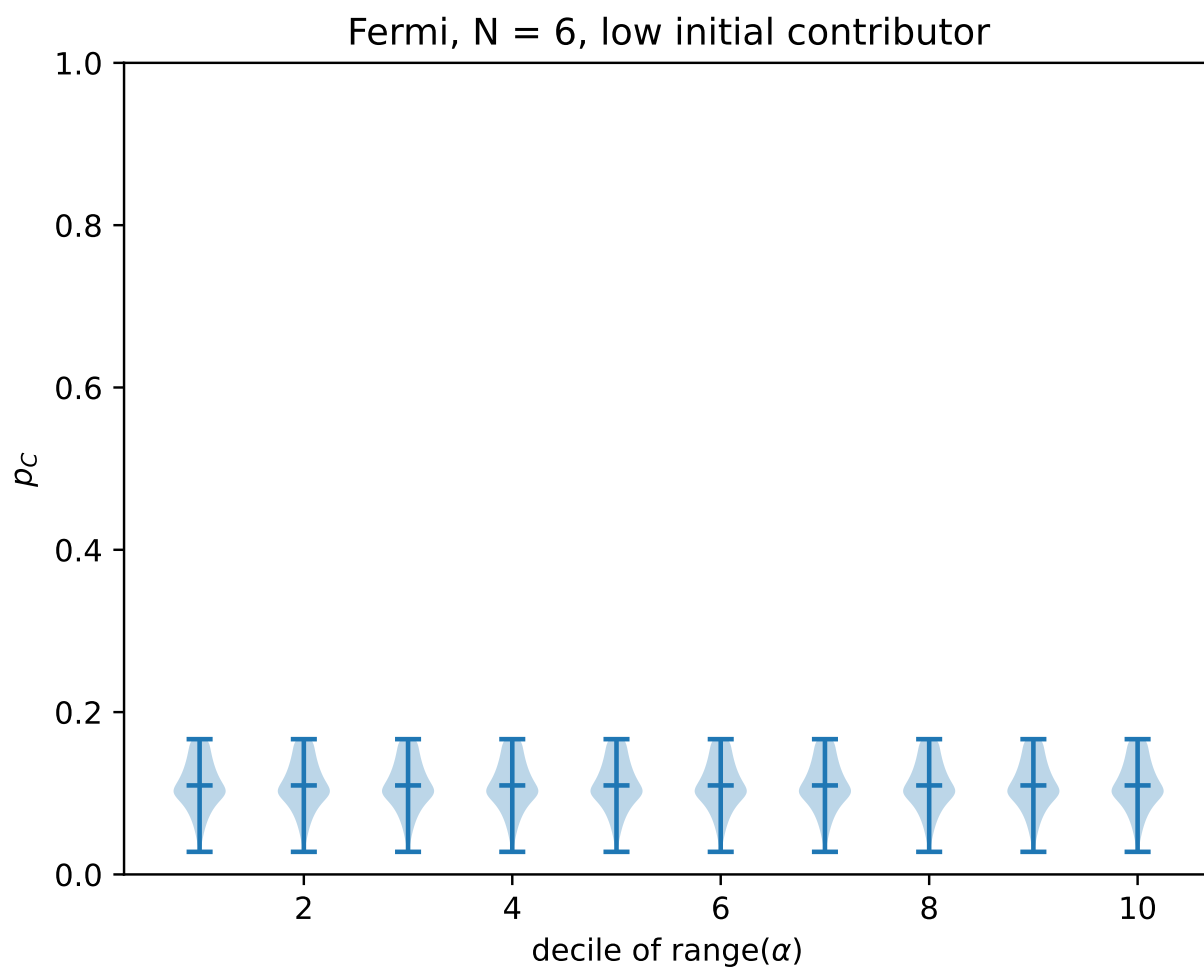


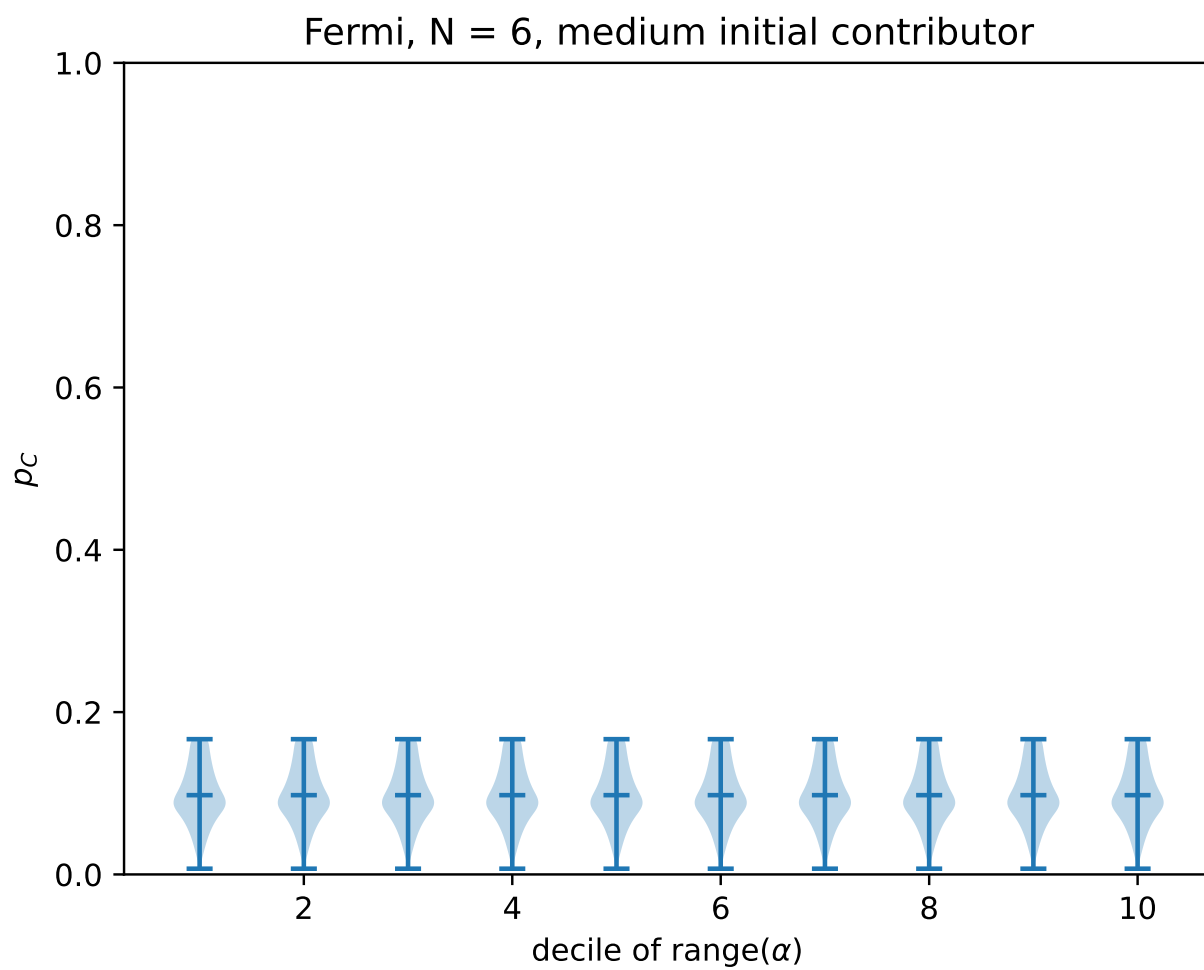


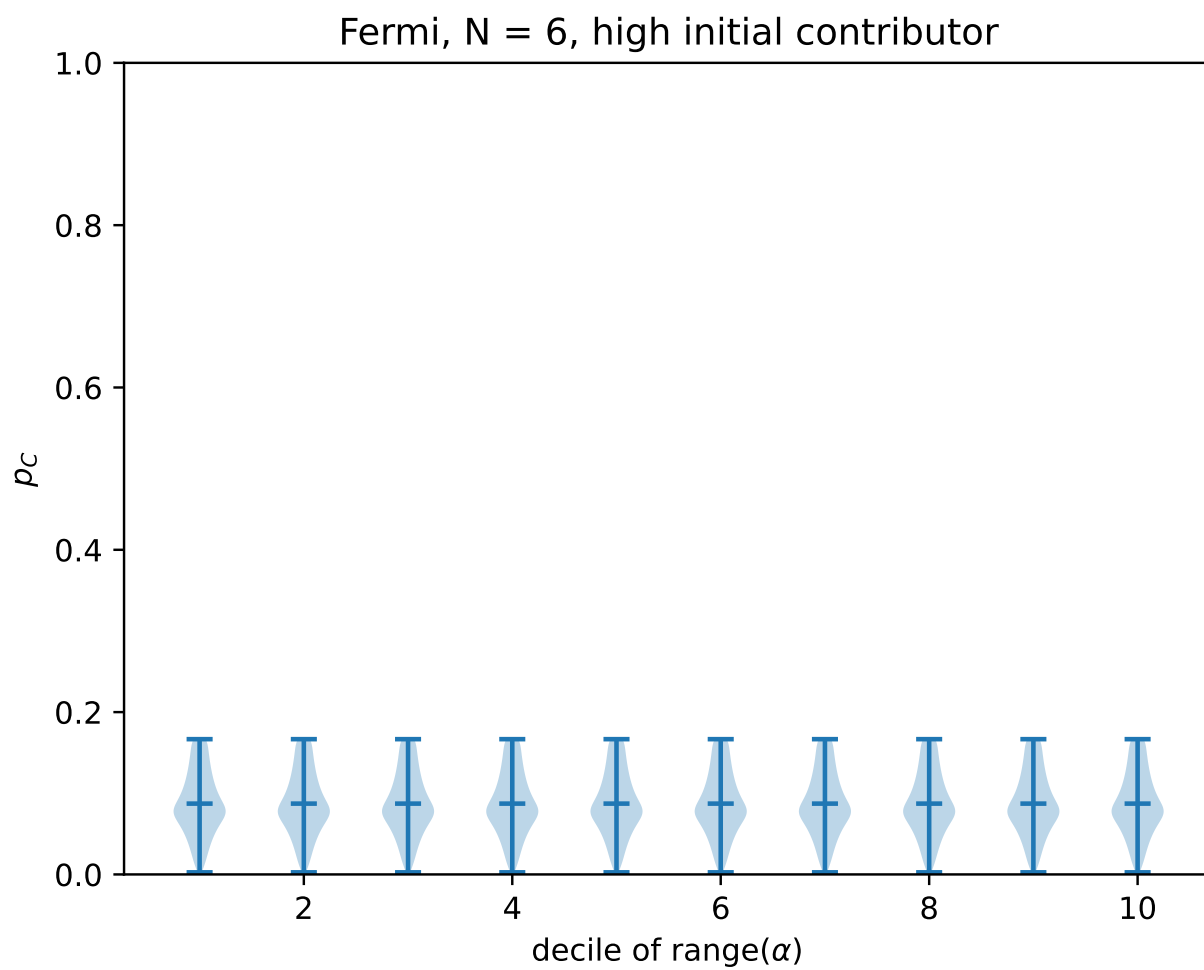


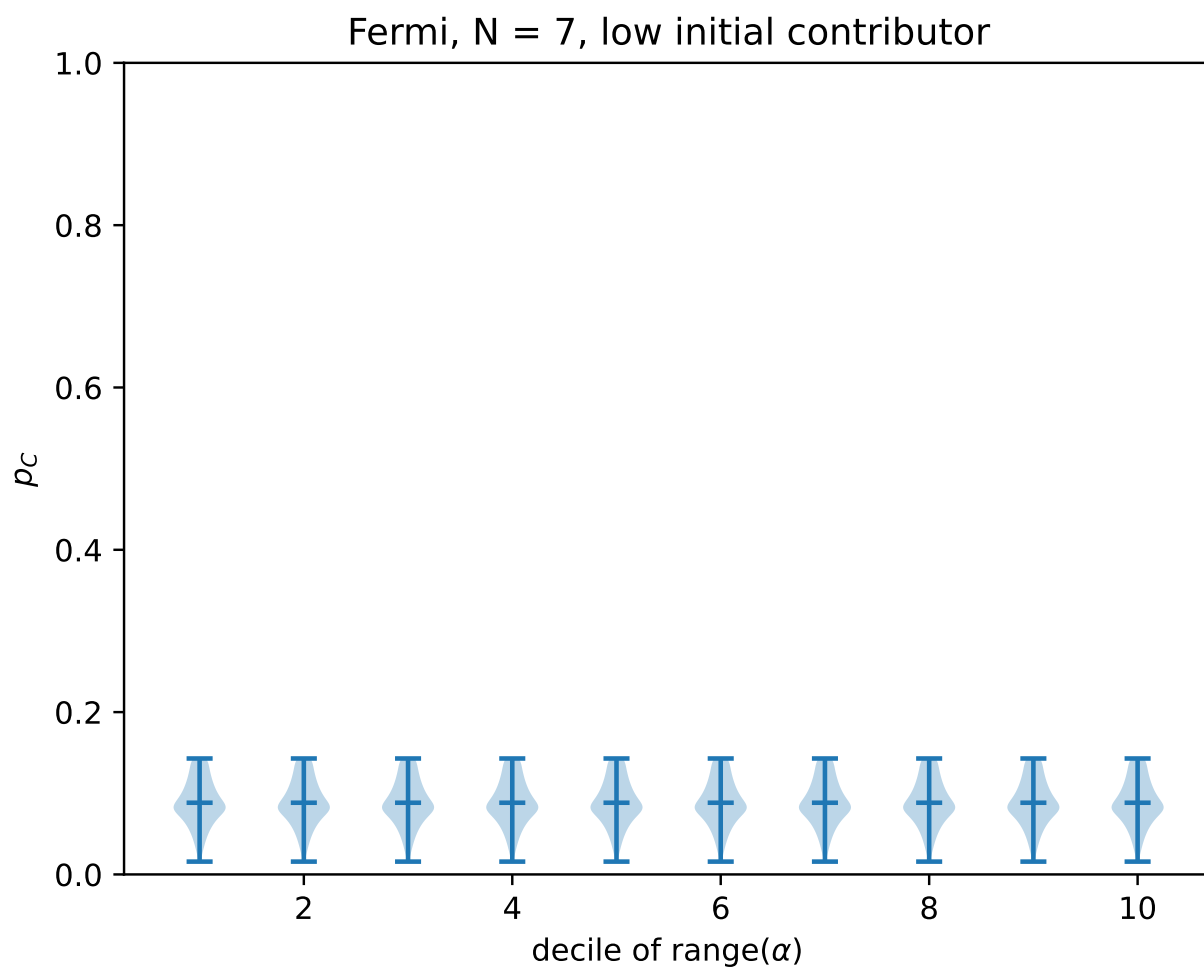


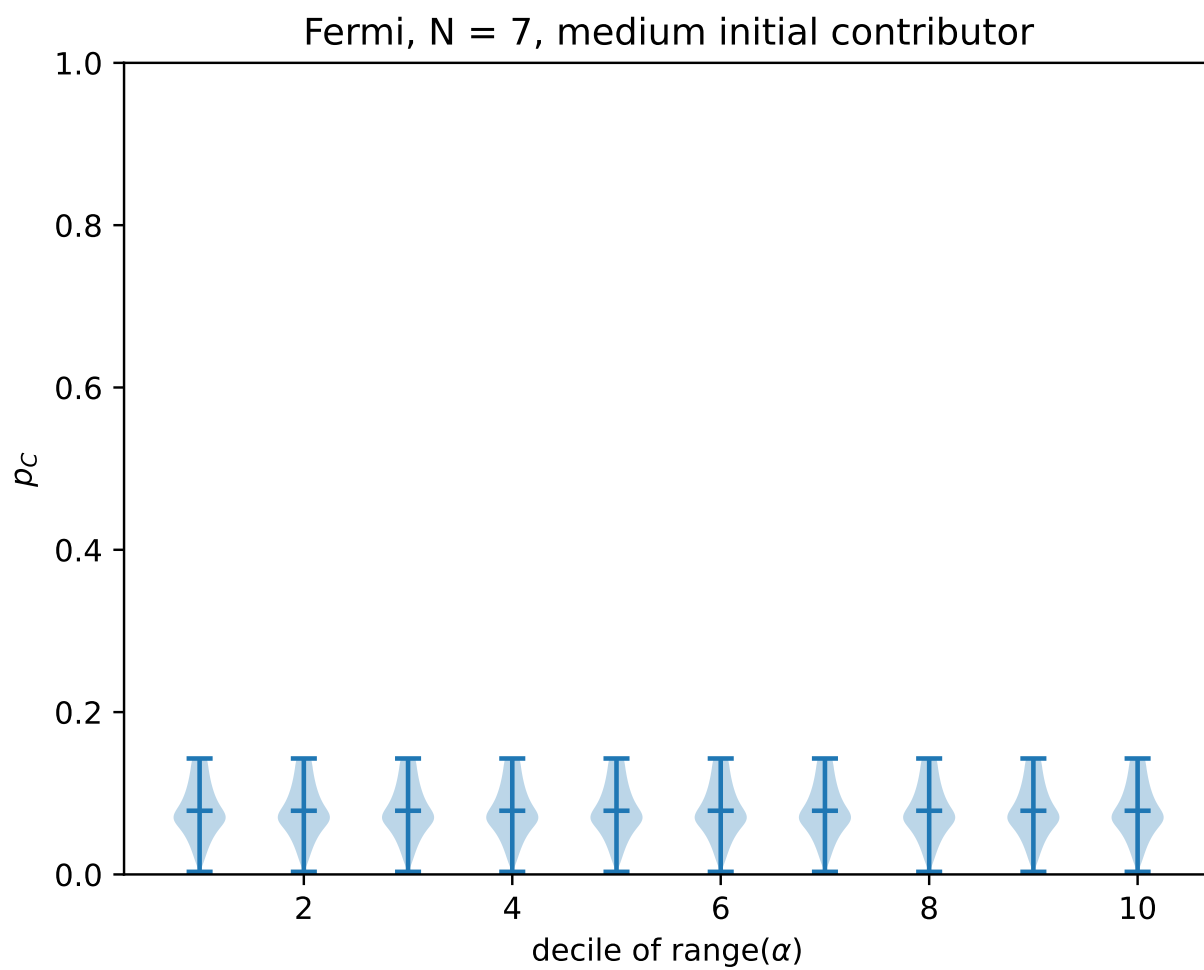


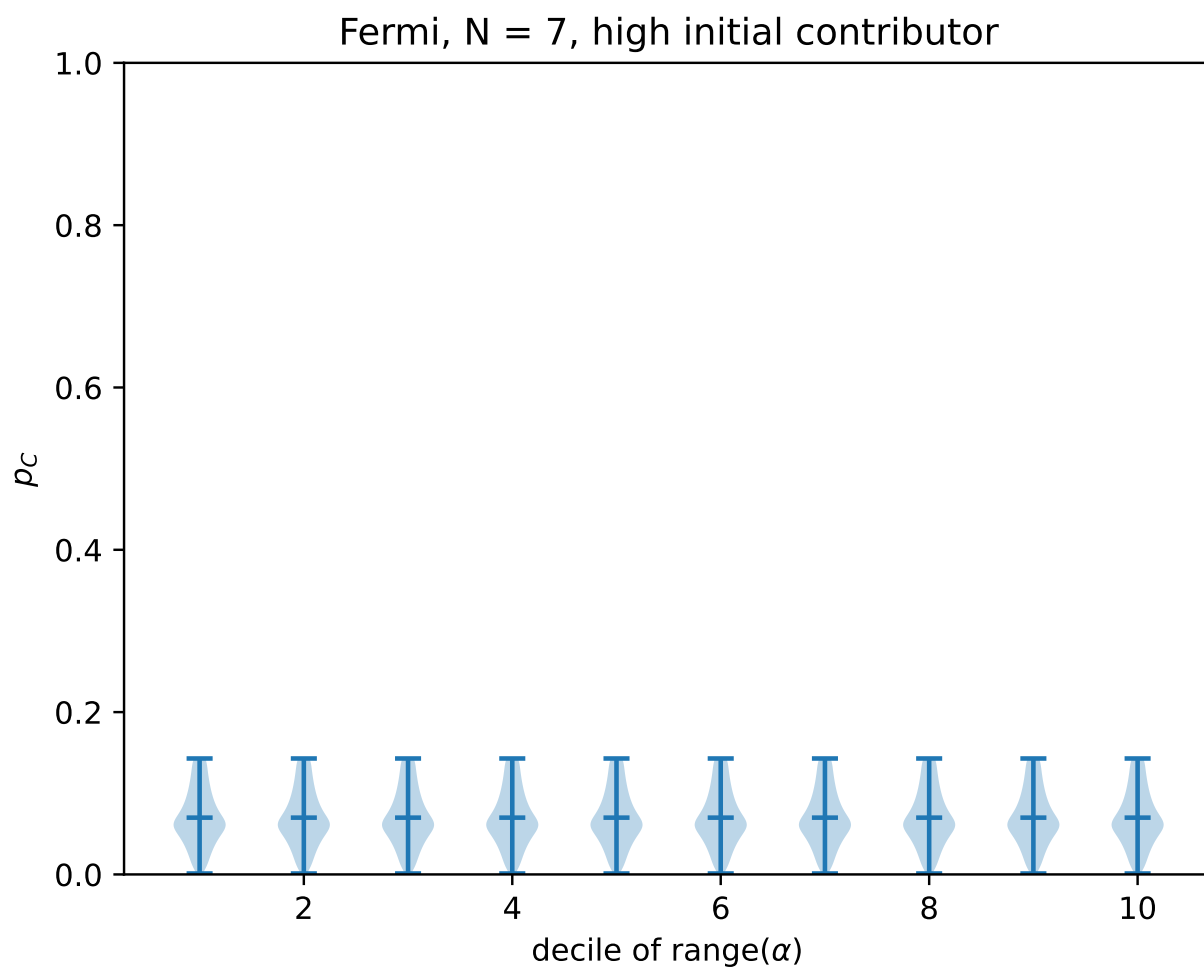


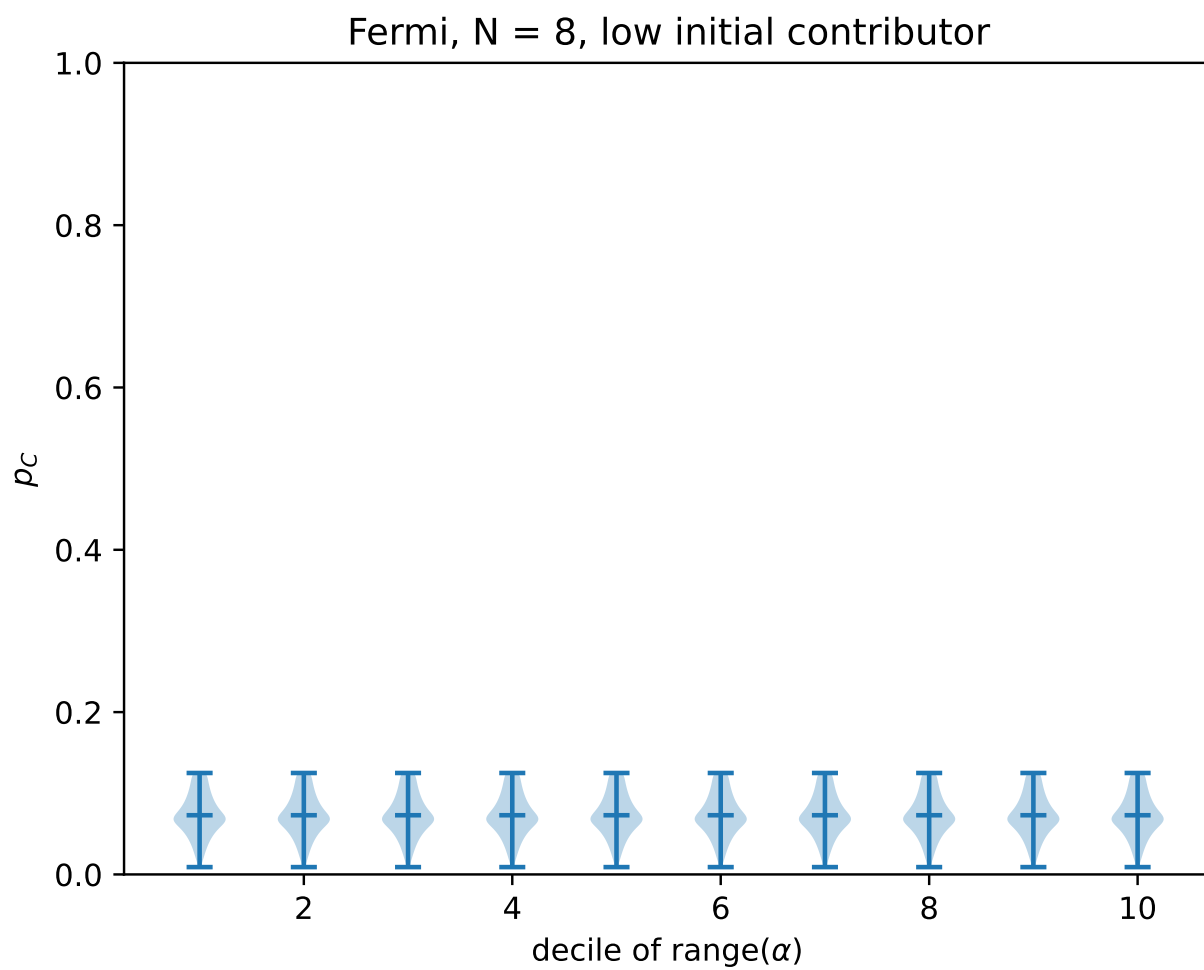


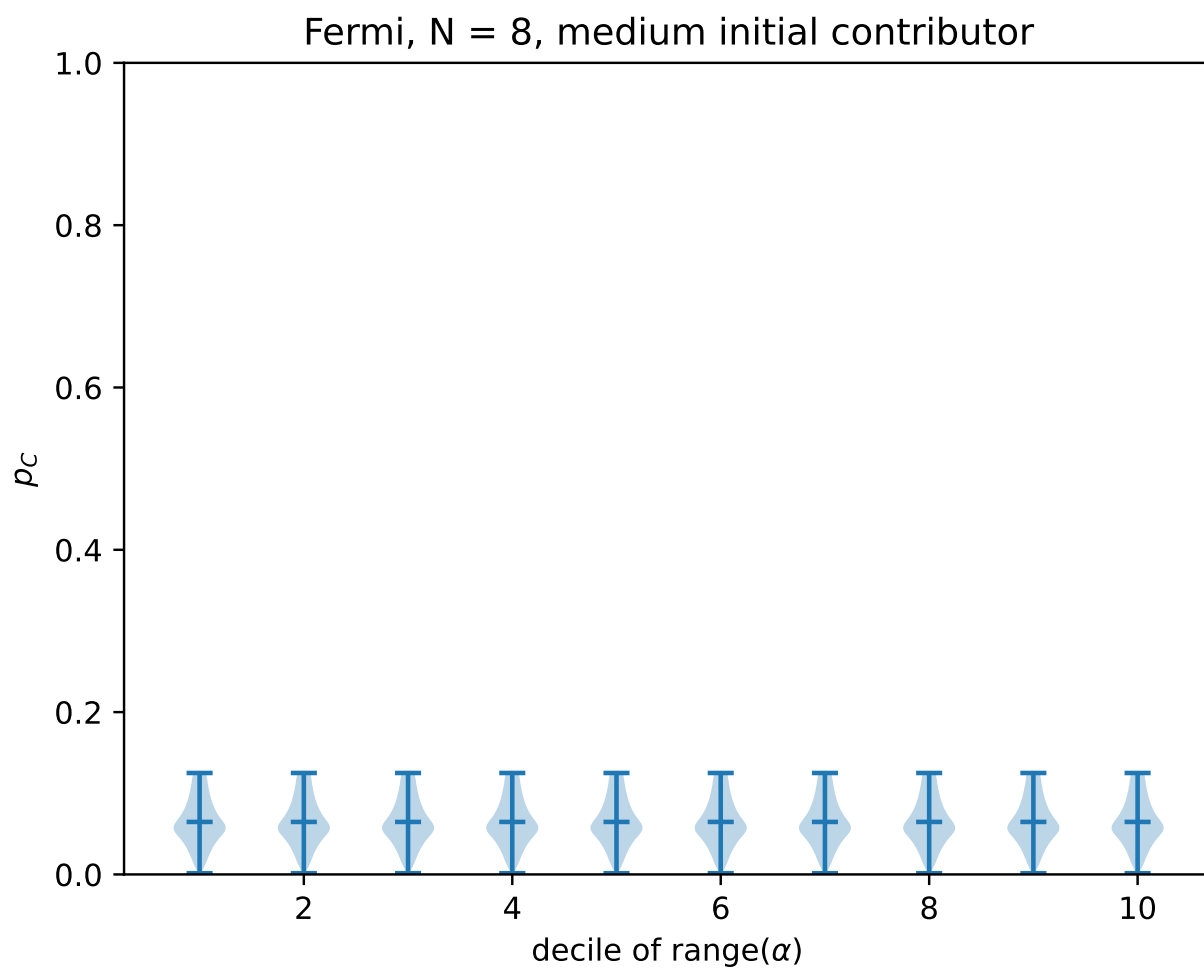


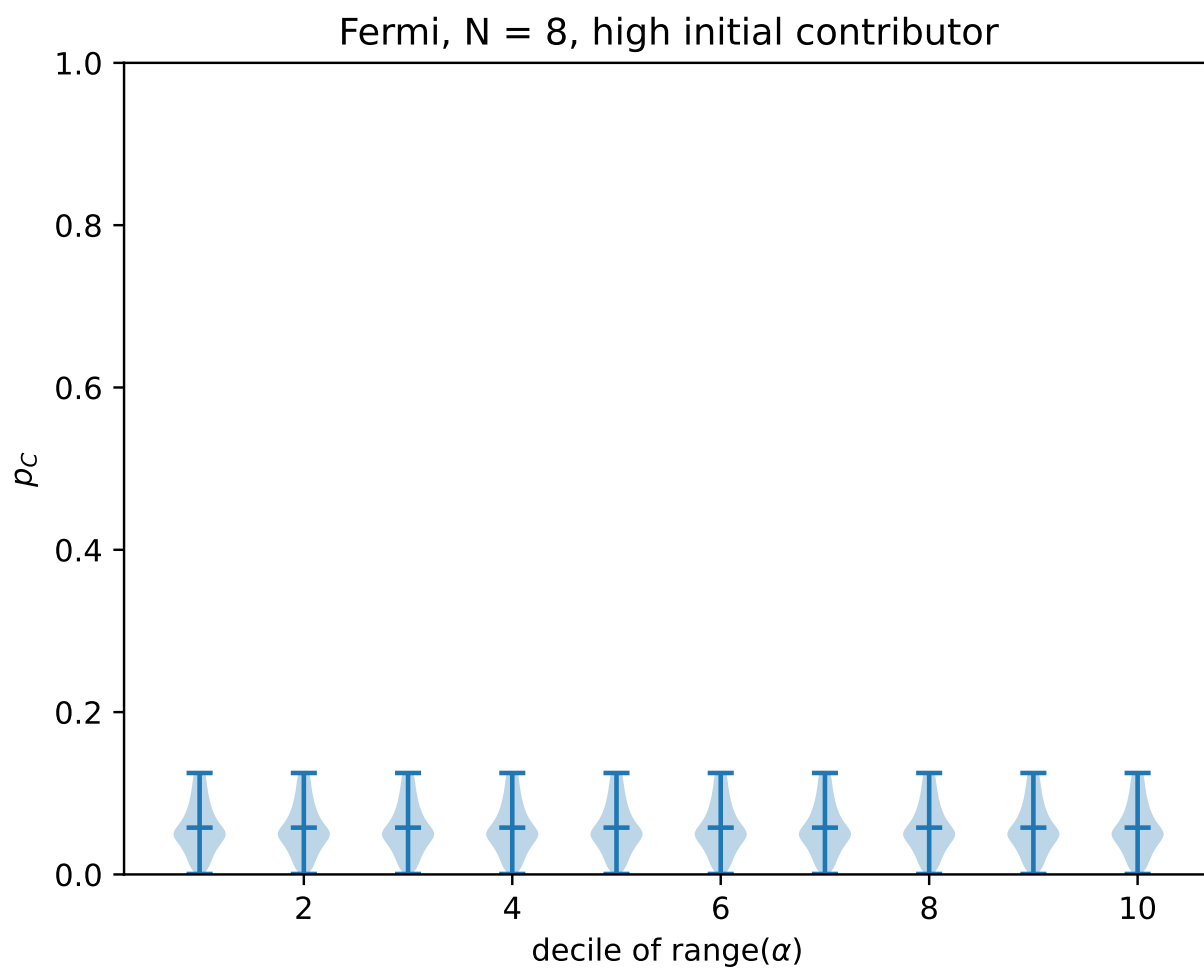


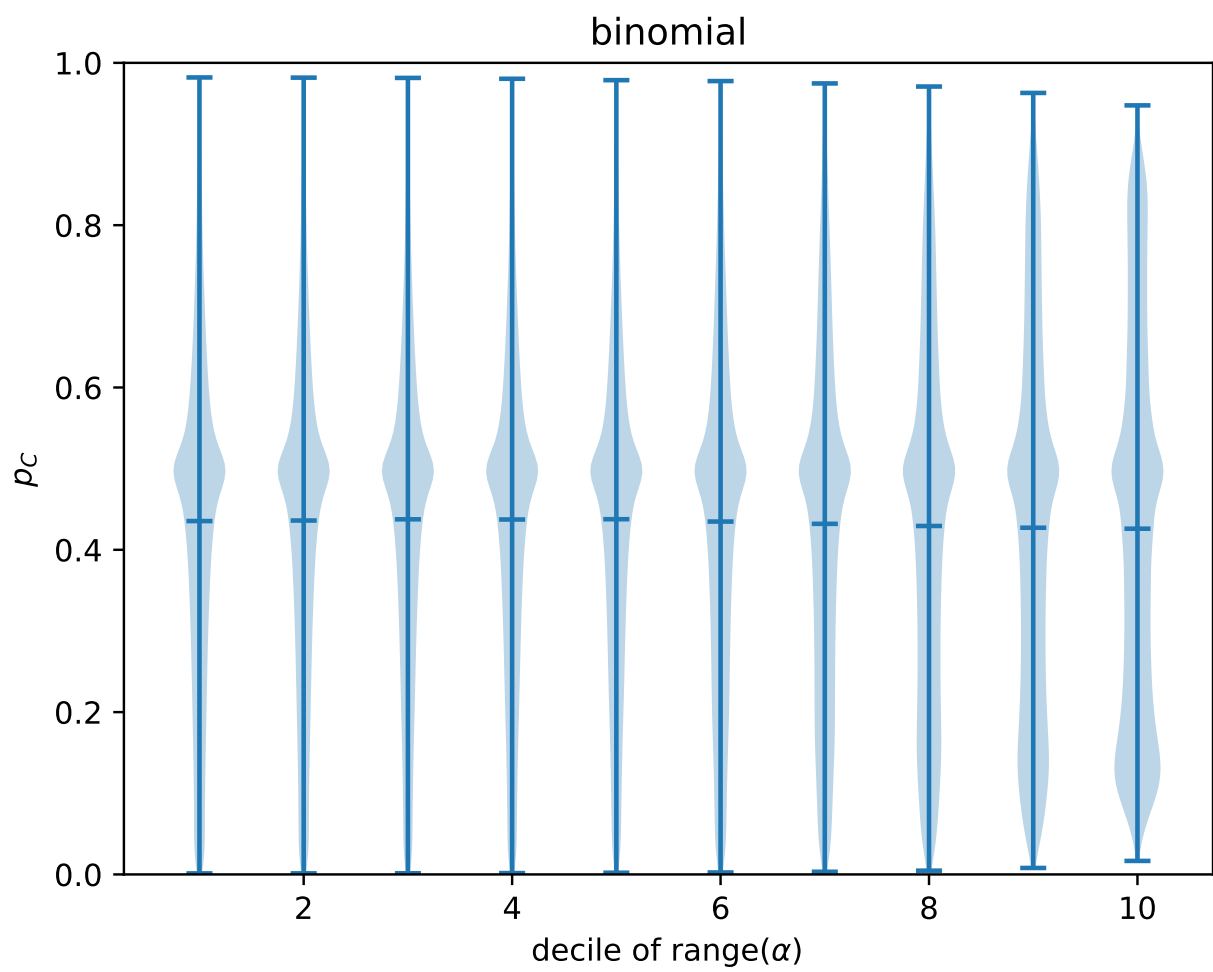


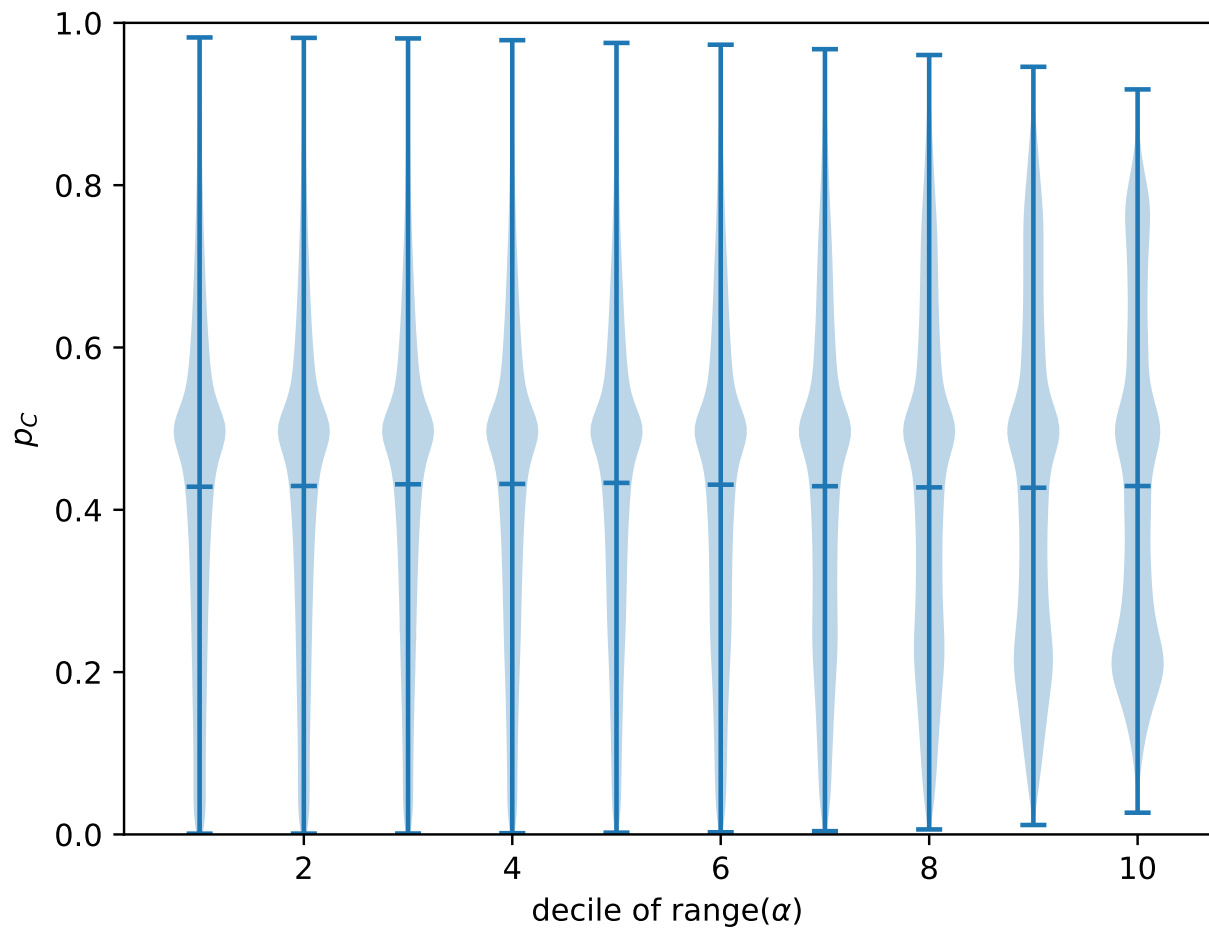


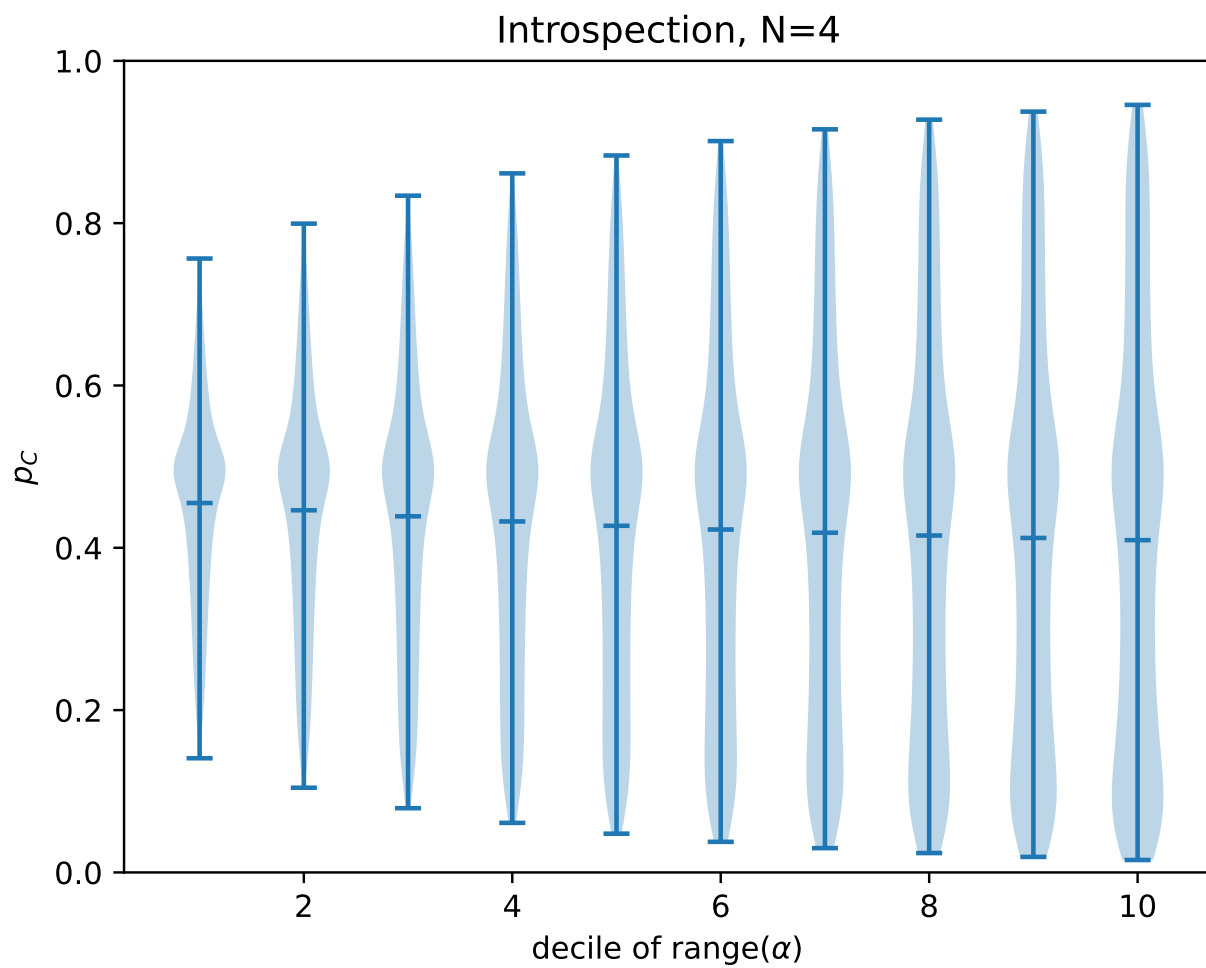


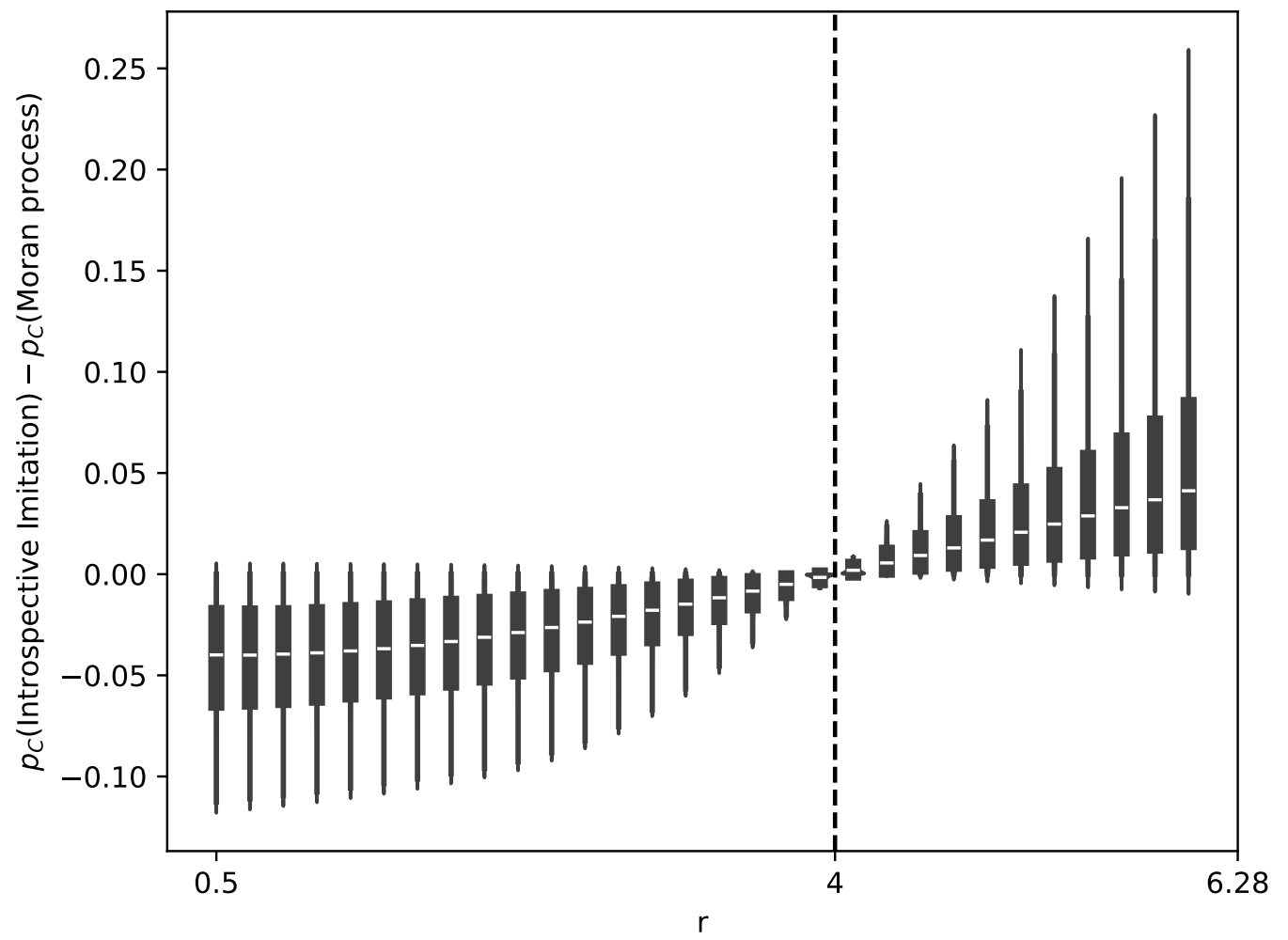


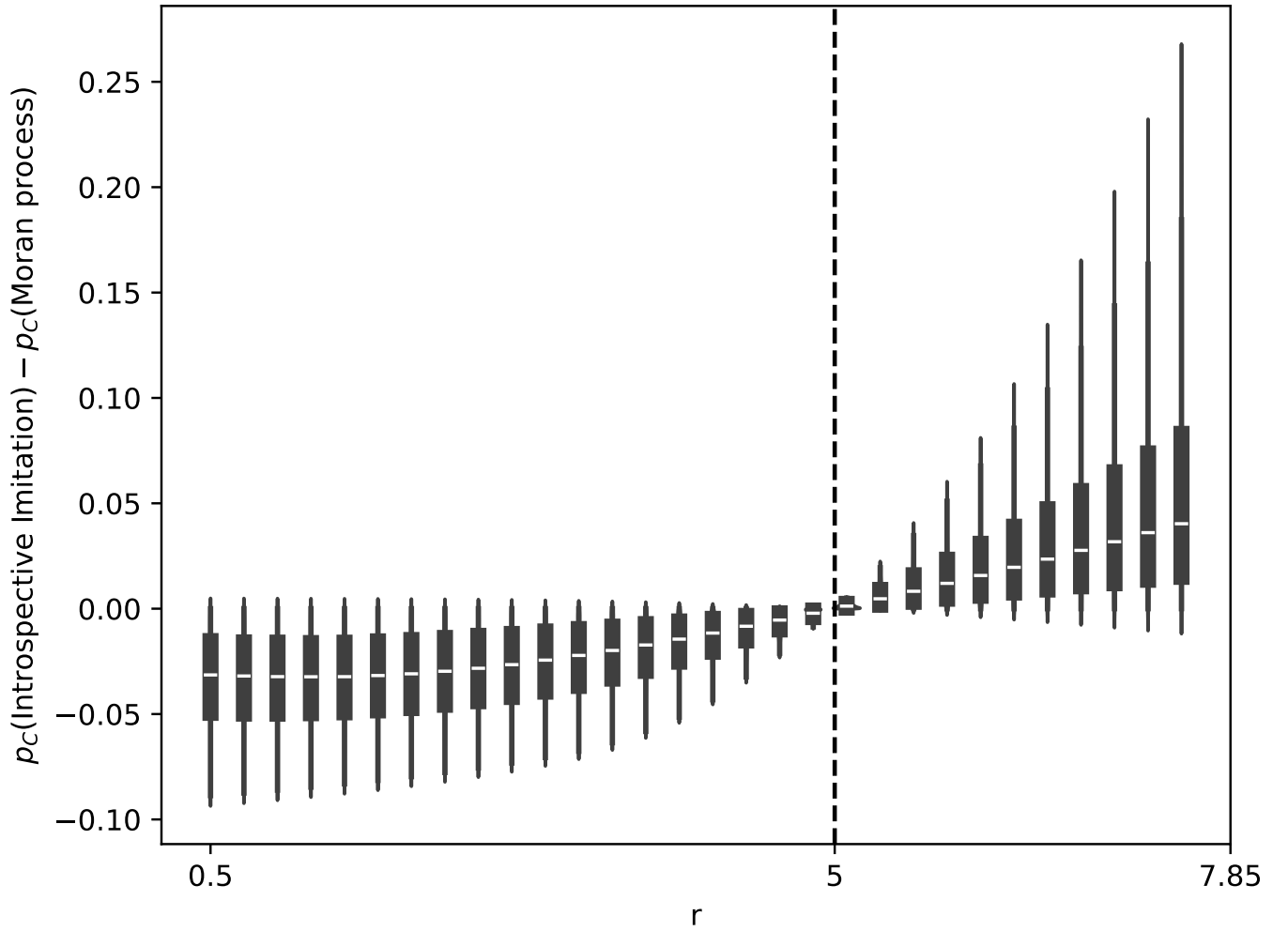












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